

Composite Failures and Fixes: Bulk-Fill vs. Traditional Techniques

Group 7

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Faculty Mentor: Dr. Comisi



Behavioral Objectives



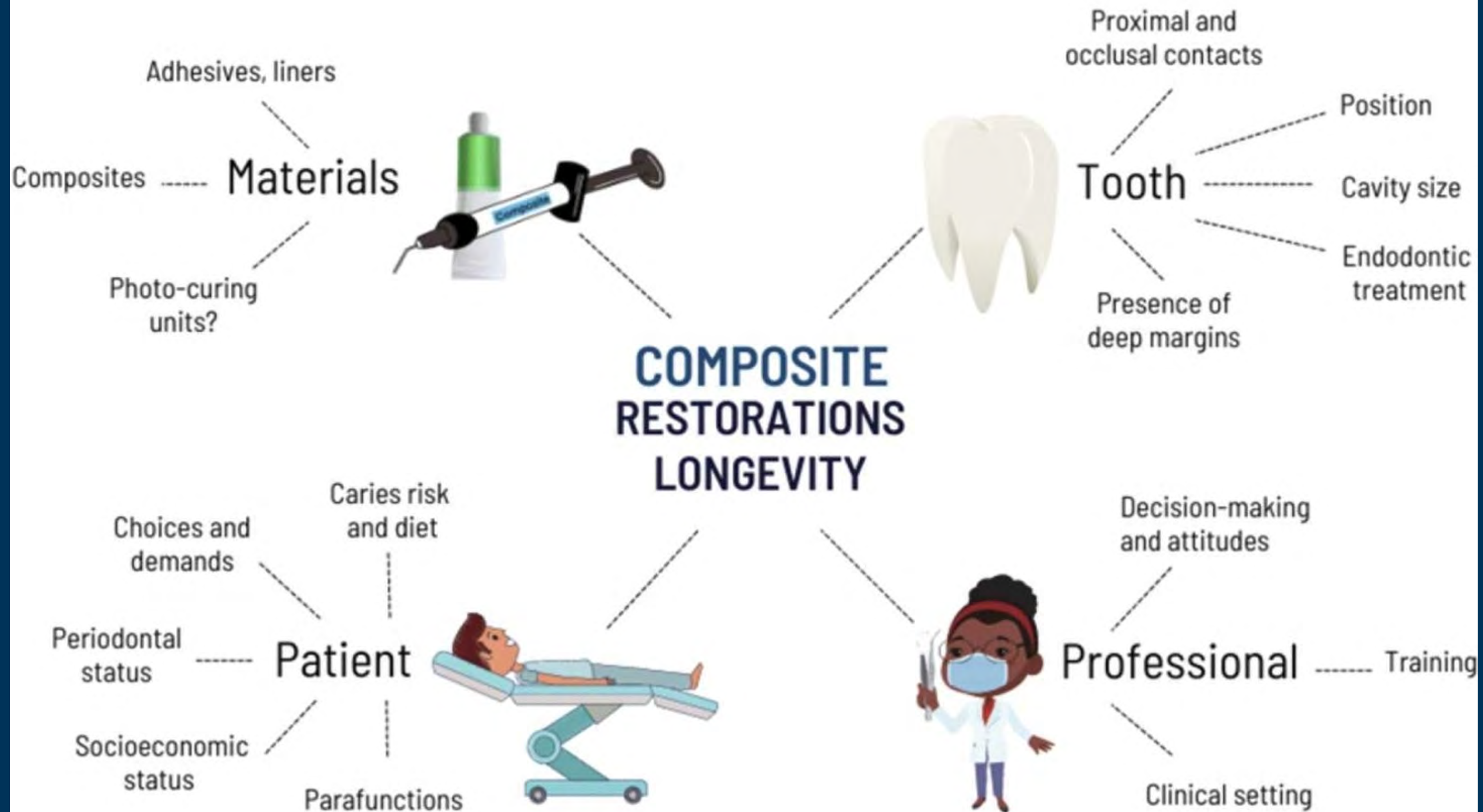
1. **Identify** the biological and mechanical factors that contribute to the failure of posterior composite restorations, including secondary caries, fracture, wear, and marginal breakdown.
2. **Differentiate** between conventional incremental layering and bulk-fill composite techniques by explaining their effects on polymerization shrinkage, stress distribution, and marginal adaptation.
3. **Explain** how material properties (filler type, translucency, depth of cure, and modulus of elasticity) influence the long-term clinical performance of posterior composites.
4. **Compare and contrast** the advantages and limitations of bulk-fill and traditional layering techniques based on current scientific evidence, focusing on marginal integrity, restoration longevity, and clinical efficiency.
5. **Apply** a clinical decision tree to **select** appropriate placement techniques and materials for posterior composites, considering patient risk factors, cavity design, and occlusal demands.



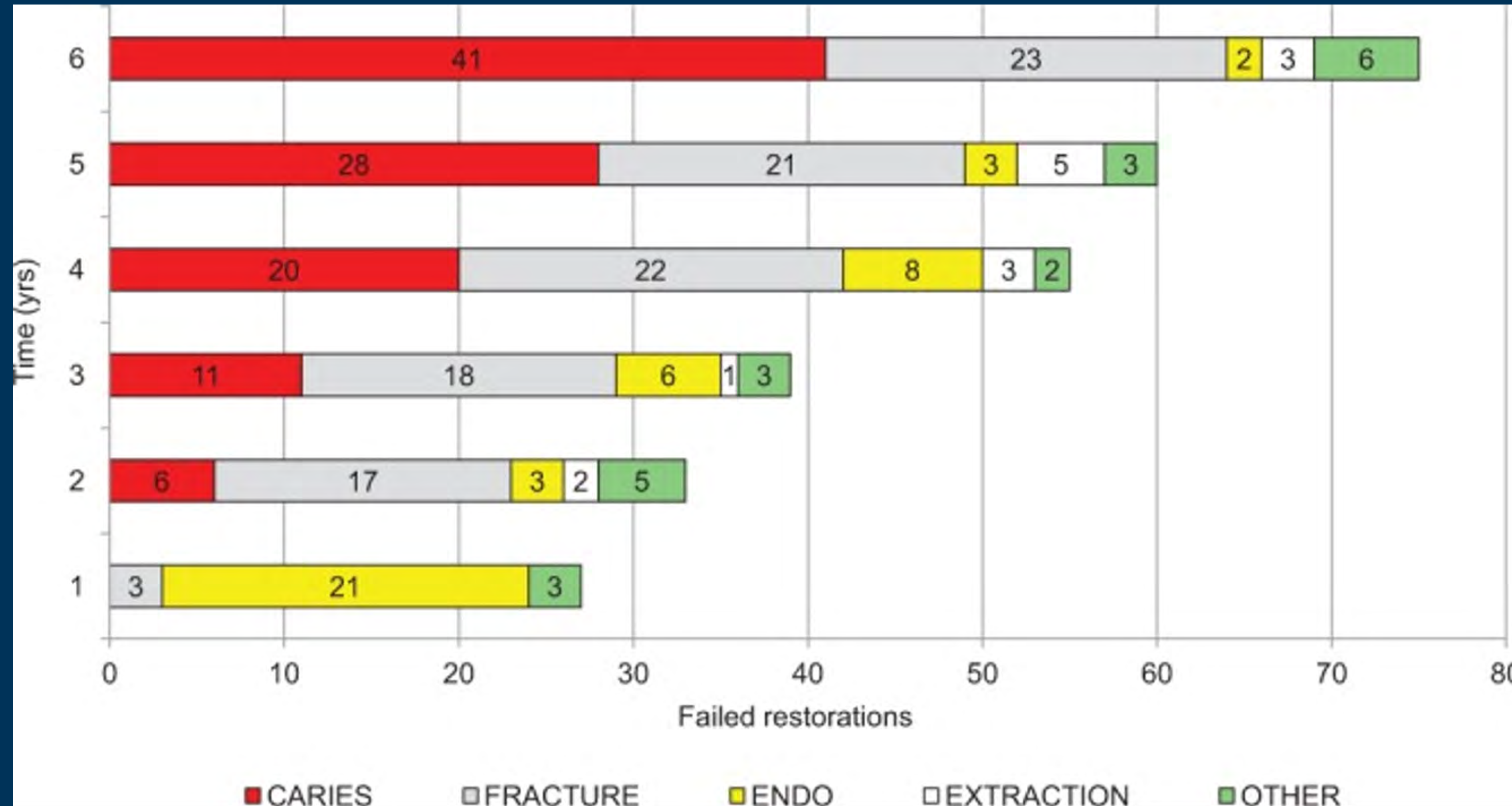
Identify the biological and mechanical factors that contribute to the failure of posterior composite restorations, including secondary caries, fracture, wear, and marginal breakdown.

Bhavi Patel & Isha Mittal

Factors that may affect the longevity of resin composite restorations



Common Causes of Failure?



- 1.8% failure at 5 years
- 2.4% failure at 10 years
- 4.6% in high caries risk patients
- 1.6% in low caries risk patients
- When comparing mean annual failure rate with amalgam
 - Composite: 2.9%
 - Amalgam: 1.6%

Search Strategy



P: Patients with posterior composite restorations

I: Biological and mechanical factors

C: No risk factors

O: Secondary caries, marginal breakdown, fracture, or wear

Databases used

Pubmed
Google Scholar

Keywords

secondary caries, composite, failure,
causes, fracture, wear

Highest level of evidence

Systematic reviews

Pubmed search string

1. ((secondary caries) AND (causes))
2. ((shrinkage) AND (composite failure))
3. ((secondary caries) AND (composite))
4. (resin composite) AND (fracture)
5. (resin composite) AND (wear)

Search results

1. 81 results, meta analysis, review, systematic review, 10 years
2. 11 results, meta analysis, review, systematic review, RCT, clinical trial, 10 years
3. 191 results, meta analysis, review, systematic review, RCT, clinical trial, 10 years
4. 133 results, meta analysis, review, systematic review, 10 years
5. 66 results, meta analysis, review, systematic review, 10 years



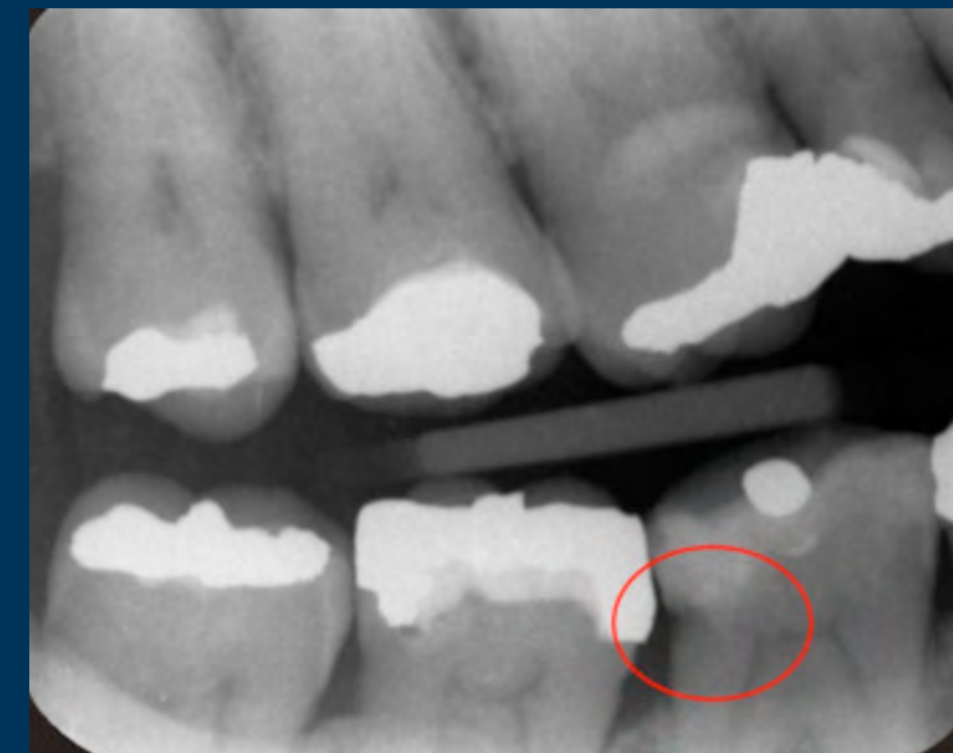
Primary Caries

Secondary Caries

- On 'fresh' tooth structure
- Not adjacent to any restoration



- Adjacent to an existing restoration on the same tooth
- can NOT be present on a tooth with no restorations



- Breakdown of tooth structure by cariogenic bacteria
- Formation depends on OH of patient

The Great Debate - Stain vs Secondary Caries



Stain - discoloration at the margin of restorations

- Stain is the first clinical sign of marginal debonding and breakdown
 - caused by microleakage
 - beginning of secondary caries
 - secondary caries usually caused by remaining primary caries

Versus

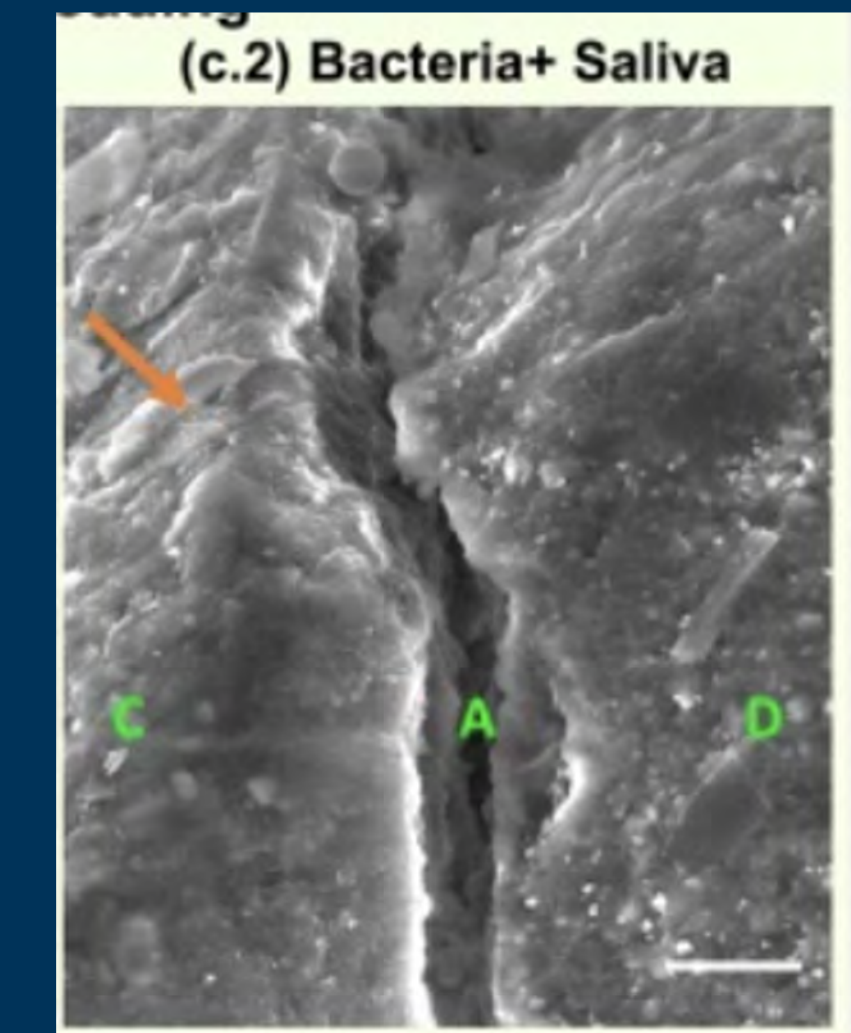
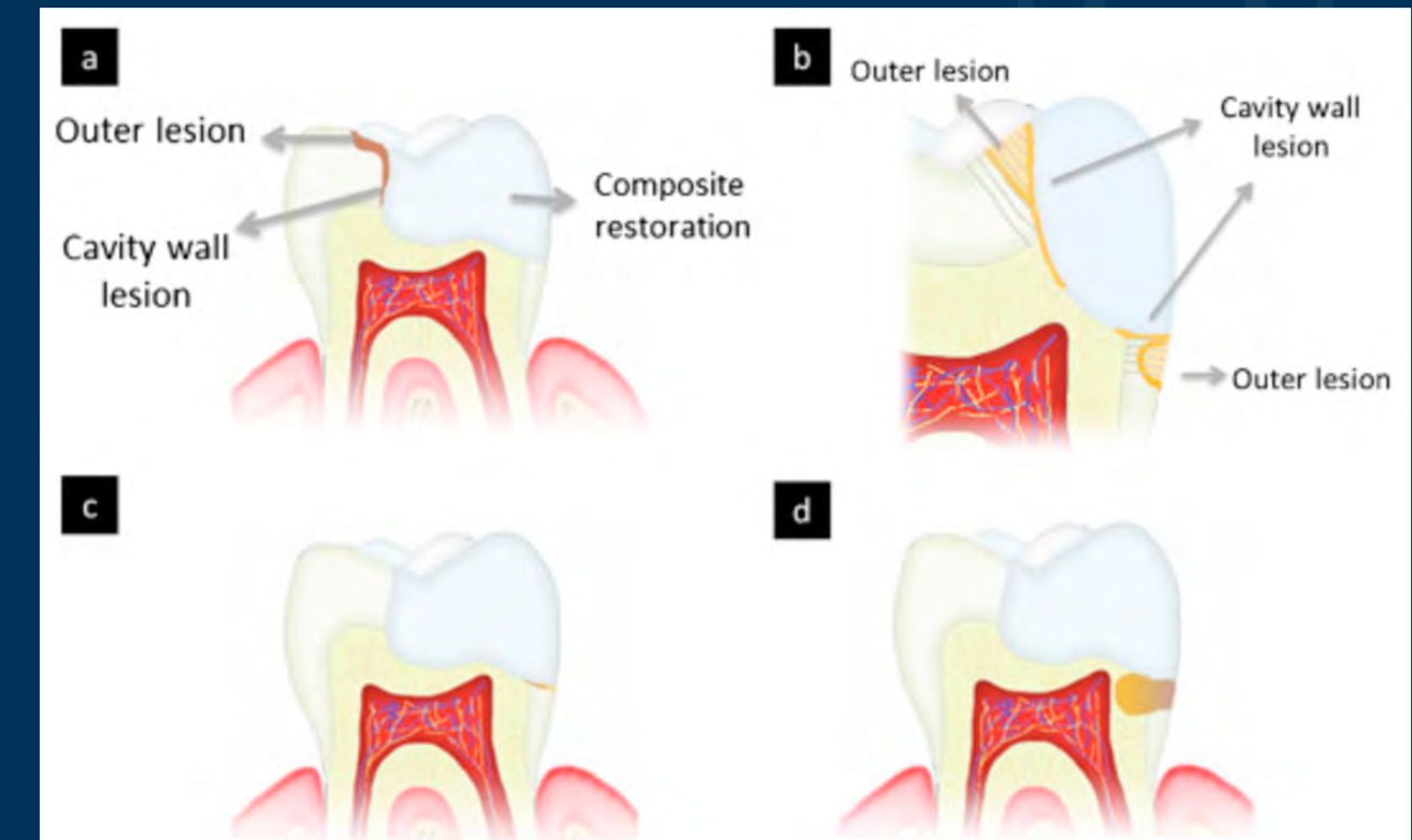


- True secondary caries has cavitation and high caries activity
 - Staining alone does not equate to secondary caries
 - Based on more of a “wait and watch” ideology
 - no longer a “when in doubt, take it out” mentality
 - need radiographic proof and/or symptoms

Radiographic proof of demineralization? Caries risk of patient?

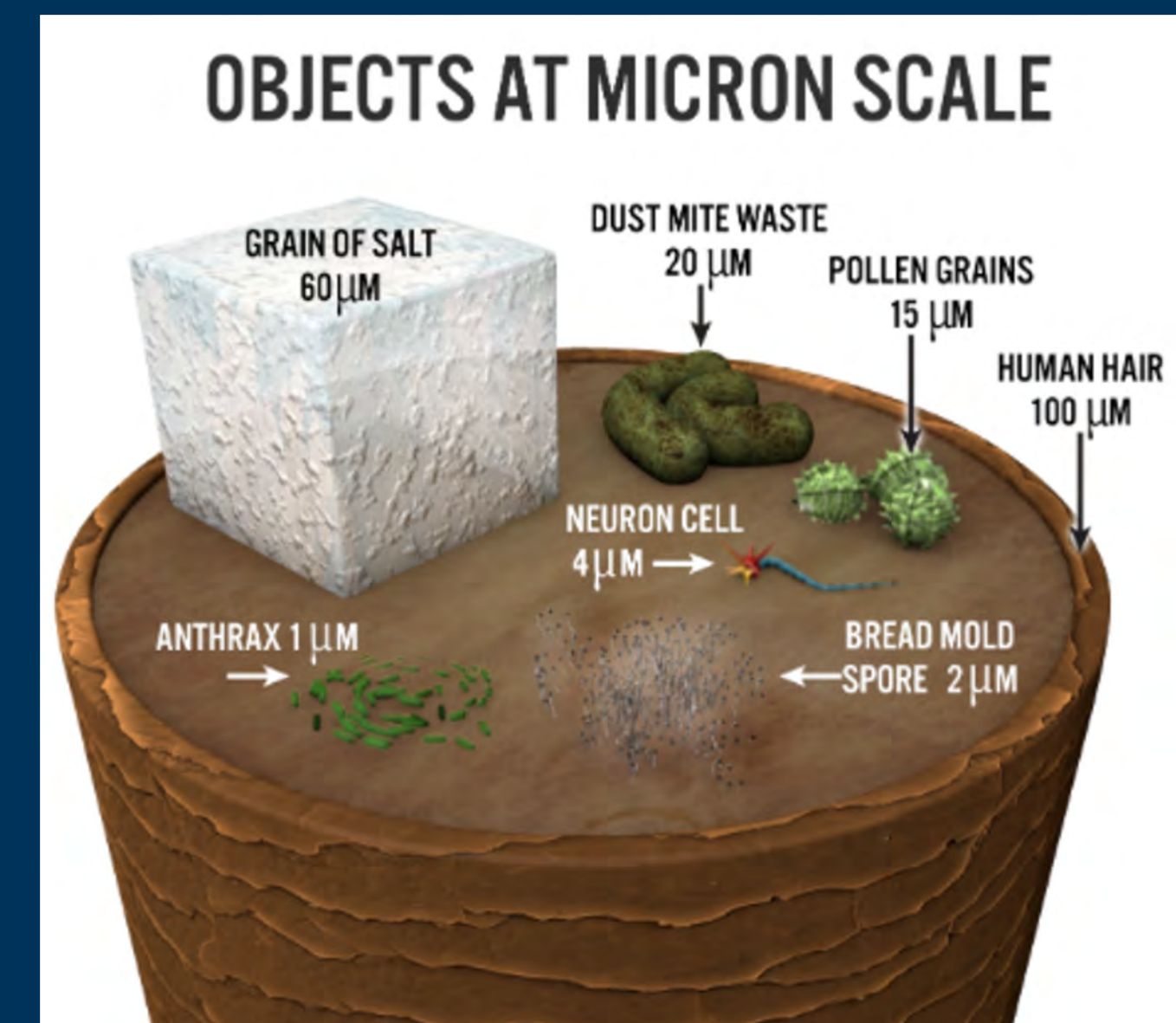
Marginal Breakdown

- Gaps in the margin (tooth-composite interface) that can lead to leakage, post op sensitivity, and secondary caries
- Can occur from polymerization shrinkage
 - once resin composite is cured, it goes from a monomer bound loosely by Van der Waal forces to a tightly bound polymer via covalent bonds -> composite bonds tightly to itself and pulls away from the tooth surface = deficient margin
 - Avg = 2-6% (Higher C factor = Higher shrinkage)



Marginal Breakdown

- Can be felt tactically with explorer - do not stick into occlusal but along margins
- Study by Kuper et. al showed that a gap as small as 60 microns can lead to secondary caries formation
 - Cariogenic bacteria easily fit into that gap (*S. mutans* 0.5-0.75 microns in size)
 - produce acids and esterases that will break down tooth structure and resin composite
 - Dental explorer tip is around 68 microns in width



What is Wear?

- Results from weakening/loss of composite matrix material
- A chronic process compared to acute breakage seen in fracture
- Quantified as height/volume/weight reduction



Ritter, A. V. (2008). Posterior composites revisited. *Journal of Esthetic & Restorative Dentistry*, 20(1).

What is Fracture?

- Physical breakage or splitting of the restoration and/or surrounding tooth structure
 - Assessed with scanning electron microscopy
- May originate within the composite restoration, at the tooth-restoration interface, or propagate through the tooth itself
- 36% of resin-composite restoration failures within the first 6 years are due to fracture
- Fractures, as opposed to secondary caries, increasingly cause failure

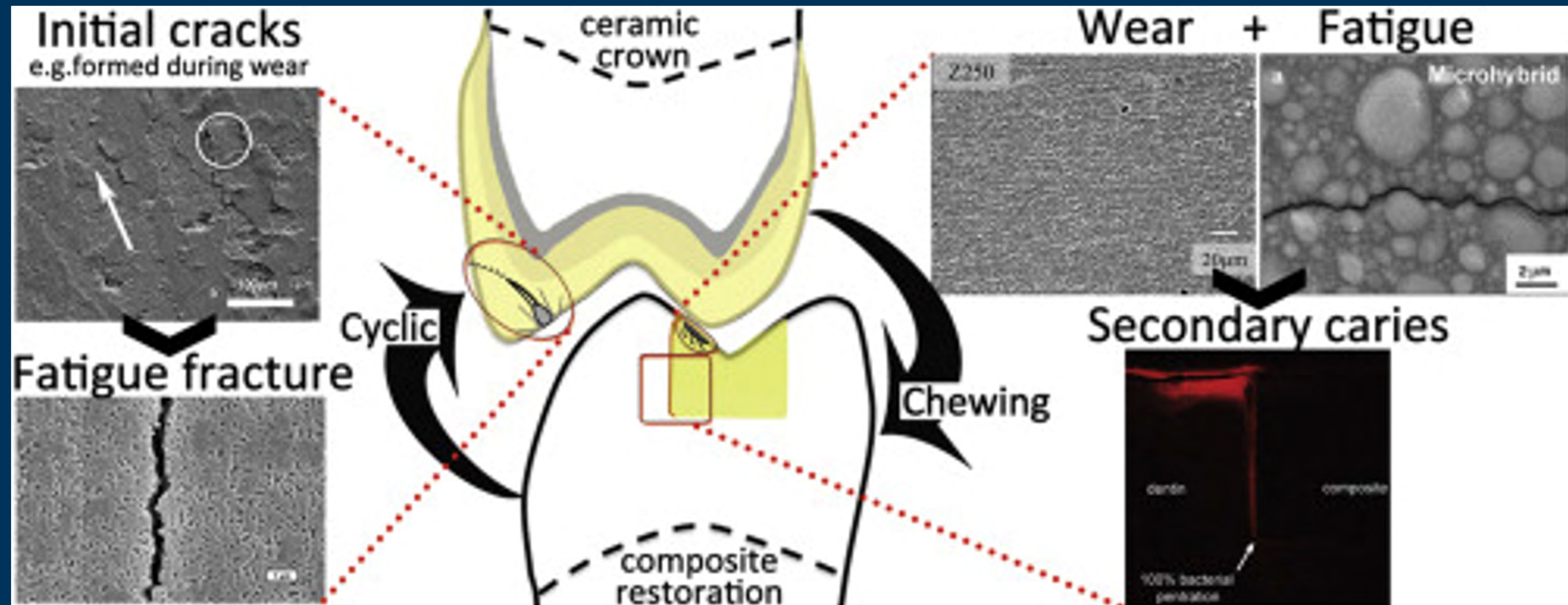


pbdc.ae/cases/re-restoration-of-tooth-26-with-fractured-old-composite-restoration/

Factors Associated with Higher Risk of Fracture and Wear



- Fatigue = fracture due to **chronic cyclic masticatory forces** (mastication, bruxism, etc.)
 - Important concept in the study of composite longevity
 - Experienced on average $> 3 \times 10^5$ times/year
 - ~822 times/day
 - Associated with higher formation of secondary caries
- Functional and parafunctional loading
 - Fracture resistance of healthy teeth: 1460 N
 - Composite restored teeth: 771 N
 - Forces generated from bruxism can exceed 800 N
- Shrinkage/stress from polymerization → marginal gaps, debonding, microleakage



Kruzic, J. J., et al. (2018). Recent advances in understanding the fatigue and wear behavior of dental composites and ceramics. *Journal of the mechanical behavior of biomedical materials*, 88, 504-533.

Factors Associated with Higher Risk of Fracture and Wear (Cont.)



- Stress concentration
 - Voids, sharp preparation, unpolished composite
- Poor adaptation or retention
- Complex cavity design (multi-surface, deep)

Biologic

- Biofilm and bacterial growth → acidic environment → hydrolyzing of adhesive
- Oral enzymatic activity weakens the composite matrix

Mastication Forces on Teeth & Composite Restorations

Interactive demonstration of occlusal loading effects

Note: Values shown are for demonstration purposes. Consult current literature for specific clinical data on composite wear rates and fracture thresholds.

Current Force

0 N

Cycles Completed

0

Wear Depth

0 μm



Start Animation

Pause

Reset

Speed: 4x

Enamel

Dentin

Composite Filling

Pulp

Normal occlusal loading - No pathology

Differentiate between conventional incremental layering and bulk-fill composite techniques by explaining their effects on polymerization shrinkage, stress distribution, and marginal adaptation.

Maliheh Sehati and Kardarius Jackson

Search Strategy

Databases used

Pubmed

Google Scholar

Trip Database

Galileo

Pubmed search string

1. ((bulk-fill composite) AND (layering composite)AND (Polymerization shrinkage))
2. ((bulk-fill composite) AND (layering composite)AND (Shrinkage stress))
3. ((bulk-fill composite) AND (layering composite)AND (marginal adaptation))

Search results

1.31

2. 18

3.33

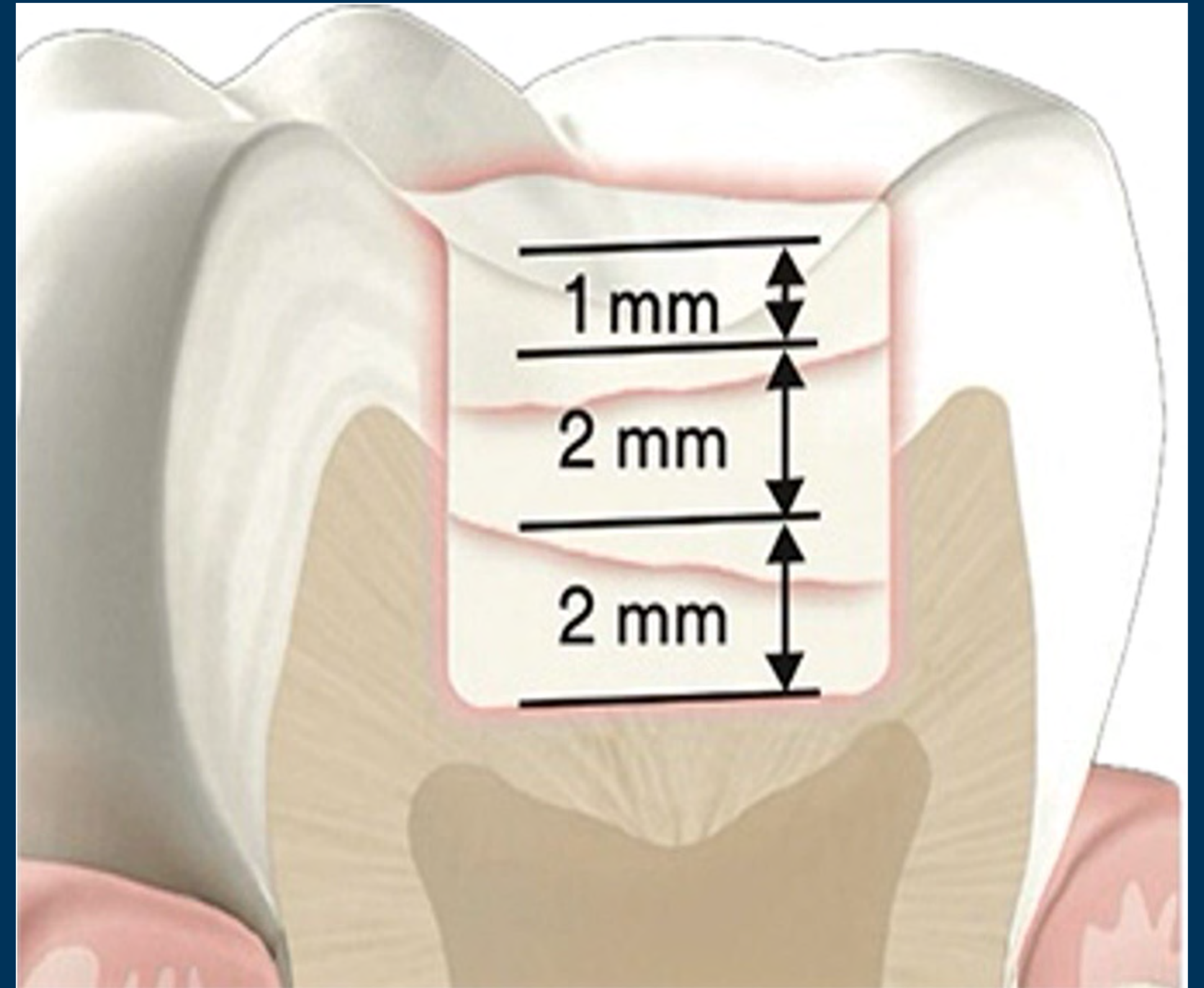
Highest level of evidence

Systematic reviews



Incremental Layering

Composite is placed in thin increments (≤ 2 mm), each layer is light-cured individually.





Why Incremental Layering?

Reduces polymerization shrinkage

Decreases shrinkage stress (lower C-factor per increment)

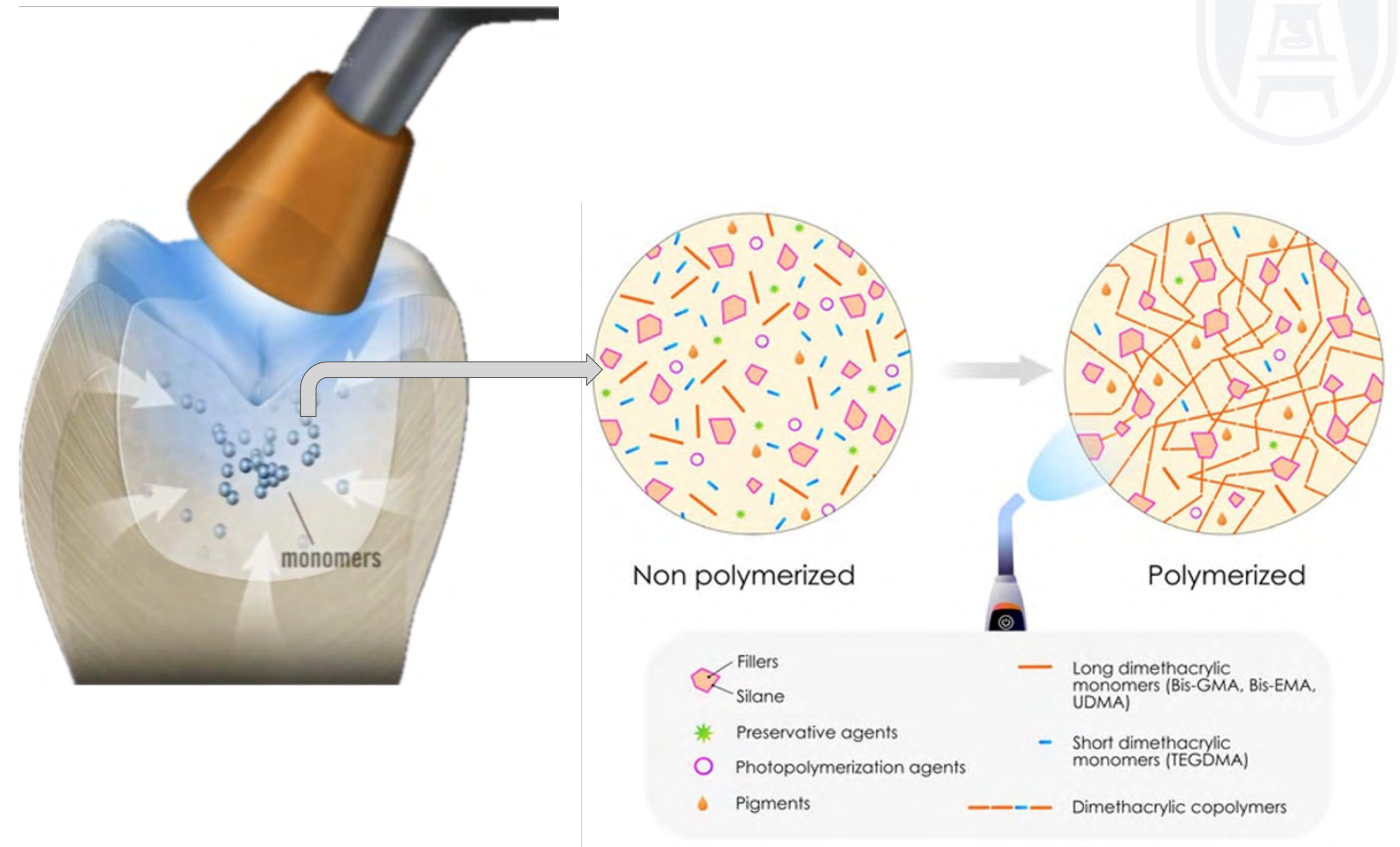
Ensures complete light curing

Improves marginal adaptation and seal



Polymerization Shrinkage

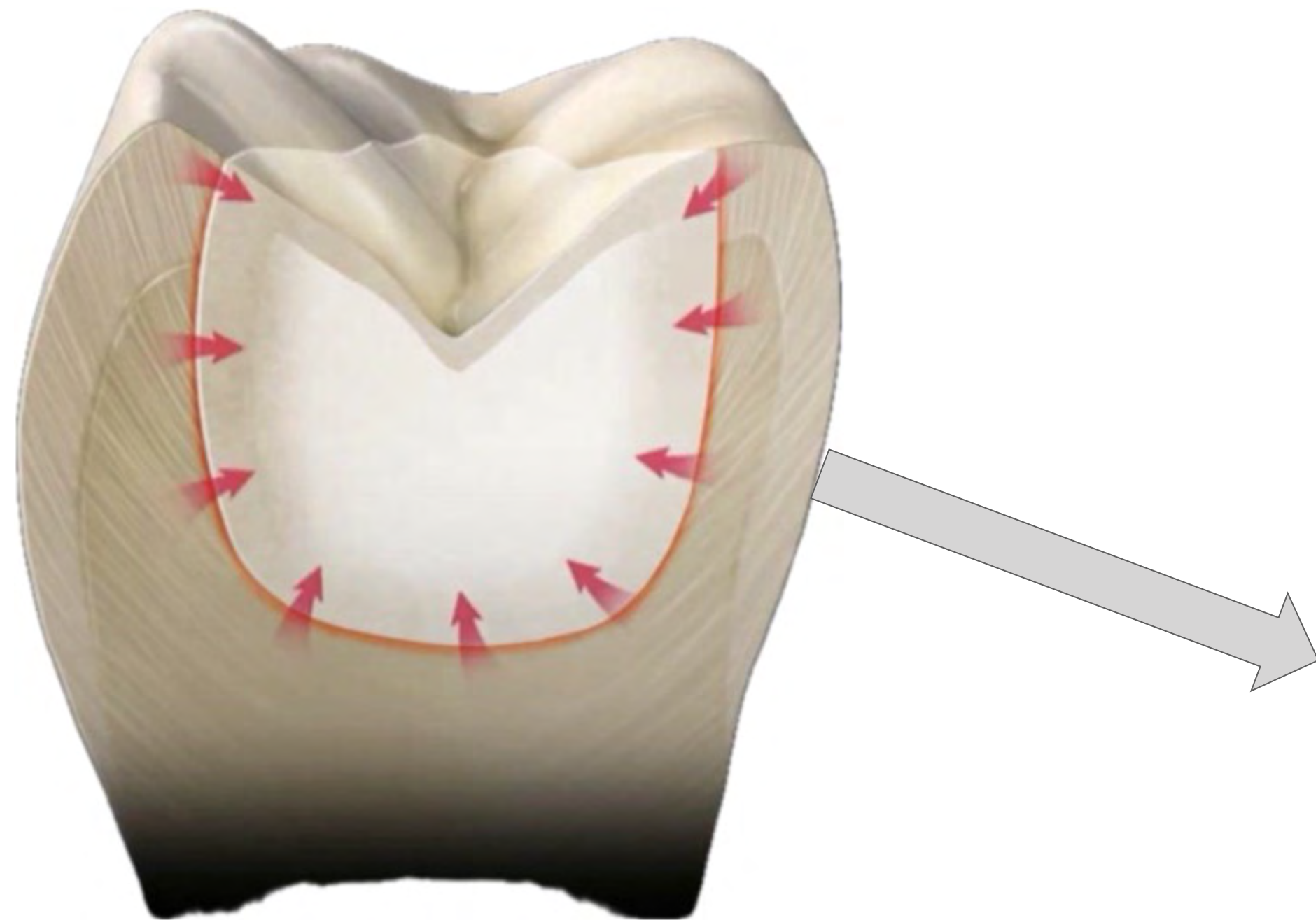
In the polymerization of a resin composite, weak, longer-distance van der Waals forces between individual monomer molecules are replaced with much stronger, shorter-distance covalent bonds to form a rigid polymer network.



Reduces free volume → volumetric shrinkage



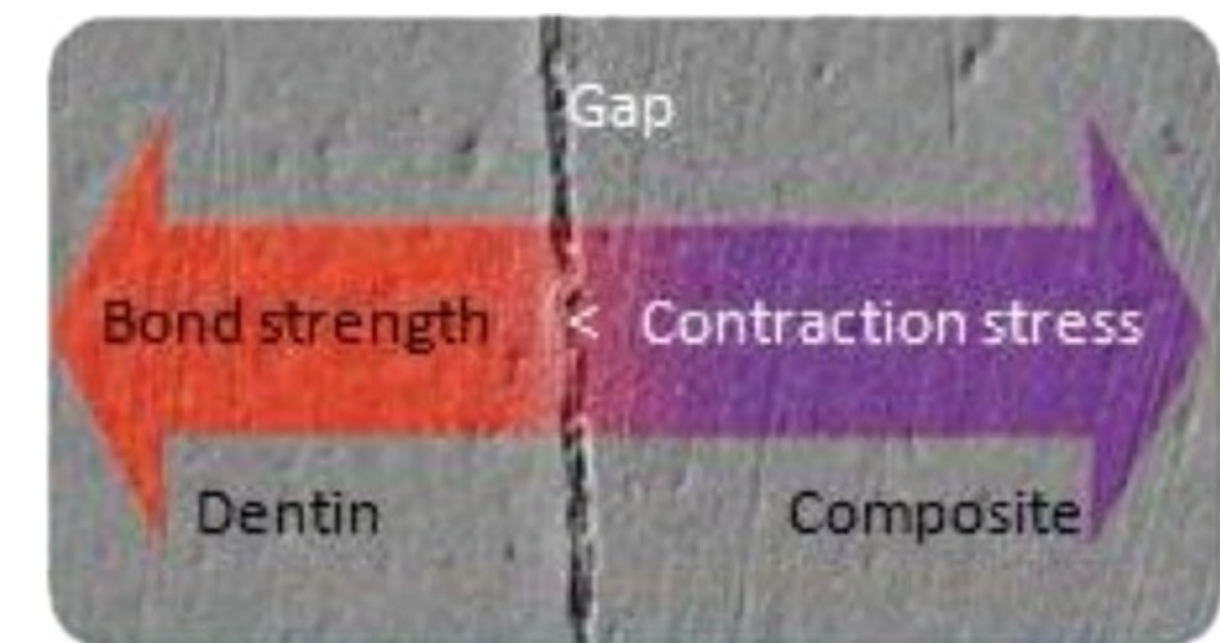
Polymerization shrinkage, stress distribution and marginal adaptation



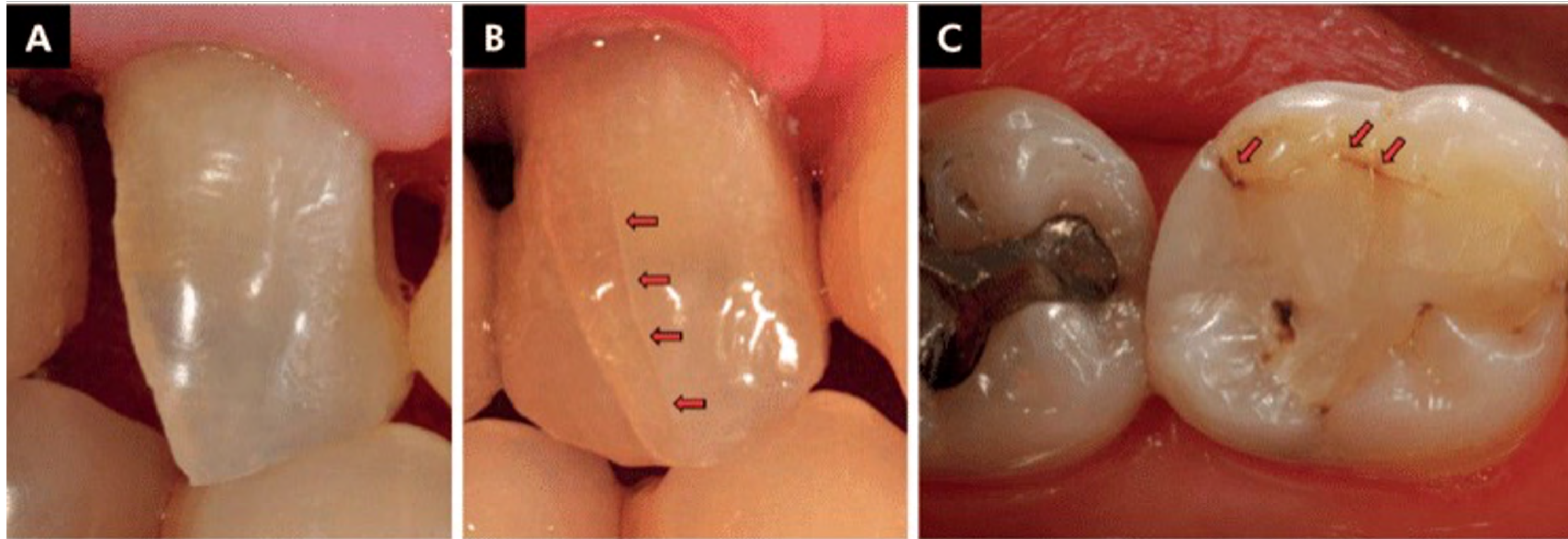
Polymerization Shrinkage induces contraction forces (Stress) at the tooth–restoration interface.



Stress can exceed the bond strength and lead to marginal gap formation and microleakage.



Clinical Signs of Polymerization Shrinkage



A. Buccal cusp after deep caries removal showing no enamel crack; B. Enamel crack (red arrows) generated immediately after incremental filling composite resin restoration; C. Marginal staining (red arrows) of the restoration after 5 years in function.

Secondary caries due to polymerization shrinkage

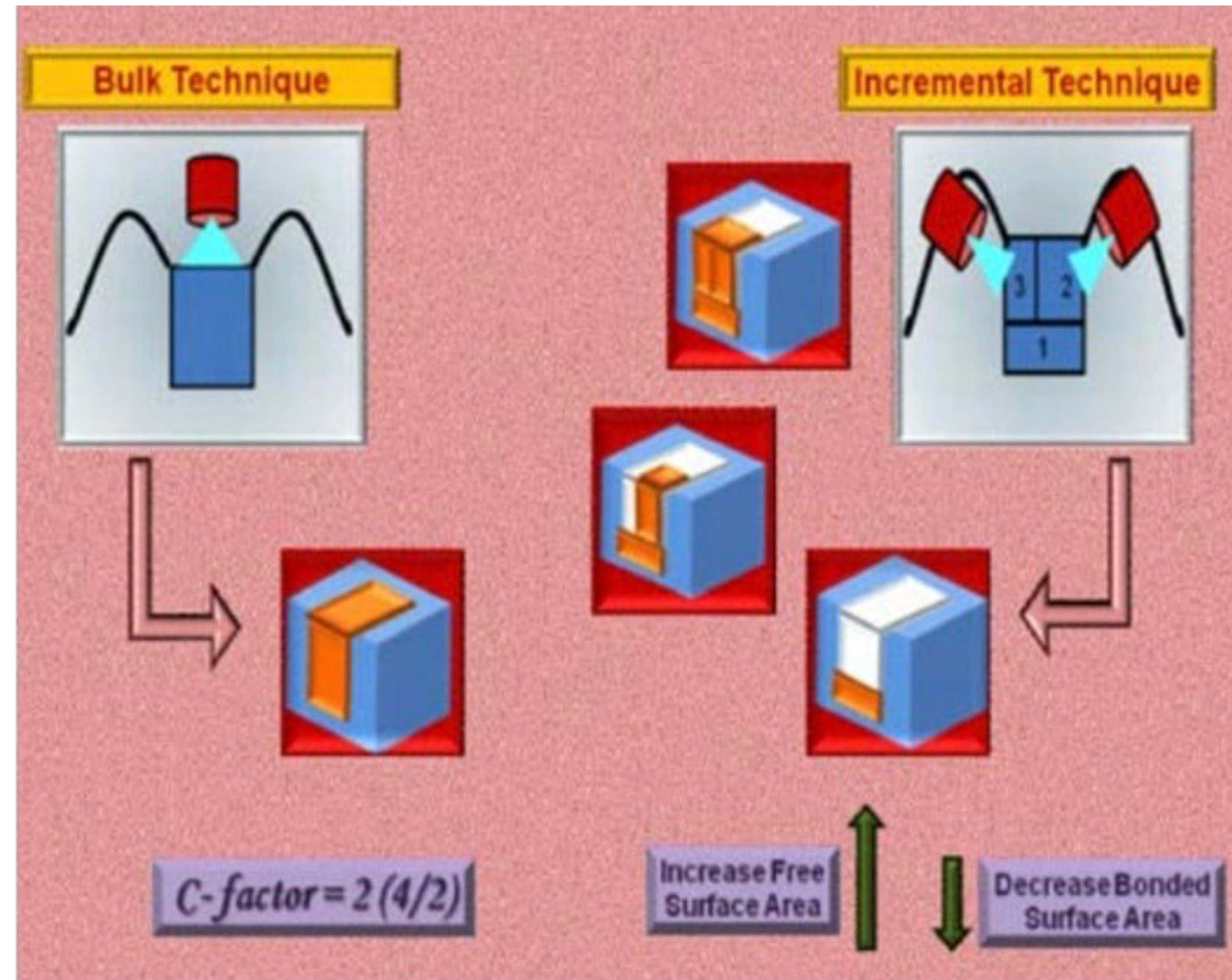


<https://www.dentsplysirona.com/en-us/discover/discover-by-topic/by-category/restorative/recurrent-caries-causes.html>

Soares CJ et al. Polymerization shrinkage stress of composite resins and resin cements – What do we need to know? *Brazilian Oral Research*. 2017;31:e62. doi:10.1590/1807-3107BOR-2017.vol31.0062.

How Incremental layering Decrease Shrinkage Stress?

A restoration that is built up in increments layering technique minimize the C-factor. The incremental technique increases the actual free surface area relative to the resin bonded surface area, along with adequate polymerization of deeper layers.

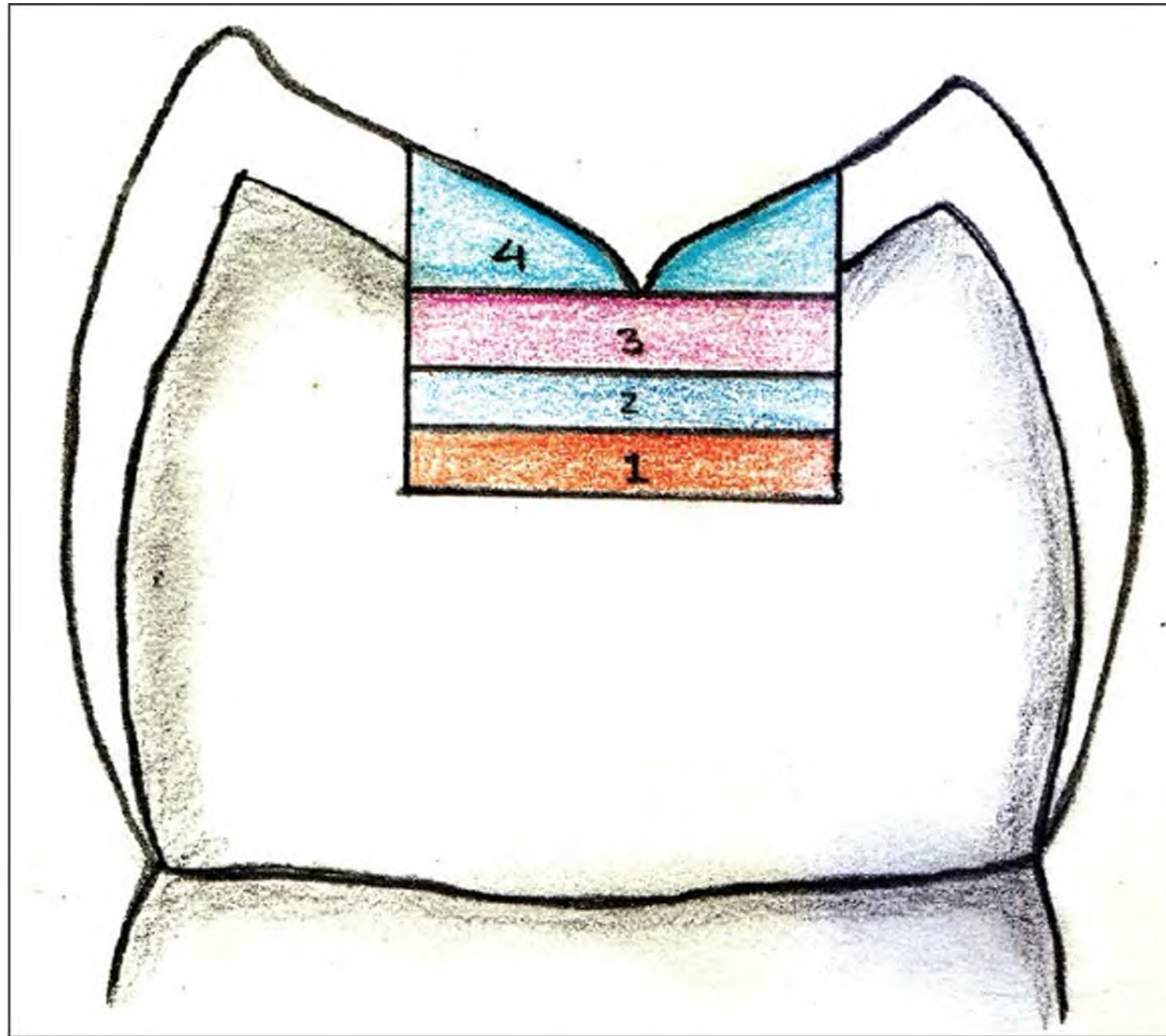


This reduces overall shrinkage stress and improving the restoration's marginal adaptation and longevity.

Various Incremental techniques

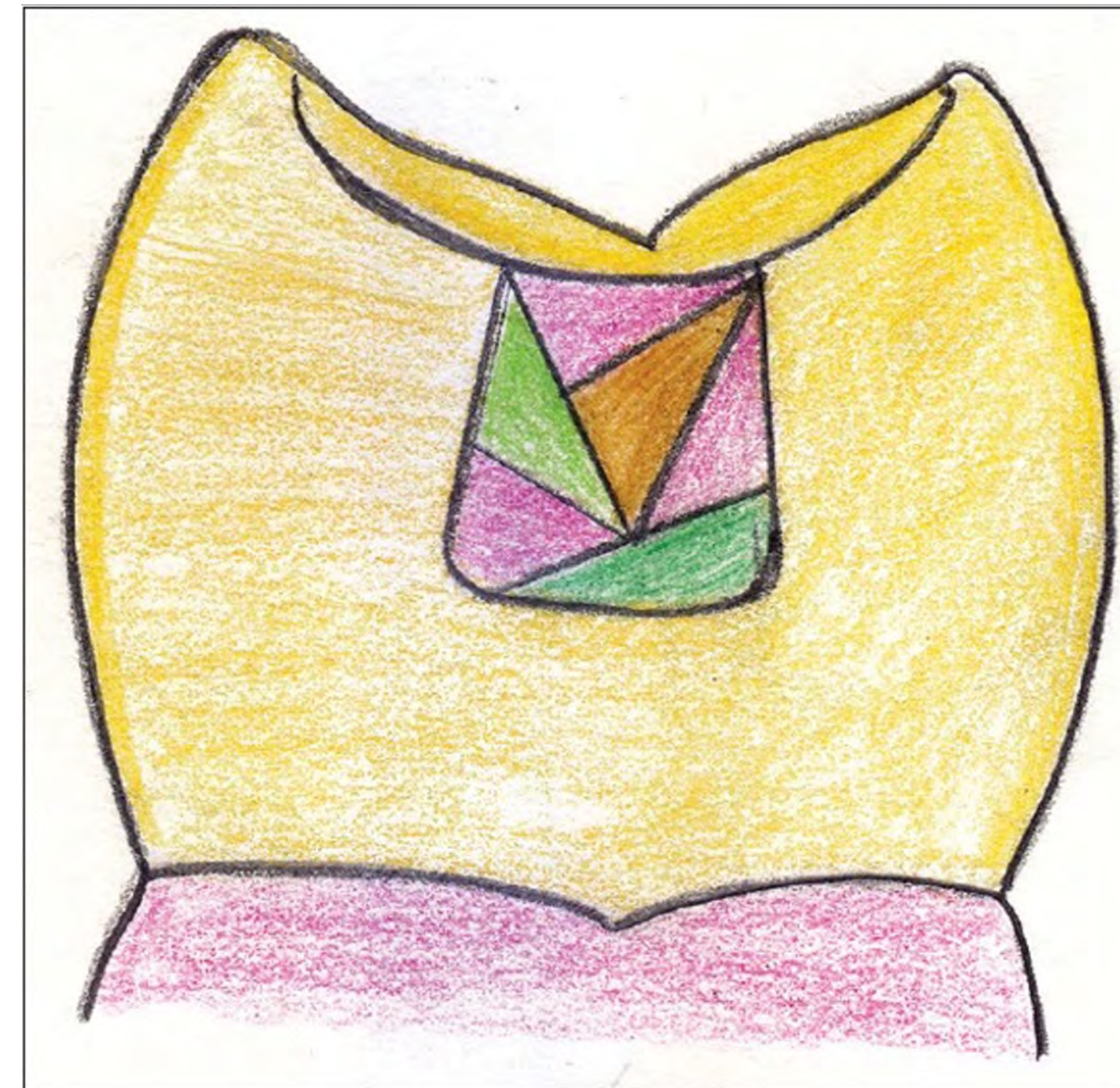


Horizontal layering technique



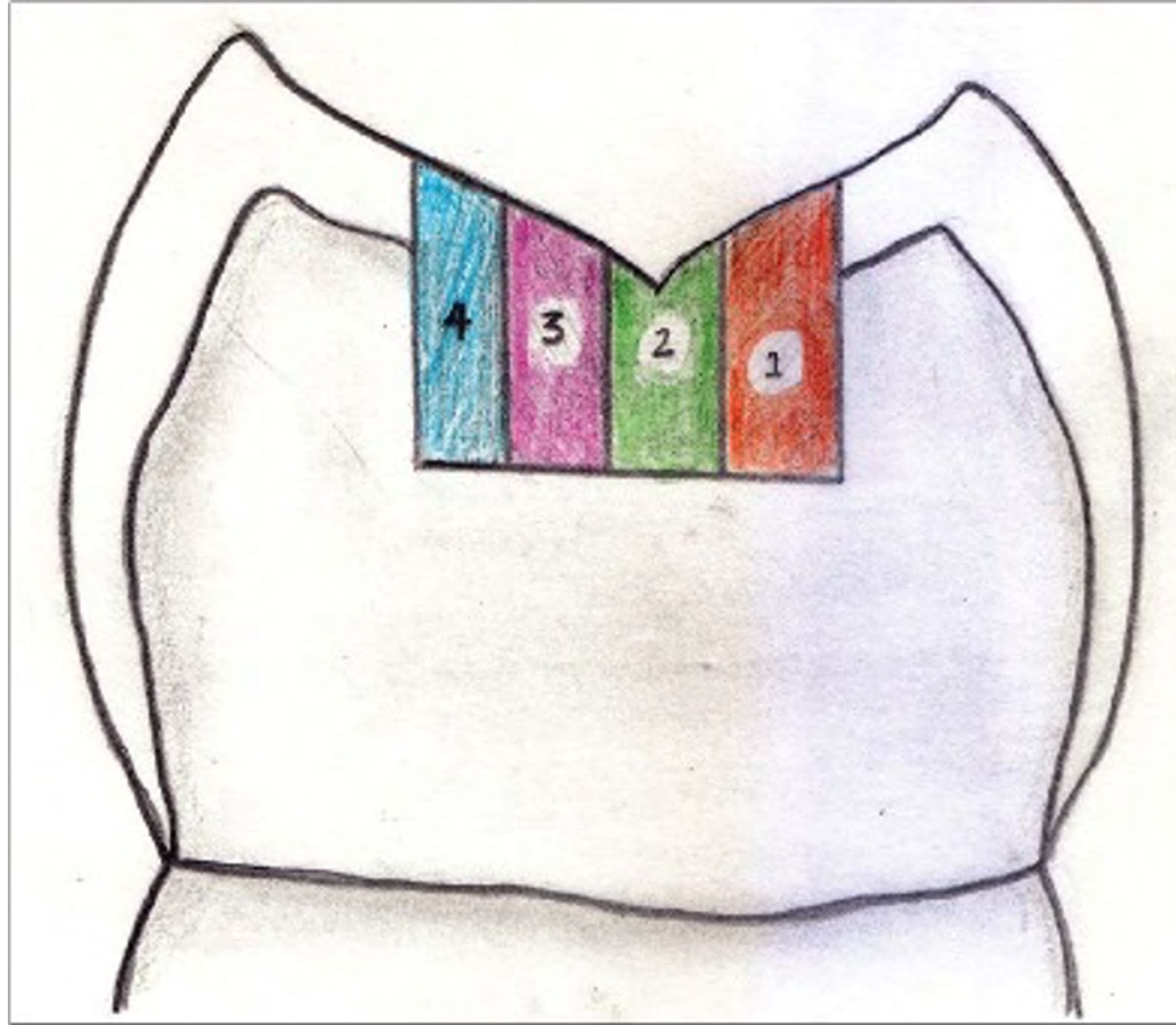
Layers <2 mm, high C-factor.

Places wedge-shaped increments; reduces C-factor.



Oblique layering technique

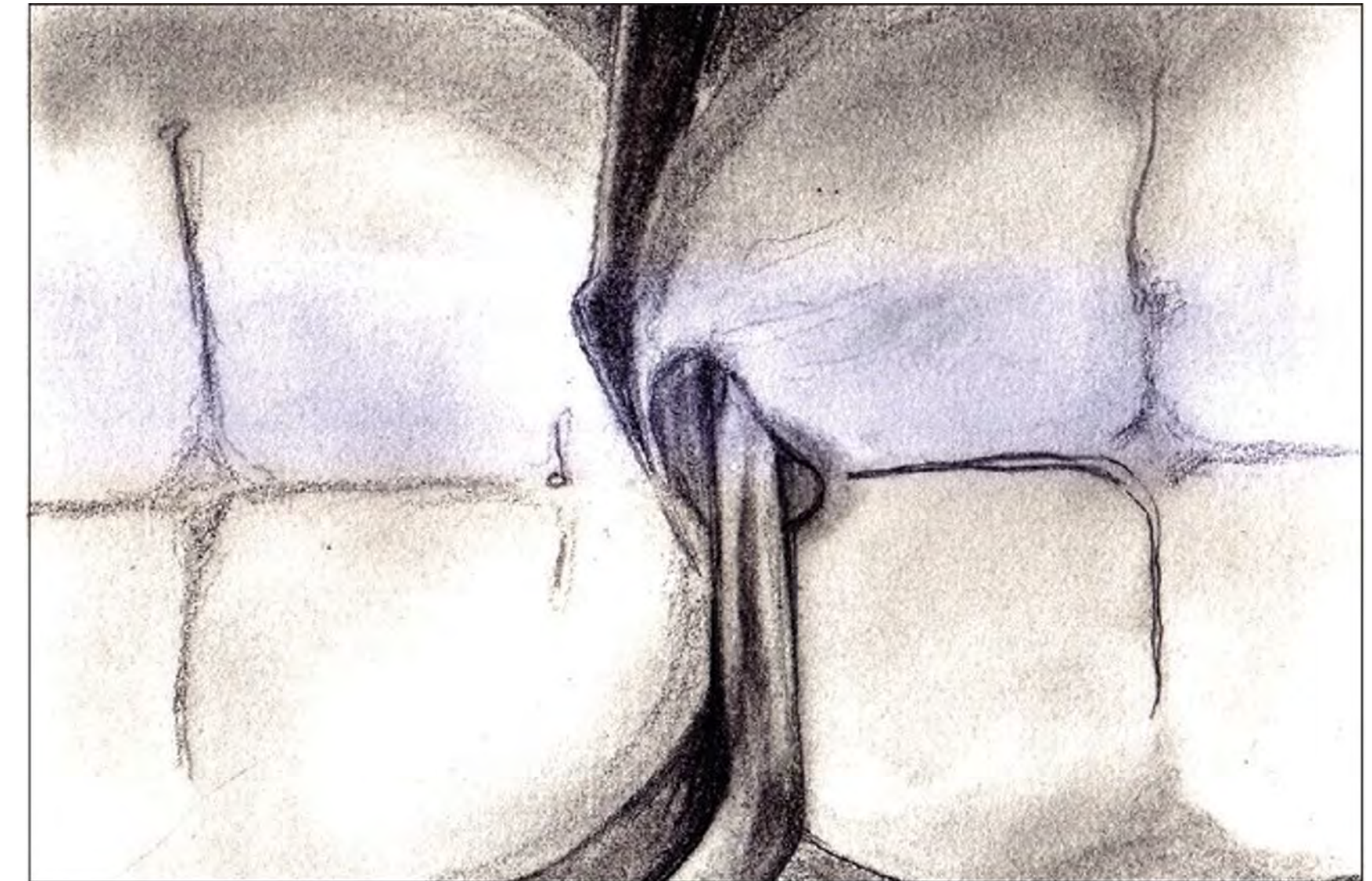
Vertical layering technique



Composite is placed vertically.

The curing light is applied from behind that wall, so the composite shrinks toward the wall. Minimizes gaps at gingival wall; lowers sensitivity.

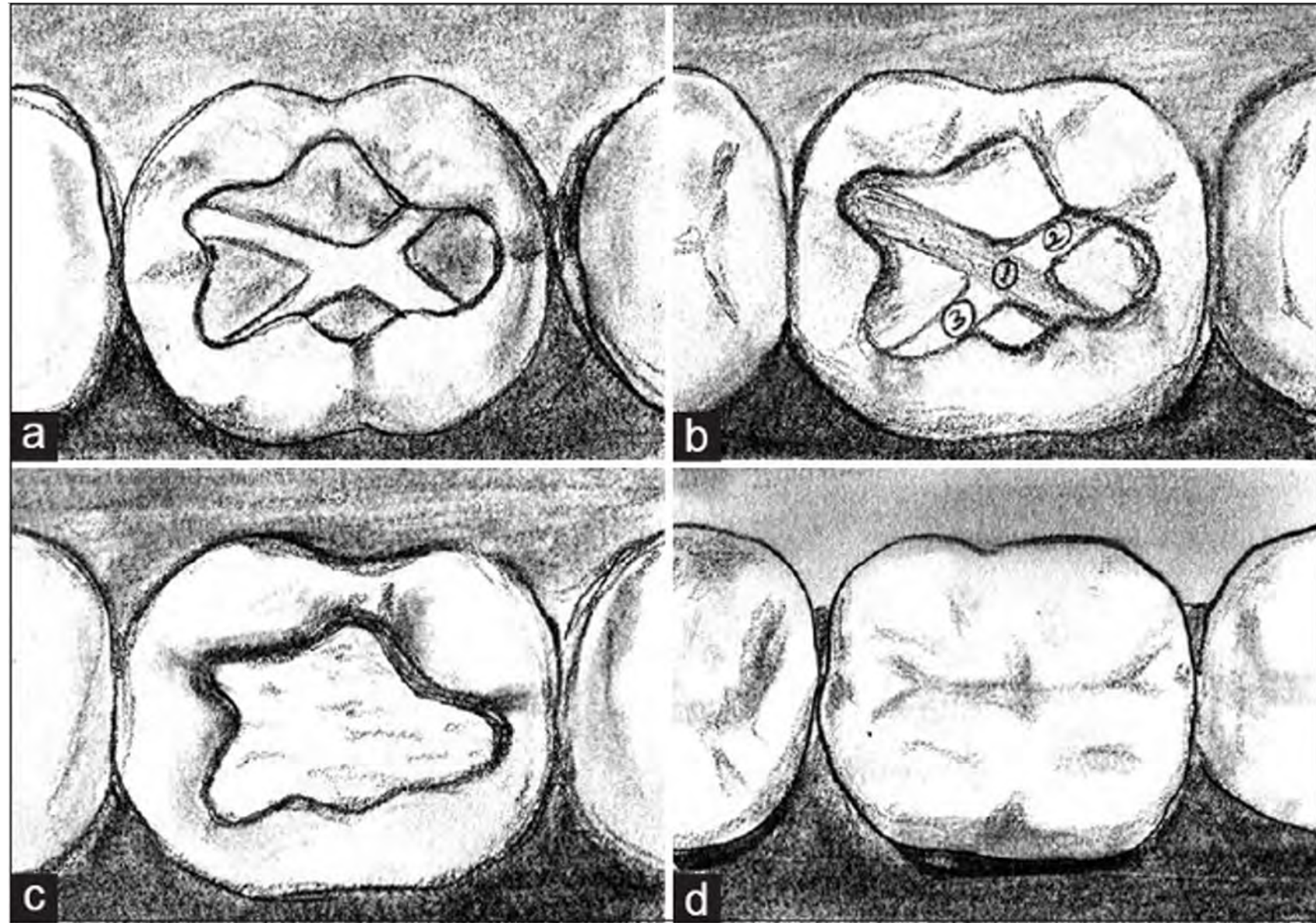
proximal contact is built first; the internal curing of this layer can strengthen the composite and cut down cervical gap that could improve marginal seal.



Centripetal build up technique

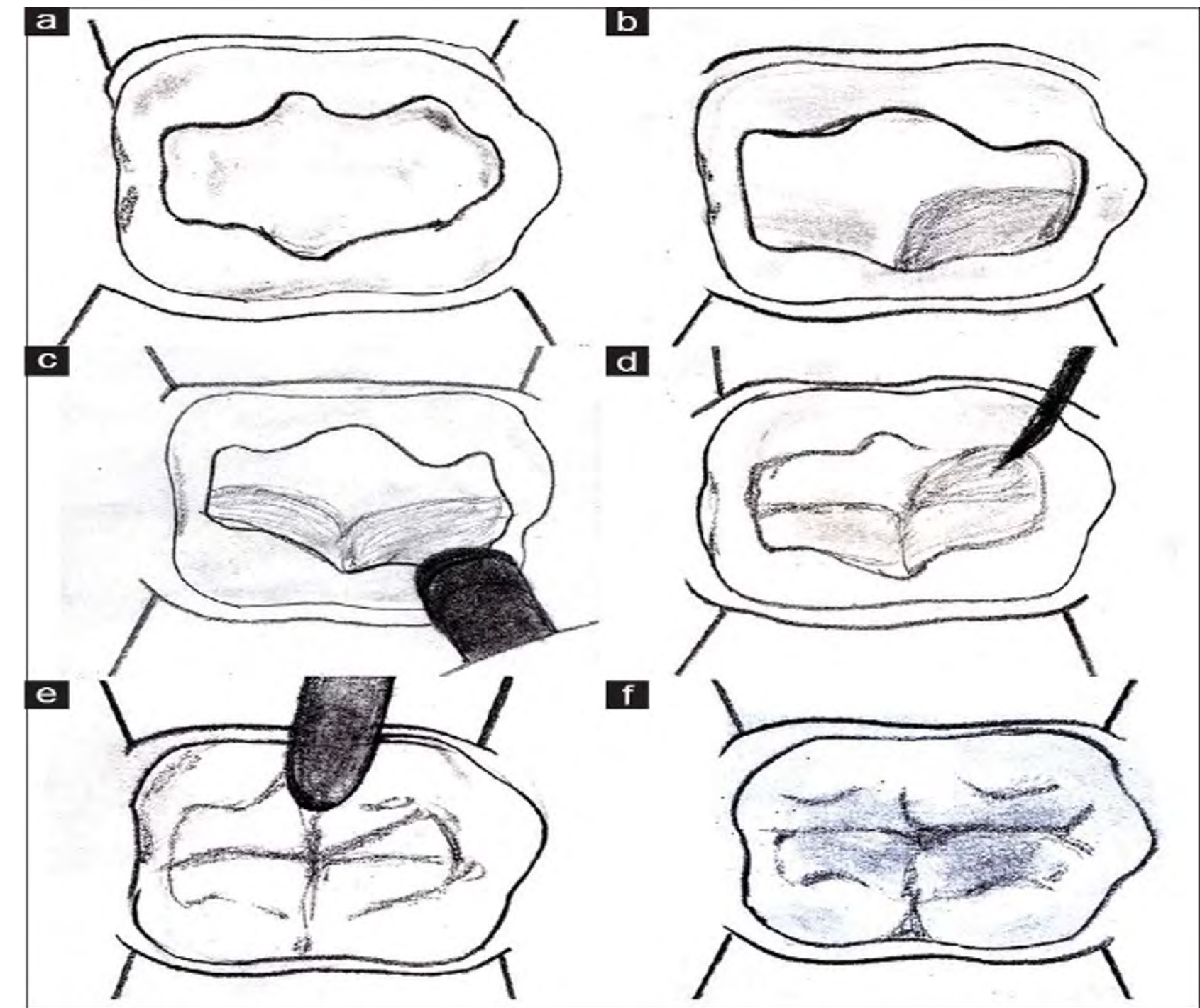


Split-increment horizontal layering technique



2 diagonal cuts split uncured composite in 4 triangle and cured. One cut filled and cured, 1/2 of the other diagonal cut filled and cured. Other 1/2 of the cut filled and cured. Same procedure followed for the next horizontal incrementals. This reduce C-factor ratio from 5, to 0.5.

Each cusp restored one at a time up to the level of the occlusal enamel. Initially time-consuming, but reduce finishing time by precise attention to reconstruction of natural morphology.



Successive cusp build up technique



incremental layering minimizes shrinkage & improves stress distribution



It is widely accepted that incremental filling decreases shrinkage stress as a result of reduced polymerization material volume.



► J Conserv Dent. 2017 Nov-Dec;20(6):386–391. doi: [10.4103/JCD.JCD_157_16](https://doi.org/10.4103/JCD.JCD_157_16)

Incremental techniques in direct composite restoration

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PMCID: PMC5799982 PMID: [29430088](#)

Abstract

Polymerization shrinkage is one of the dental clinician's main entanglements when placing resin-based composite restorations. None of the method can assure a perfectly sealed



Marginal adaptation



Capacity of a restorative substance to closely adhere to the cavity walls while minimizing the presence of gaps at the interface between the tooth and the restoration. (Kaur et al., 2024)

Placing composite in small increments improves adaptation to cavity walls and reduces the chance of voids or marginal gaps. (Chandrasekhar, V., et al., 2017).

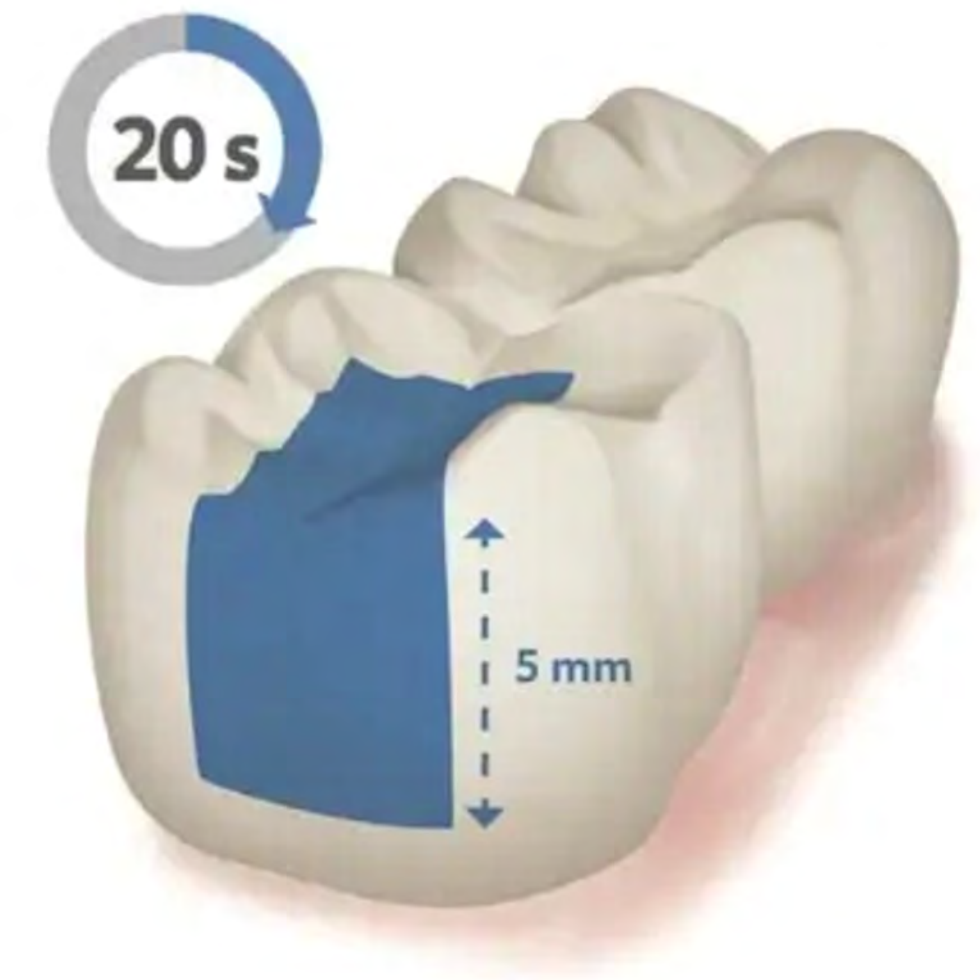
Margin integrity is influenced by cavity design, location of margins, choice of restorative material, and placement technique. (Taylor & Lynch, 1993)

Bulk-fill composite: Overview & Technique



How do Bulk-fill (*Bulk cure*) composites change the game?

- Allow 4-5 mm increments due to enhanced light transmission from increased translucency and optimized resin filler properties.
- Designed to minimize polymerization shrinkage stress by incorporating low-shrinkage monomers and optimized filler systems for stress relief (lower viscosity).
- Simplify placement to reduce chair time.
- Reduce layering steps by lowering the risk of voids and contamination from blood and saliva between increments.



<https://nexusmedo.com/product/nano-bulk-fill-composite/>

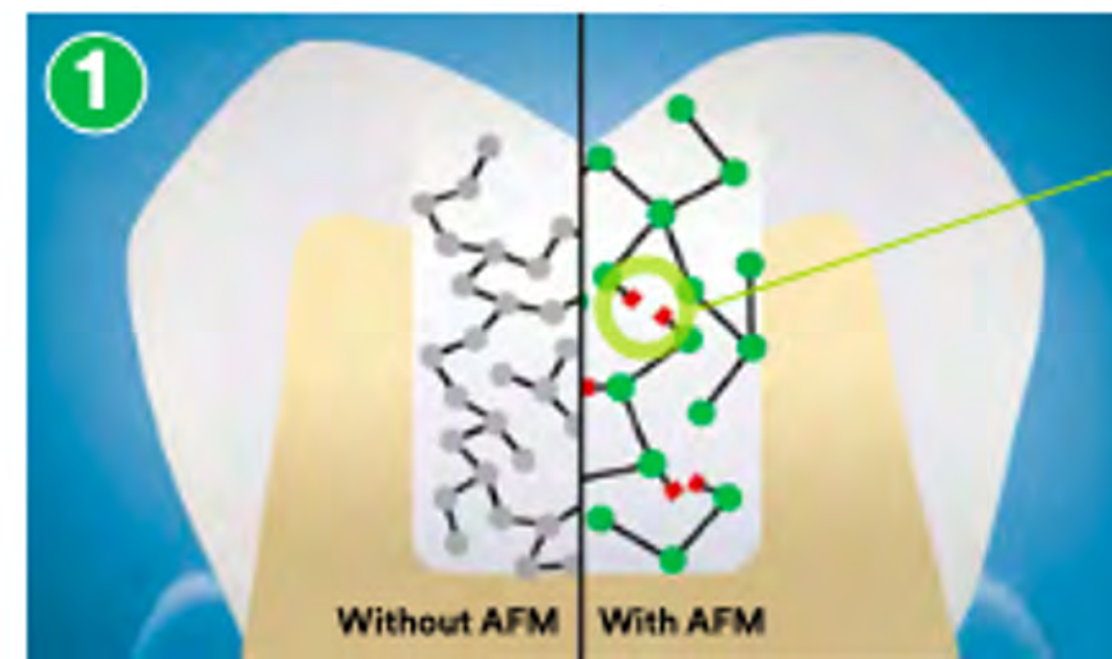
Polymerization shrinkage - The Core Issue



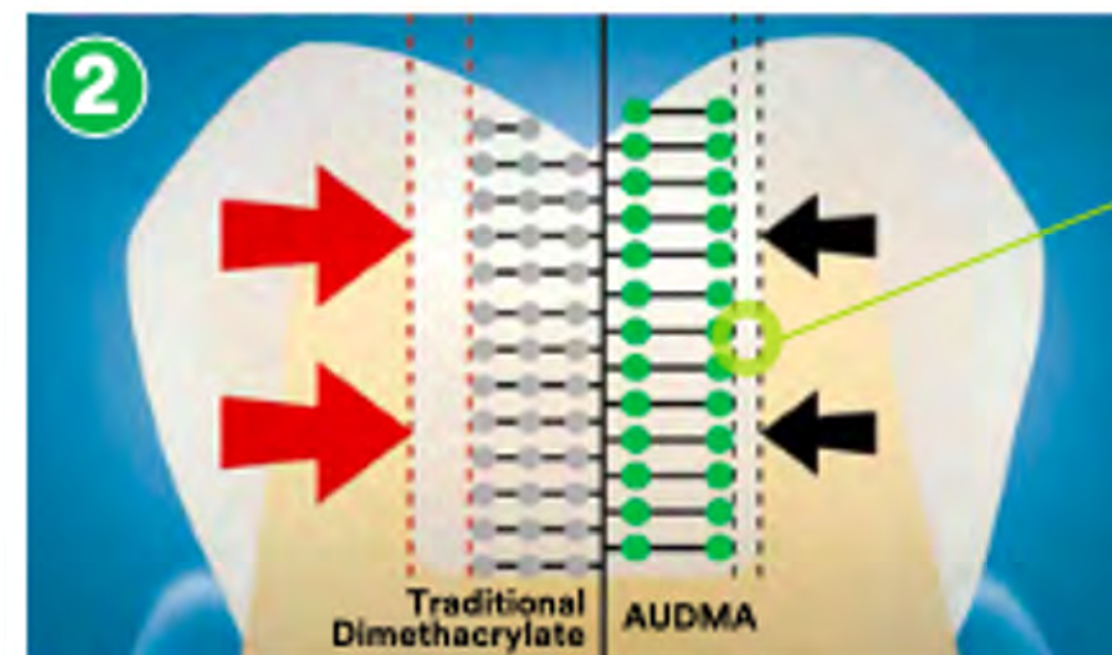
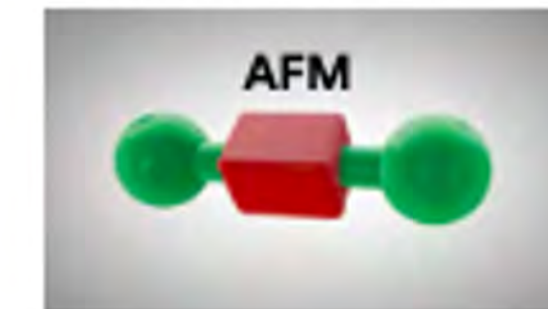
- Polymerization shrinkage occurs as monomers cross-link during photopolymerization, causing volumetric contraction.
- This contraction generates **shrinkage stress**, exerting force against cavity walls.

3M Filtek One Bulk fill Restorative

Here's how it works.



The innovative component of the first resin is an addition-fragmentation monomer (AFM). The unique feature of this resin is that, during polymerisation, the central group **can fragment to relieve stress**. The fragments can then re-polymerise in a lower stress state.



The other resin component is aromatic urethane dimethacrylate (AUDMA). Because it's a larger monomer than found in traditional dimethacrylates, it limits the number of shrinkage zones. That helps **reduce the amount of shrinkage and stress** that occurs during polymerisation.



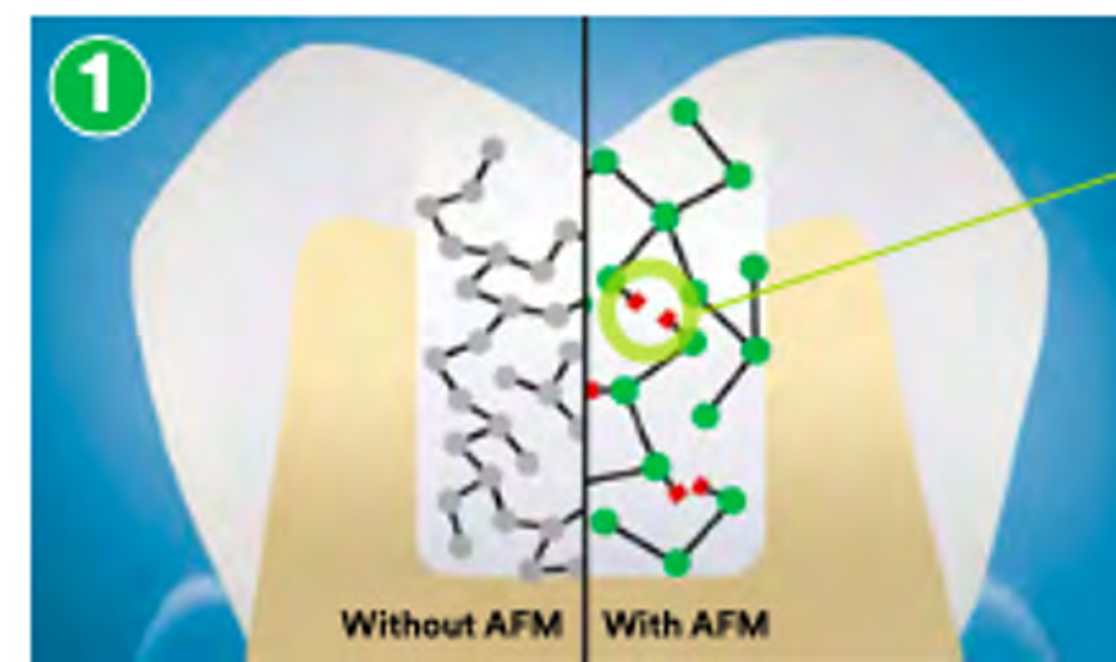
Polymerization shrinkage - The Core Issue



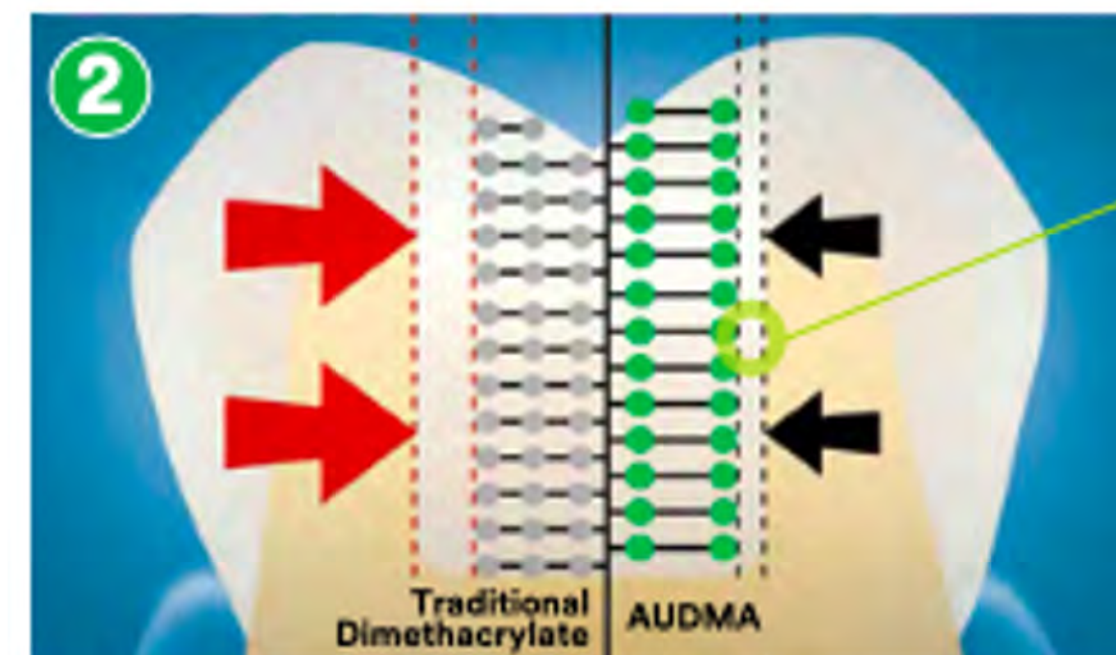
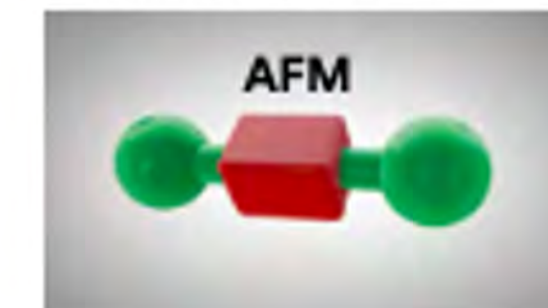
- Once polymerized into a 3D cross-linked matrix, composite loses its flexibility.
- Shrinkage stress can reach **3-5 MPa**, potentially causing marginal gaps, debonding, cuspal deflection, and sensitivity.

3M Filtek One Bulk fill Restorative

Here's how it works.



The innovative component of the first resin is an addition-fragmentation monomer (AFM). The unique feature of this resin is that, during polymerisation, the central group **can fragment to relieve stress**. The fragments can then re-polymerise in a lower stress state.



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Polymerization shrinkage - The Core Issue



Bulk fill composites aim to reduce stress by:

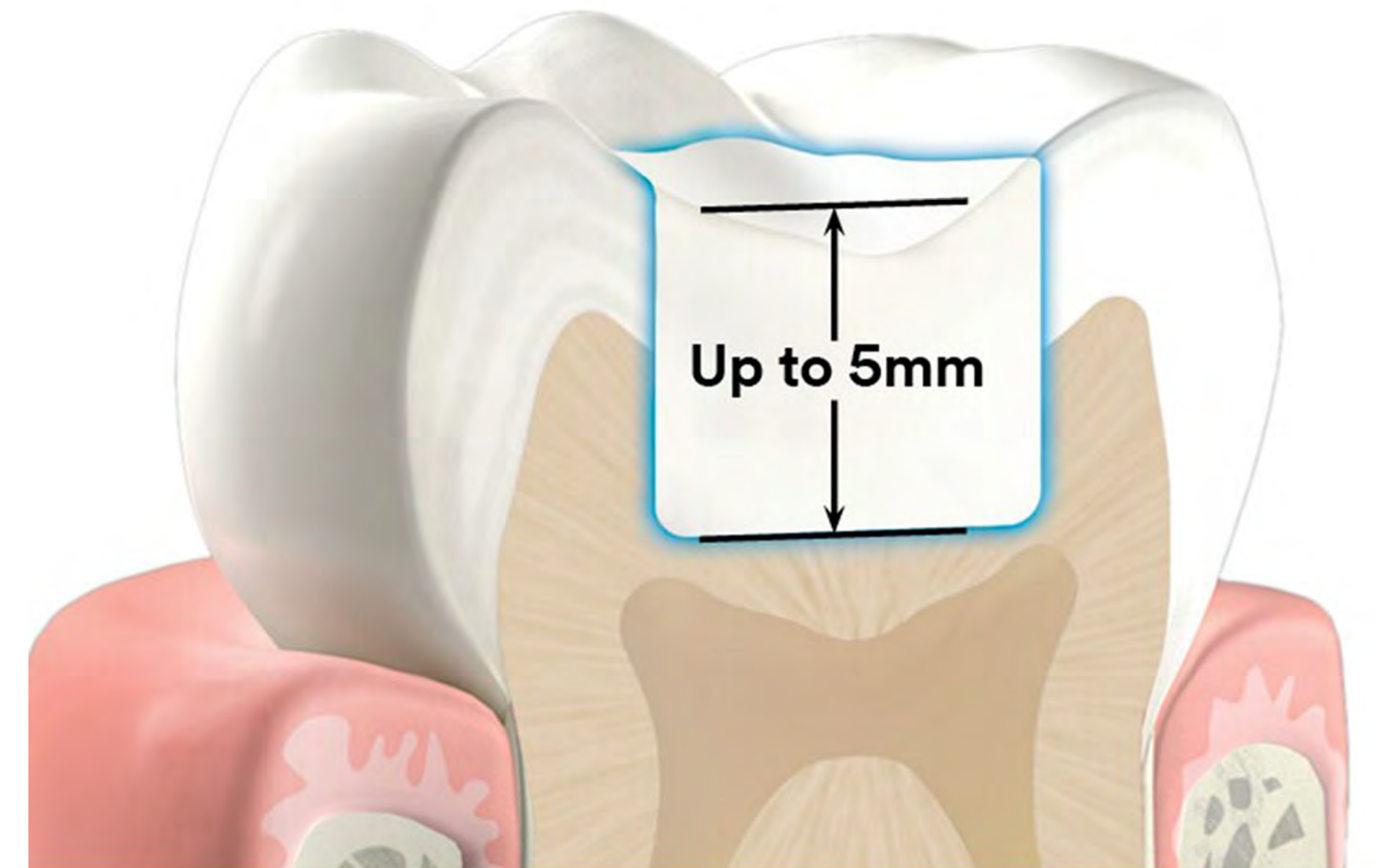
- Incorporating high molecular weight monomers

-Larger monomer size results in less volumetric contraction and reduced polymerization shrinkage.

- Adding stress-relieving resin matrices with lower elastic modulus

-Early during polymerization these matrices allow **viscoelastic flow**, dissipating internal stress before composite reaches its rigid phase.

Simple One-Step Placement
Stress relief enables 5mm depth of cure



Polymerization shrinkage - The Core Issue



Bulk fill composites aim to reduce stress by:

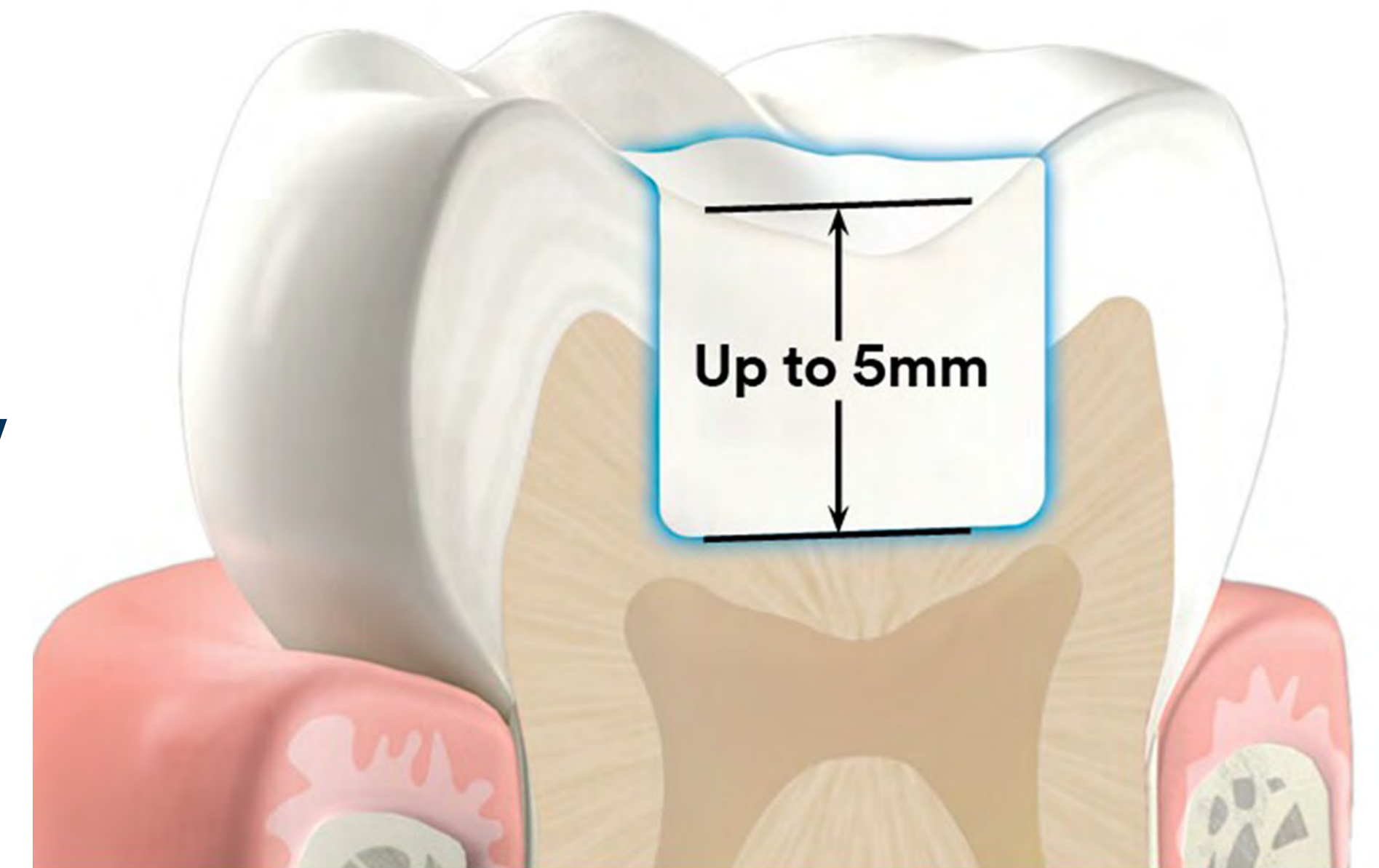
- **Extending curing depth through optimized photoinitiators**

-Optimized initiators improve the depth of cure, reducing the risk of under-cured composite at deeper layers.

- **Improving optical translucency**

-Increased translucency allows deeper light penetration, improving the degree of conversion at the base of the restoration.

Simple One-Step Placement
Stress relief enables 5mm depth of cure



Stress Distribution & Marginal Adaptation



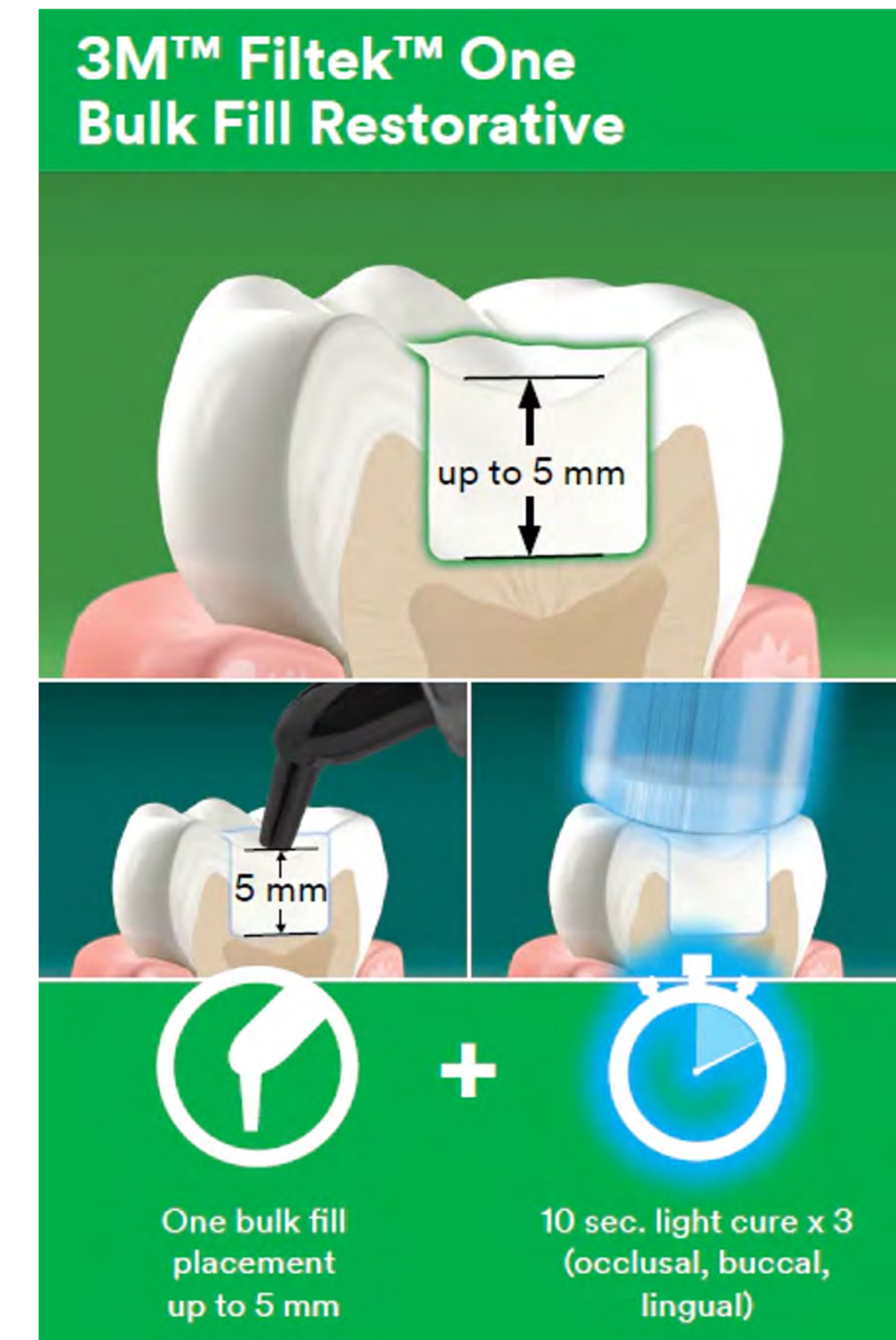
Better stress distribution = Stronger margins

****Reducing shrinkage makes curing easier and changes how the forces are distributed inside the tooth****

- Comparable or slightly improved marginal integrity vs incremental layering (Sengupta et al., 2023; Boaro et al., 2019)

****Proper curing and adhesive selection remain critical for sealing margins effectively****

While material science has improved marginal adaptation, the **OPERATOR'S TECHNIQUE** still determines long-term success!



Stress Distribution & Marginal Adaptation



Better stress distribution = Stronger margins

- Bulk-fill composites absorb and distribute stress more evenly due to their specialized chemistry, however lower filler content results in **lower wear resistance**.

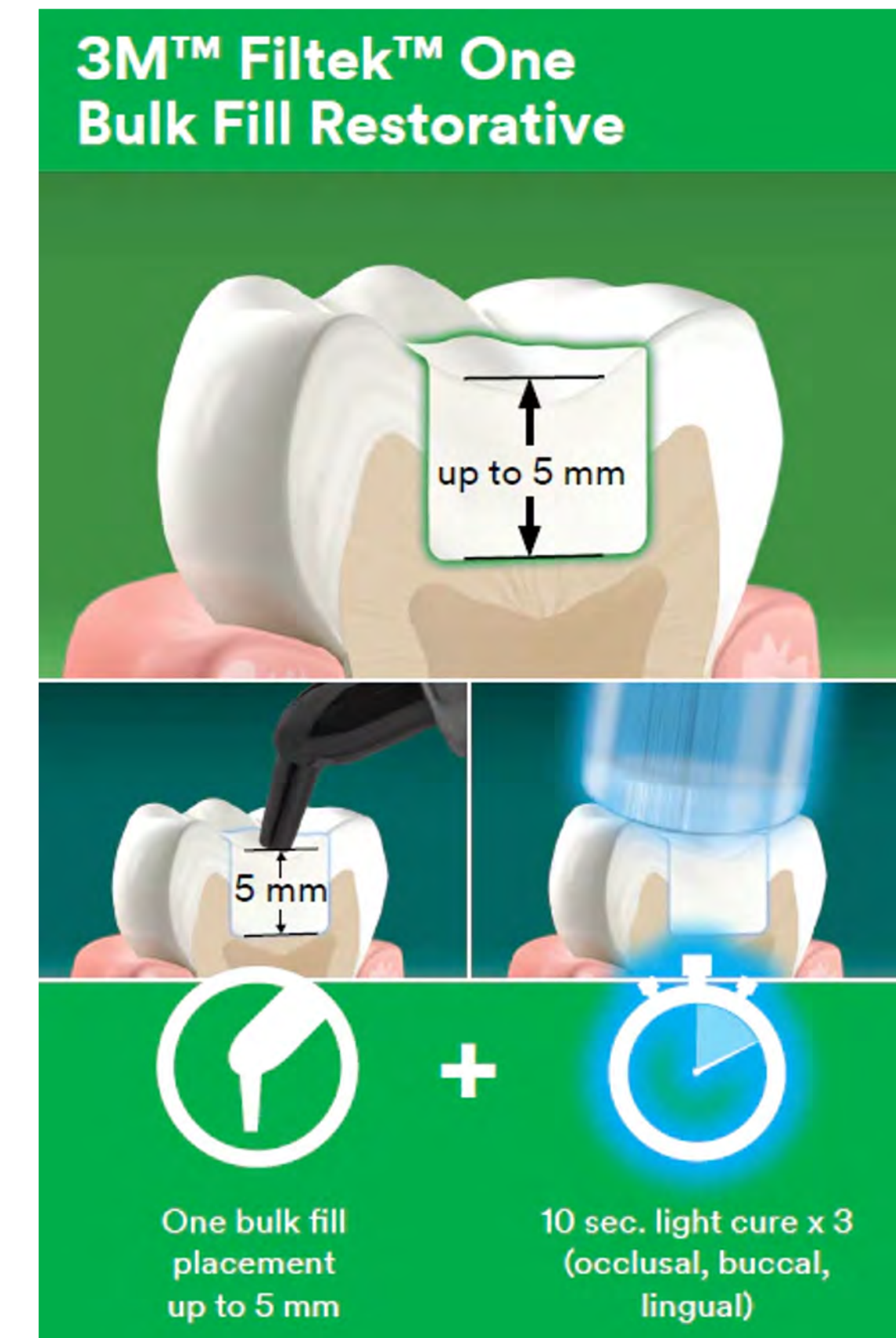
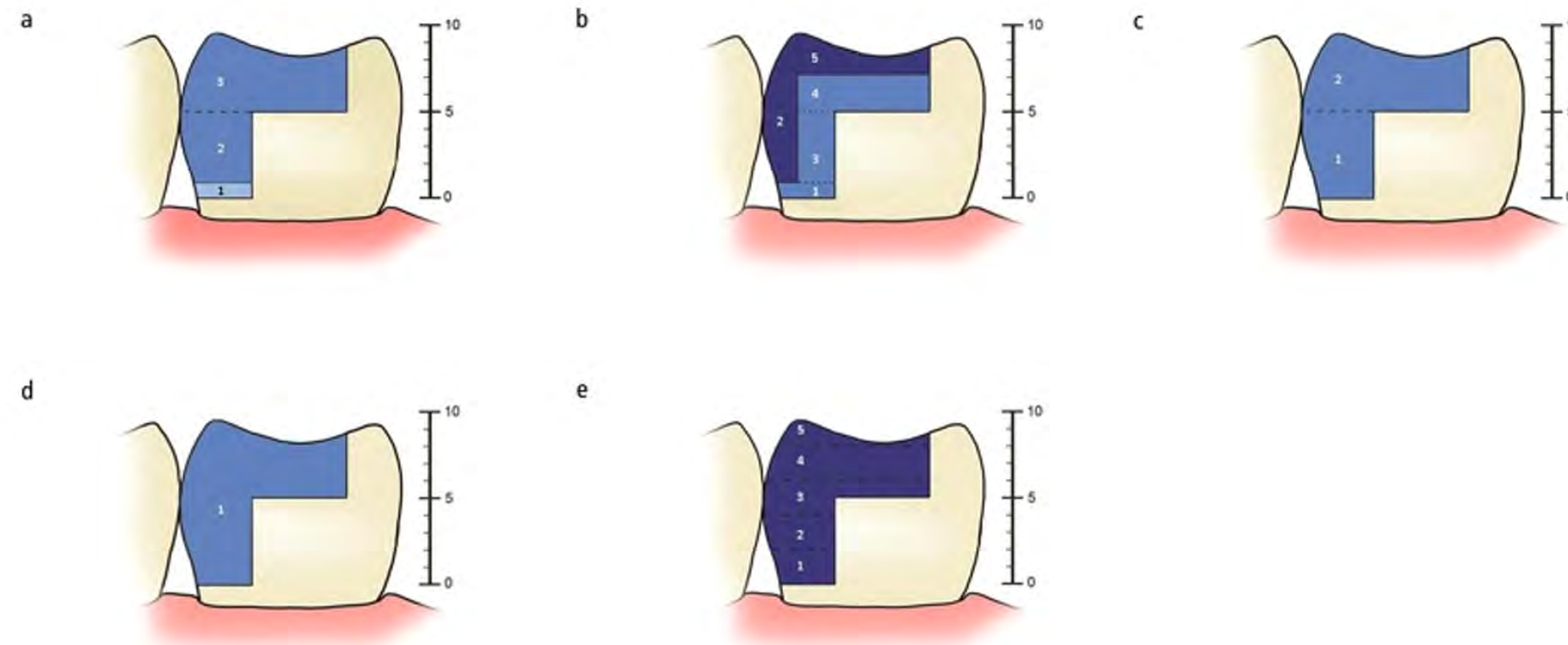


Figure 1: a) Application of a bulk-fill RBC in 4 mm layers with an optional conventional flowable RBC liner that is, 3M ESPE - Filtek Bulk-Fill Posterior Restorative; Ivoclar Vivadent- Tetric EvoCeram Bulk-Fill; Voco - xtra fil.

From: [Bulk-fill resin-based composite restorative materials: a review](#)



b) Application of a bulk-fill base RBC in 4 mm layers with a marginal ridge and occlusal capping layer of conventional RBC. Note that a 1 mm seal is shown at the base of the cavity with the bulk-fill base RBC that is, Dentsply - SDR; 3M ESPE - Filtek Bulk-Fill Flowable; Heraeus Kulzer - Venus Bulk-Fill; Ivoclar Vivadent - Tetric EvoFlow Bulk-Fill; Voco - xtra base. c) Application of a sonic activated bulk-fill RBC in 5 mm layers that is, Kerr - SonicFill 2. d) Application of a dual cure bulk-fill RBC in a single increment layer that is, Coltene - Fill Up; Parkell - HyperFil. As these RBCs typically have inferior aesthetics a capping layer of conventional RBC can be incorporated to improve these attributes. e) Application of conventional RBC in 2 mm increments that is, Kerr - Herculite XRV

Incremental Layering vs Bulk-Fill

Technique	Incremental Layering	Bulk-Fill Composite	
Goal	Reduce polymerization shrinkage stress by curing small volumes and lowering the effective C-factor per layer.	Reduce chair time and simplify procedure by placing one large increment; shrinkage stress managed by material formulation (translucency, photoinitiators, stress-relaxation chemistry).	Arbildo-Vega et al., 2020 (Clinical Effectiveness of Bulk-Fill and Conventional Resin Composite Restorations: Systematic Review and Meta-Analysis.) https://pmc.ncbi.nlm.nih.gov/articles/PMC7464794/
Marginal adaptation	Smaller increments improve wall contact and proximal form; fewer voids.	Comparable marginal adaptation when proper adhesive protocol and curing are used.	Arbildo-Vega et al., 2020 (Clinical Effectiveness of Bulk-Fill and Conventional Resin Composite Restorations: Systematic Review and Meta-Analysis.) https://pmc.ncbi.nlm.nih.gov/articles/PMC7464794/
Polymerization shrinkage stress distribution	Stress distributed across layers → less localized stress at margins.	Large volume polymerizes at once; shrinkage stress depends on bulk material's chemistry.	Sengupta, Anshu, et al. 2023(The clinical performance of bulk-fill versus the incremental layered application of direct resin composite restorations: a systematic review) https://pubmed.ncbi.nlm.nih.gov/37402908/
Clinical conclusion	Technique-sensitive but advantageous in high C-factor restorations (Class I / MOD).	Comparable clinical success + reduced chair time.	Arbildo-Vega et al., 2020 (Clinical Effectiveness of Bulk-Fill and Conventional Resin Composite Restorations: Systematic Review and Meta-Analysis.) https://pmc.ncbi.nlm.nih.gov/articles/PMC7464794/

Explain how material properties (filler type, translucency, depth of cure, and modulus of elasticity) influence the long-term clinical performance of posterior composites.

Morgan Reeves

Search Strategy



Keywords

Posterior composites
Filler Type
Translucency
Depth of cure
Elastic modulus
Clinical performance

Databases used

Pubmed
Science Direct

Pubmed search string

1. ((Filler type) AND (composite) AND (Clinical performance))
2. ((Translucency) AND (Posterior composite) AND (Clinical performance))
3. ((Depth of cure) AND (Posterior composite) AND (Clinical performance))
4. ((Elastic modulus)) AND (Posterior composite) AND (Clinical performance))

Search results

1. 40 results
2. 153 results
3. 63 results
4. 198 results

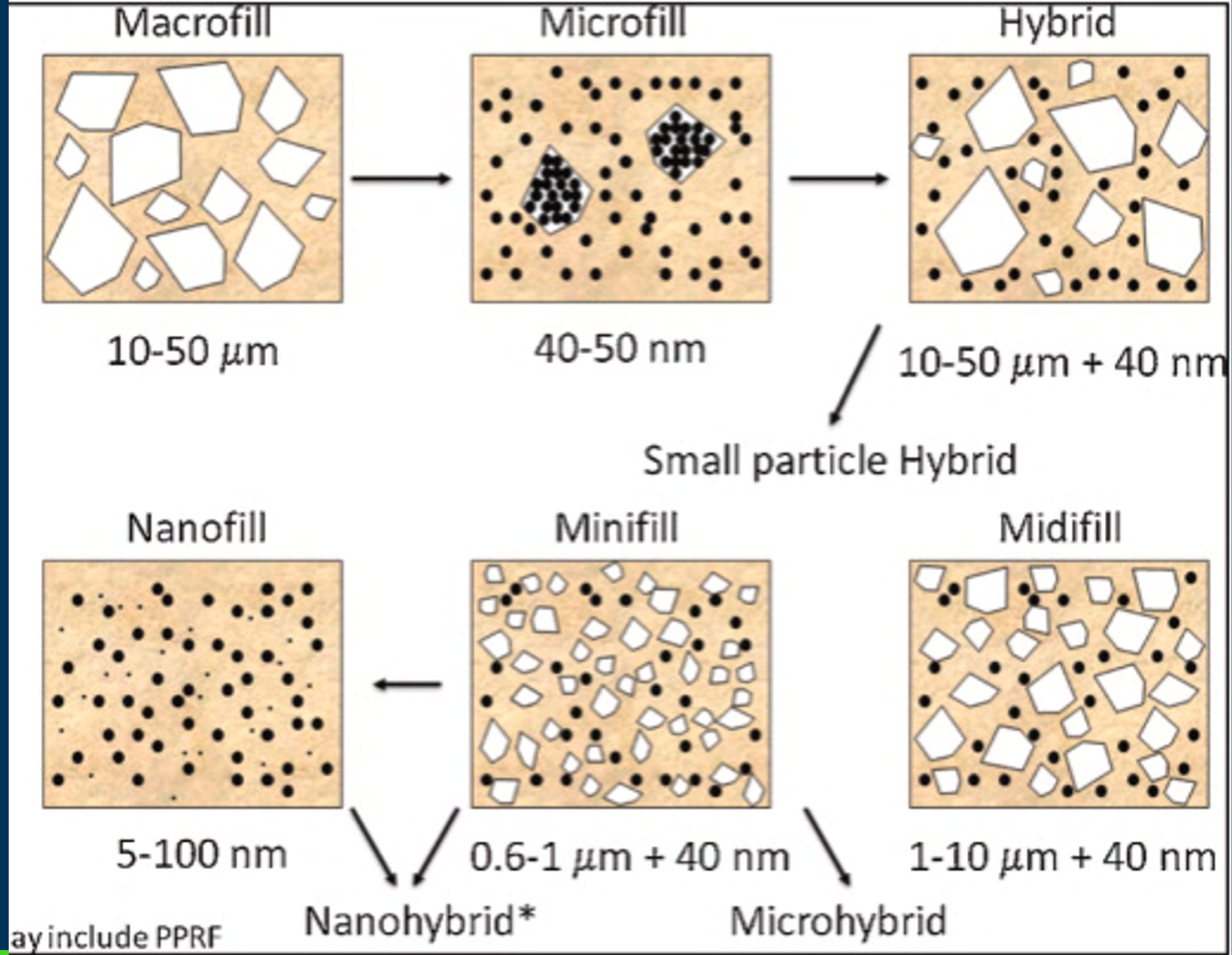
Highest level of evidence

Systematic review

Longevity of Composites: Filler Type

Composite Type	Advantages	Disadvantages
Macro-filled Composites	• Improved compressive strength, hardness, and elastic modulus. • Reduced polymerization shrinkage and thermal expansion.	• Poor wear resistance due to crack propagation at filler/resin interface • Loss of filler particles leads to poor polishability • Early discoloration and staining due to surface roughness
Micro-filled Composites	• Smooth surface with good initial polishability.	• Lower fracture resistance, stiffness, and fatigue strength • Lower wear and stress resistance
Hybrid Composites	• Combine macro- and micro-filled advantages. • High mechanical strength allows resistance to masticatory forces (suitable for posterior restorations). • Excellent optical properties and acceptable polishability (ideal for anterior use)	• Can be brittle in some cases
Nanocomposites	• High translucency and excellent polish retention • High surface quality due to small particle size. • Enhanced strength. • Superior polishability.	• May be more expensive and technique-sensitive • Agglomeration of nanoparticles





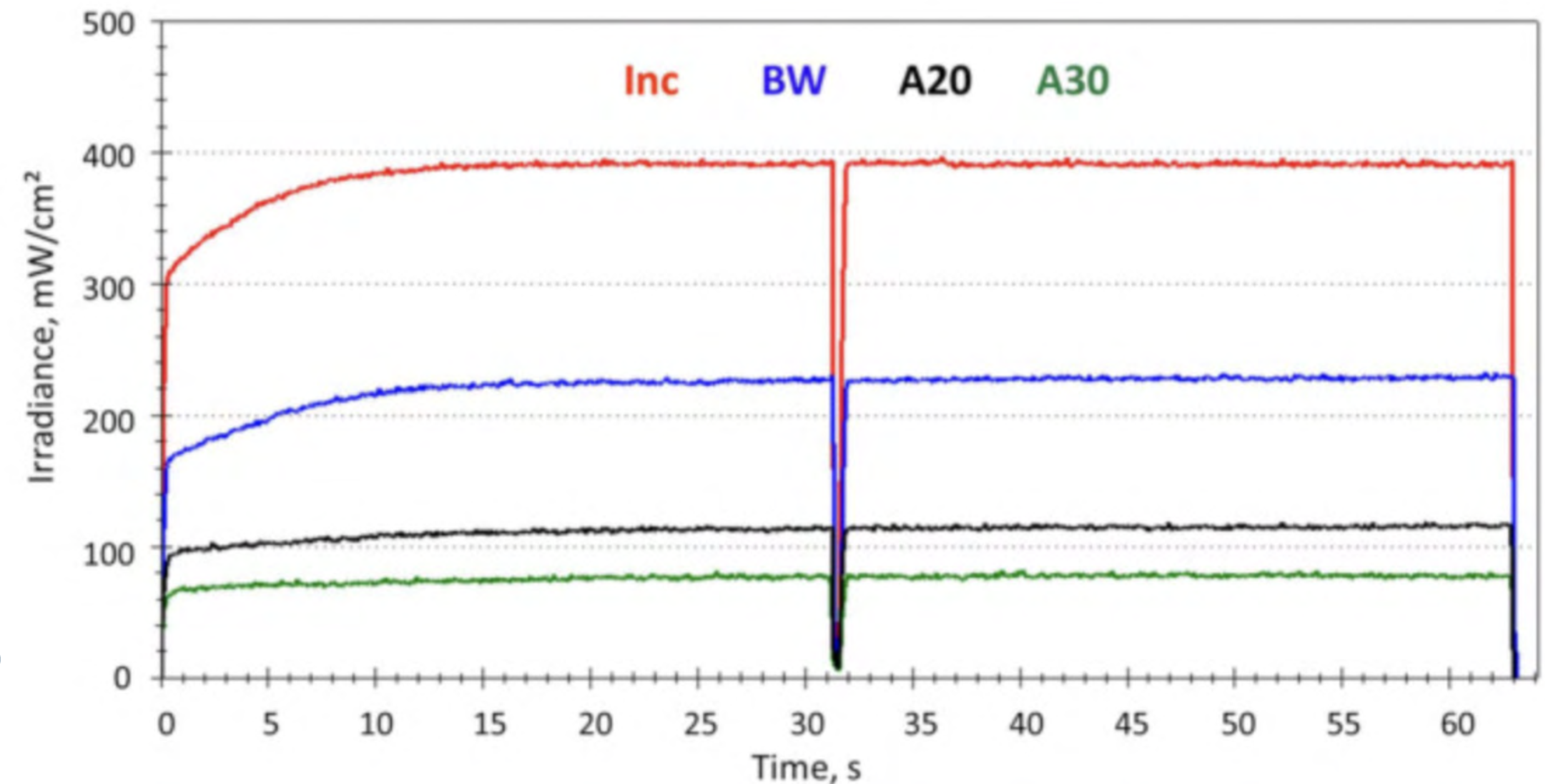
Longevity of Composites: Translucency

- Higher light interference in more opaque composites due to light scattering and reflection from filler particles and pigments.
- Less degree of conversion/polymerization at the base of the restoration in more opaque composites → Weaker restoration and poor long-term clinical performance.
- Bulk composites rely on translucency to allow more effective cure at base.

► Materials (Basel). 2023 May 16;16(10):3773. doi: [10.3390/ma16103773](https://doi.org/10.3390/ma16103773)

Light Transmission Characteristics and Cytotoxicity within A Dental Composite Color Palette

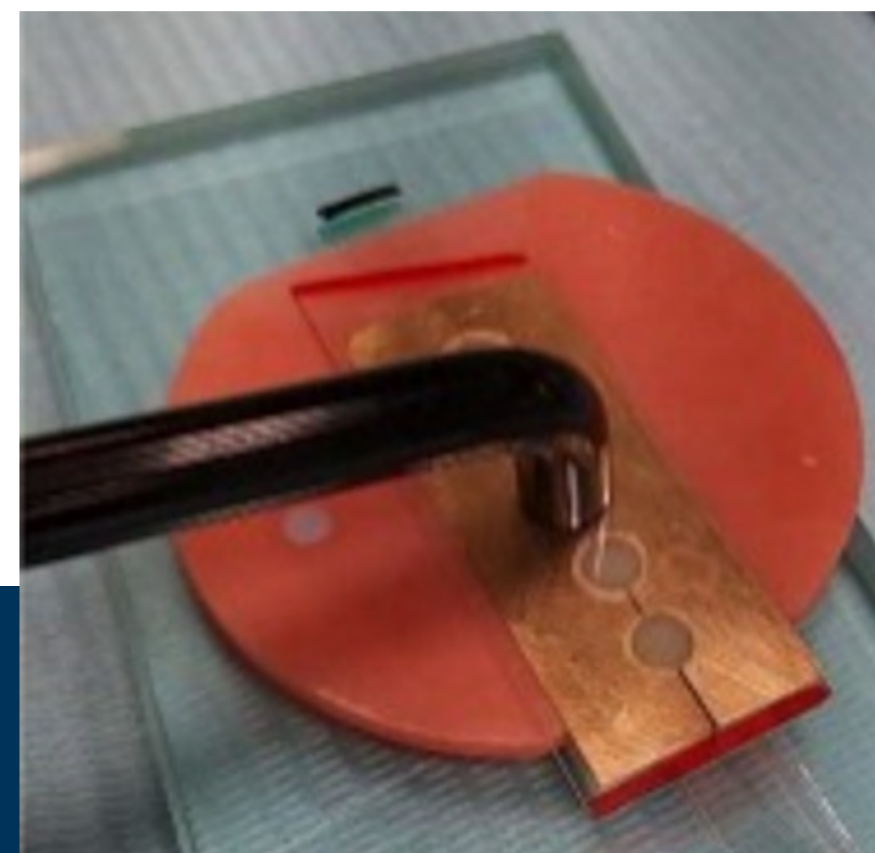
[Nicoleta Ilie](#)^{1,*}, [Andrei Cristian Ionescu](#)², [Karin Christine Huth](#)¹, [Marioara Moldovan](#)³



A20 and A30: Opaque dentin shades
Inc: Translucent enamel shade
BW: A light "bleach shade"

Longevity of Composites: Depth of Cure

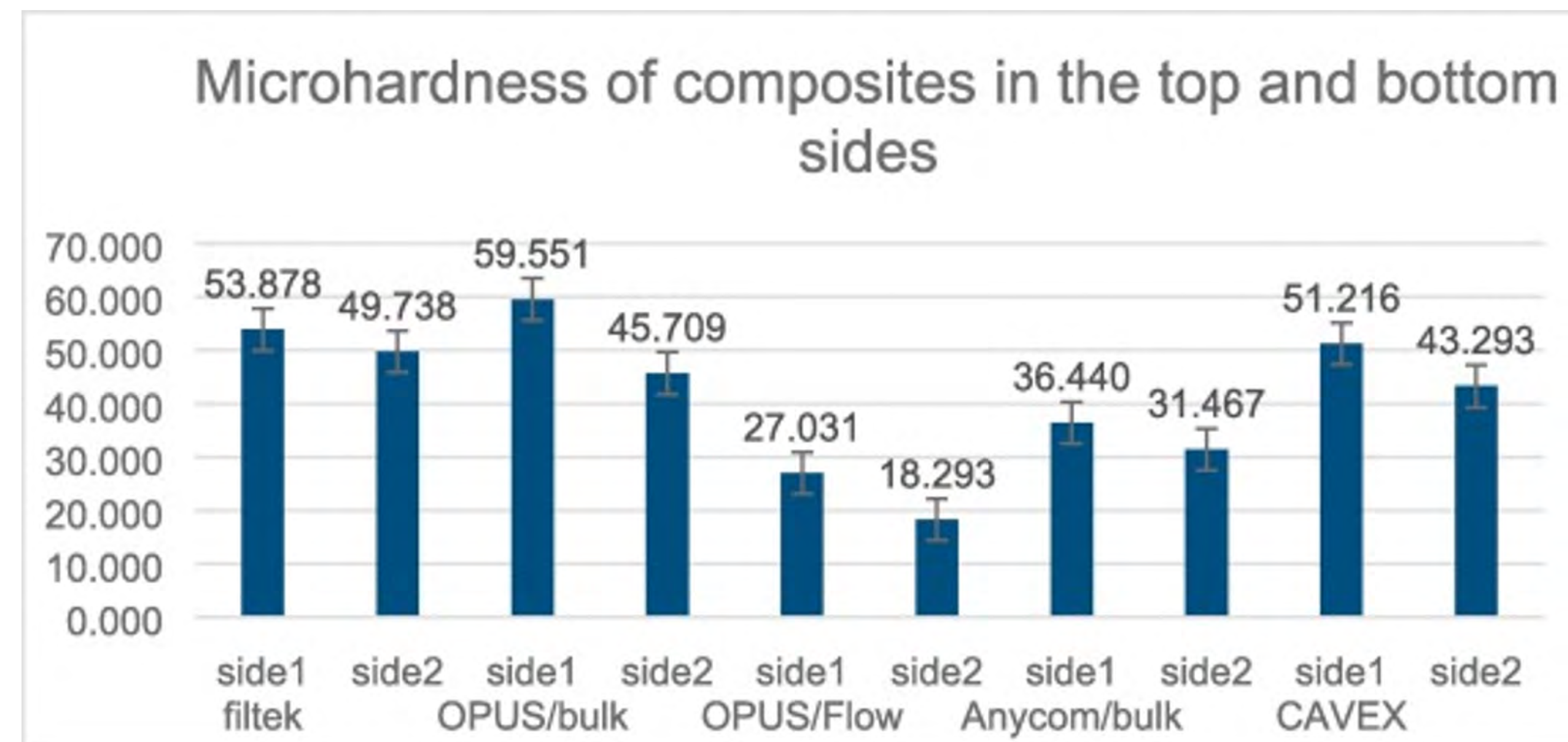
- Reduced polymerization in the deeper layers of a composite restoration lead to:
 - Decreased wear resistance, hardness, and bond strength.
 - Increased susceptibility to marginal defects.
 - Discoloration.
 - Disintegration (especially gingival floor of class II box)



Original Article

The effect of depth of cure on microhardness between bulk-fill and hybrid composite resin material

Ahmed A. Elhejazi ^a ✉, Asim Alosimi ^b, Faisal Alarifi ^b, Abdullah almuqayrin ^b



Side1: Top, Side2: Bottom

Longevity of Composites: Modulus of Elasticity



- Modulus must be high enough to withstand occlusal forces but low enough to allow composite to flex during tooth flexure.
 - High elastic modulus for Class 1 restorations experiencing masticatory forces.
 - Low elastic modulus for Class V restorations.
- Higher Modulus of elasticity leads to increased polymerization shrinkage stress → gap formation at margins → recurrent decay, increased post-operative sensitivity, and discoloration.
- The optimal elastic modulus of composite matches that of the material (dentin/enamel) being replaced, especially that of dentin.
 - Composites with elastic modulus similar to dentin exhibit less polymerization shrinkage and gap formation.

$$\text{Elastic modulus} = \frac{\text{Stress}}{\text{Strain}}$$

Longevity of Composites: Putting it all Together



Poor long-term clinical performance:

1. Micro-filled composite (smaller filler size, more resin matrix) used in posterior teeth → Increased wear, fracture risk, and susceptibility to secondary caries.
1. Opaque composites → Decreased polymerization at base of restoration → Increased leakage of bacteria (secondary caries), early failure due to weak restoration, and potential toxicity from remaining monomers.
1. Depth of cure beyond 2mm (conventional composite) and 4mm (bulk-fill composite) → Decreased polymerization at base of restoration → Decreased strength and increased disintegration leading to early failure of restoration and secondary caries.
1. Composite with low elastic modulus (compared to dentin) → deforms under masticatory forces → early failure of restoration
1. Composite with high elastic modulus (compared to dentin) → marginal gap formation from polymerization stress → Recurrent caries and early failure of restoration.



Longevity of Composites: Putting it all Together



Great long-term clinical performance:

1. Macro-filled composite (larger filler size), nanocomposite, hybrid composites used in posterior teeth → Restorations experience less wear and tear from the increased strength of these composites.
1. More translucent composites → Increased polymerization throughout the entirety of restoration → Stronger restoration and a better seal at margins reducing the susceptibility of recurrent caries.
1. Depth of cure at or below 2mm (conventional composite) and 4mm (bulk-fill composite) → Increased polymerization at base of restoration → Increased strength and increased seal at margins.
1. Composite with similar elastic modulus to dentin → Strong enough to withstand masticatory forces (Decreased risk of fractured restoration) and less gap formation from polymerization shrinkage (Decreased risk of secondary caries)



Compare and contrast the advantages and limitations of bulk-fill and traditional layering techniques based on current scientific evidence, focusing on marginal integrity, restoration longevity, and clinical efficiency.

Clara Castresana and Wes James

Search Strategy

Keywords:

Dental composite	Polishability
Bulk-fill	Color stability
Conventional fill	Efficiency
Incremental fill	Longevity
Marginal Integrity	Lifespan

Databases used

Pubmed
Chat GPT
Galileo

Pubmed search string

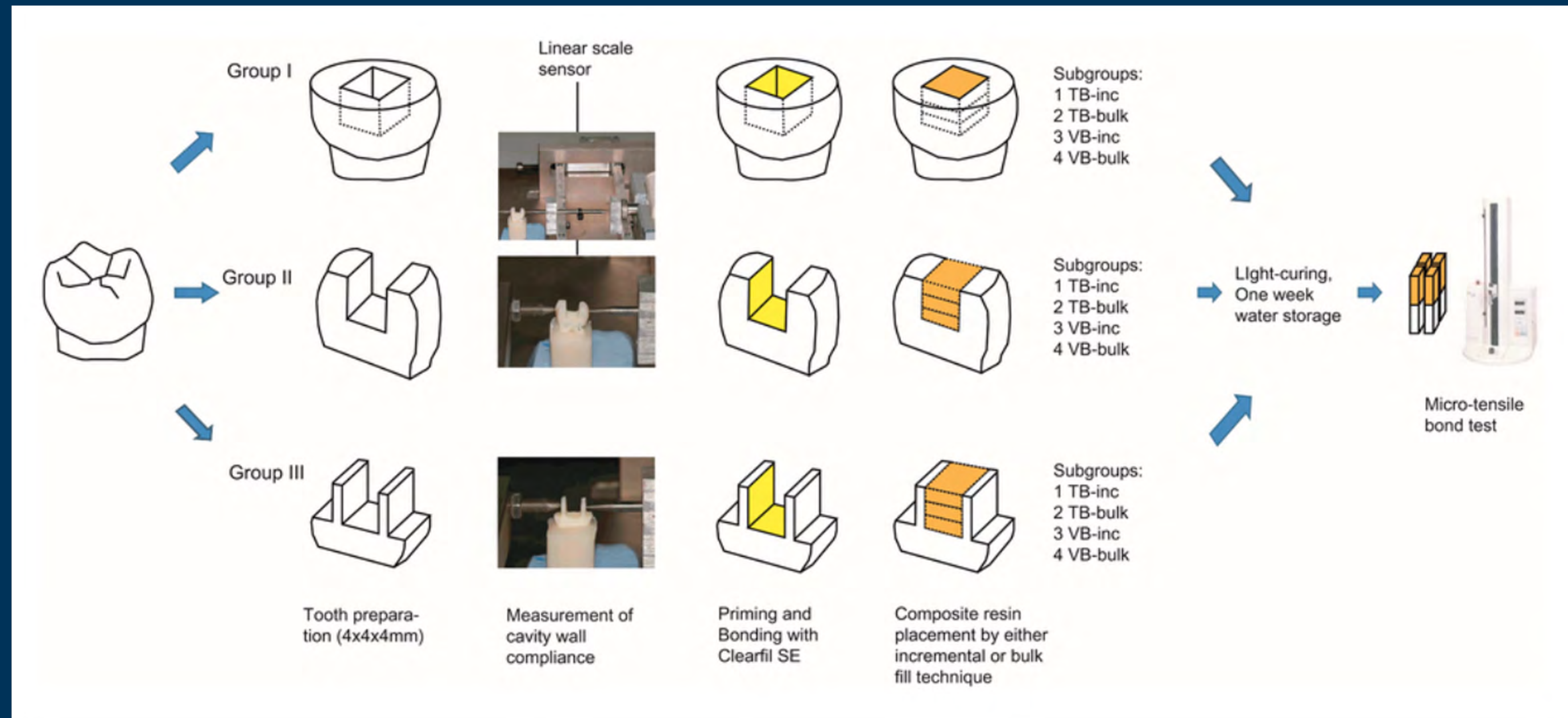
Filtered for Systematic reviews and within 10 years:
((Dental Composite) AND (Bulk Fill)) AND
(Conventional Composite Layering Technique) - 77
results, 1 systematic review
((Dental Composite) AND (Bulk Fill)) - 690 results, 17
systematic reviews
((Dental Composite) AND (Conventional Composite
Layering Technique)) - 108 results, 4 systematic
reviews
((Incremental Fill) AND (composite) AND (efficiency))
- 7 results
((Incremental fill) OR (bulk fill) AND (composite)) -
1,354 results
((Bulk fill) AND (polishing) AND (resin composites) - 7
results
((Color) AND (bulk fill restorations))- 85 results

Filtered for reviews with no time limit:
(Bulk fill) AND ("Dental materials : official publication
of the Academy of Dental Materials"[Journal]) -

Highest level of evidence

Systematic reviews and
Meta-analyses

Marginal Integrity Conventional Fill



Clinical, radiographic and biochemical assessment of proximal cavities restored with composite resin using incremental vs. bulk packing techniques: One-year randomized clinical trial

[Monaliza Maher Abdelaziz](#), [Shereen Fathy](#), [Amany Ahmed Alaraby](#), [Wessam Ibrahim Shehab](#) & [Marwa Mohamed Temirek](#) ✉

[BMC Oral Health](#) **24**, Article number: 1162 (2024) | [Cite this article](#)



Table 4 Inter and intragroup comparisons of MMP-9

Interval	Mean ± SD (ng\ml)		Test statistic	p-value
	Group (I)	Group (II)		
Baseline	16.39 ± 5.70 ^C	16.82 ± 5.43 ^C	0.13	0.897
7 days	114.75 ± 15.87 ^A	90.89 ± 10.28 ^A	3.09	0.011*
3 months	50.50 ± 8.43 ^B	36.67 ± 6.68 ^B	3.15	0.010*
Test statistic	149.47	180.37		
p-value	<0.001*	<0.001*		

Marginal Integrity

Bulk Fill



Meta-Analysis > Clin Oral Investig. 2019 Jan;23(1):221-233. doi: 10.1007/s00784-018-2429-7. Epub 2018 Mar 28.

Clinical performance of bulk-fill and conventional resin composite restorations in posterior teeth: a systematic review and meta-analysis

Conclusions: The present systematic review and meta-analysis indicate similar clinical performances of bulk-fill and conventional resin composites over a follow-up period of 12 to 72 months.

Clinical significance: Based on the results of this study, the bulk-fill resin composites could be an alternative for direct restorations in posterior teeth. However, clinical trials of longer duration are required.

- Strengths - Meta-analysis, class 1 and 2
- Weaknesses - only posterior teeth tested, follow-up avg 2.75 yrs, product and technique heterogeneity

> Evid Based Dent. 2023 Sep;24(3):143. doi: 10.1038/s41432-023-00905-4. Epub 2023 Jul 4.

The clinical performance of bulk-fill versus the incremental layered application of direct resin composite restorations: a systematic review

Conclusion: Bulk filled resin composite restorations demonstrated clinical outcomes similar to those of incrementally layered resin composite restorations within a review interval of 6 months to 10 years.

- Strengths - study designed around class 1 and 2 restorations, 18 RCT included, split mouth, Filtek, Tetric, and SDR
- Weaknesses - Product and technique heterogeneity

Color Stability/Polishability Conventional Fill

- Polishing experiment showed that bulk fill had the highest surface roughness, while nanofill had the lowest
 - Surface roughness → plaque accumulation and marginal discoloration

- Greater polymer conversion in some resins containing UDMA can lead to better marginal seal and less discoloration
 - incremental fills performed better



Color Stability/Polishability

Bulk Fill



Review > [Polymers \(Basel\)](#). 2020 Aug 10;12(8):1786. doi: 10.3390/polym12081786.

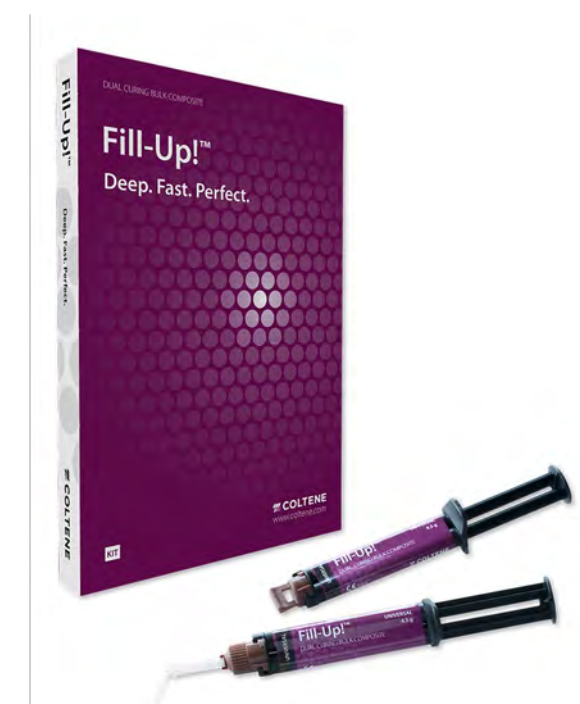
Clinical Effectiveness of Bulk-Fill and Conventional Resin Composite Restorations: Systematic Review and Meta-Analysis

- Conclusion - No significant difference between color stability, translucency, or surface texture.
 - Follow-up no longer than 10 years.

Review > [J Esthet Restor Dent](#). 2025 Oct 10. doi: 10.1111/jerd.70044. Online ahead of print.

Are Sculptable Bulk-Fill Composites Susceptible to Color Change: A Systematic Review

- Conclusion - High variability in color stability depending on brand. SonicFill and Fill-up were least susceptible. Fiber-reinforced were most susceptible.



Longevity

Conventional Fill



- Study of students doing incremental fills and bulk fills
 - Significant reduction in the microhardness from top to bottom of fill
 - All fills met the 80% microhardness of the occlusal surface parameter
 - Bulk fill restorations were found to have lower microhardness
 - Material specific property
- Incremental fills leave the possibility of air entrapment, leading to adhesive failure between layers
- Larger sizes of filler particles and irregular arrangement → wear of resin matrix, exposure of particles, increased surface roughness



Longevity

Bulk Fill



- Hardest to find evidence because of the novelty of product lines
 - Most studies with follow-up will have used products from 5-10 years ago, due to nature of the study. How can there be a 20 year follow up on a relatively new product. Rely on in-vitro studies heavily.

Meta-Analysis > Dent Mater. 2019 Oct;35(10):e249-e264. doi: 10.1016/j.dental.2019.07.007.

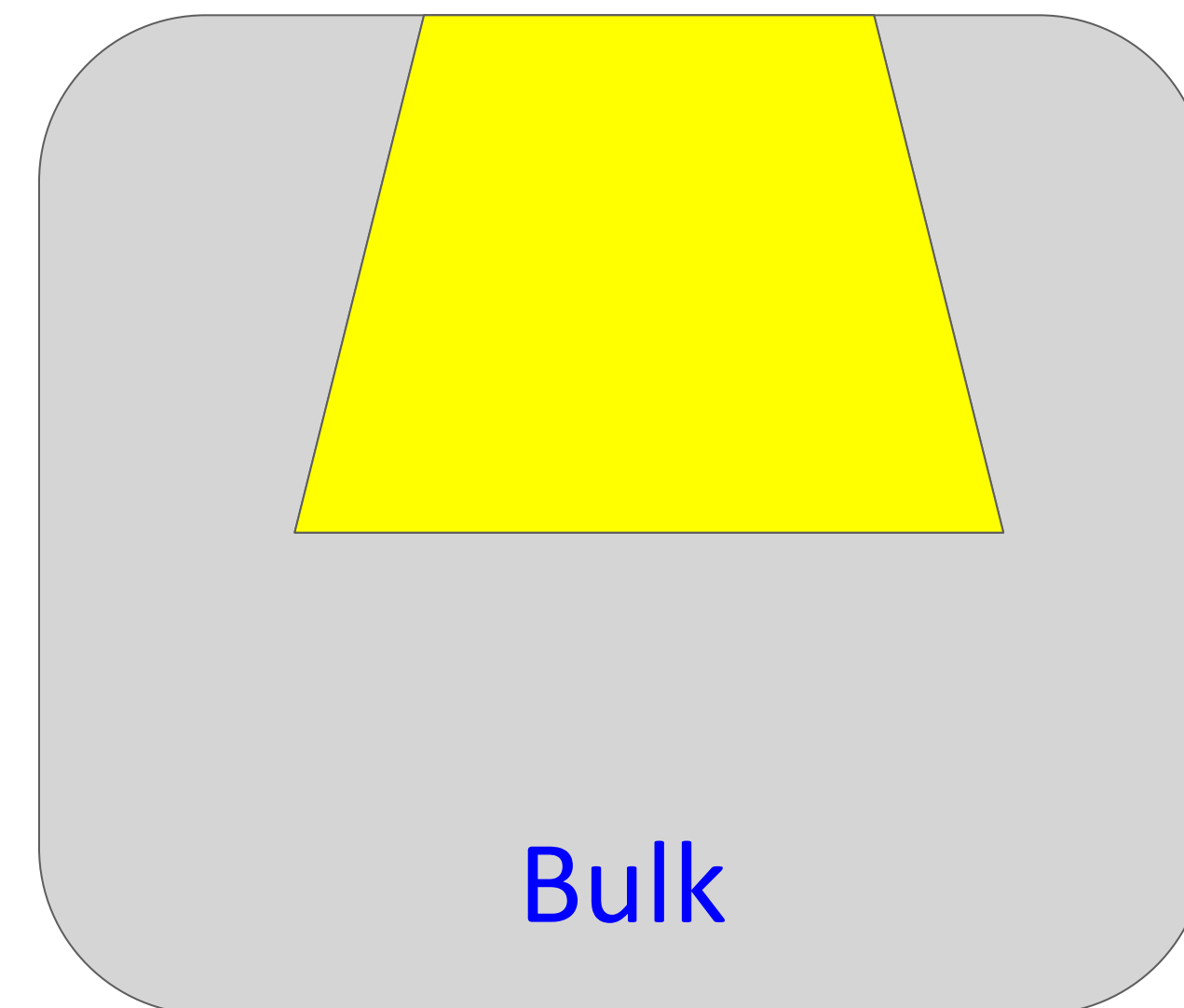
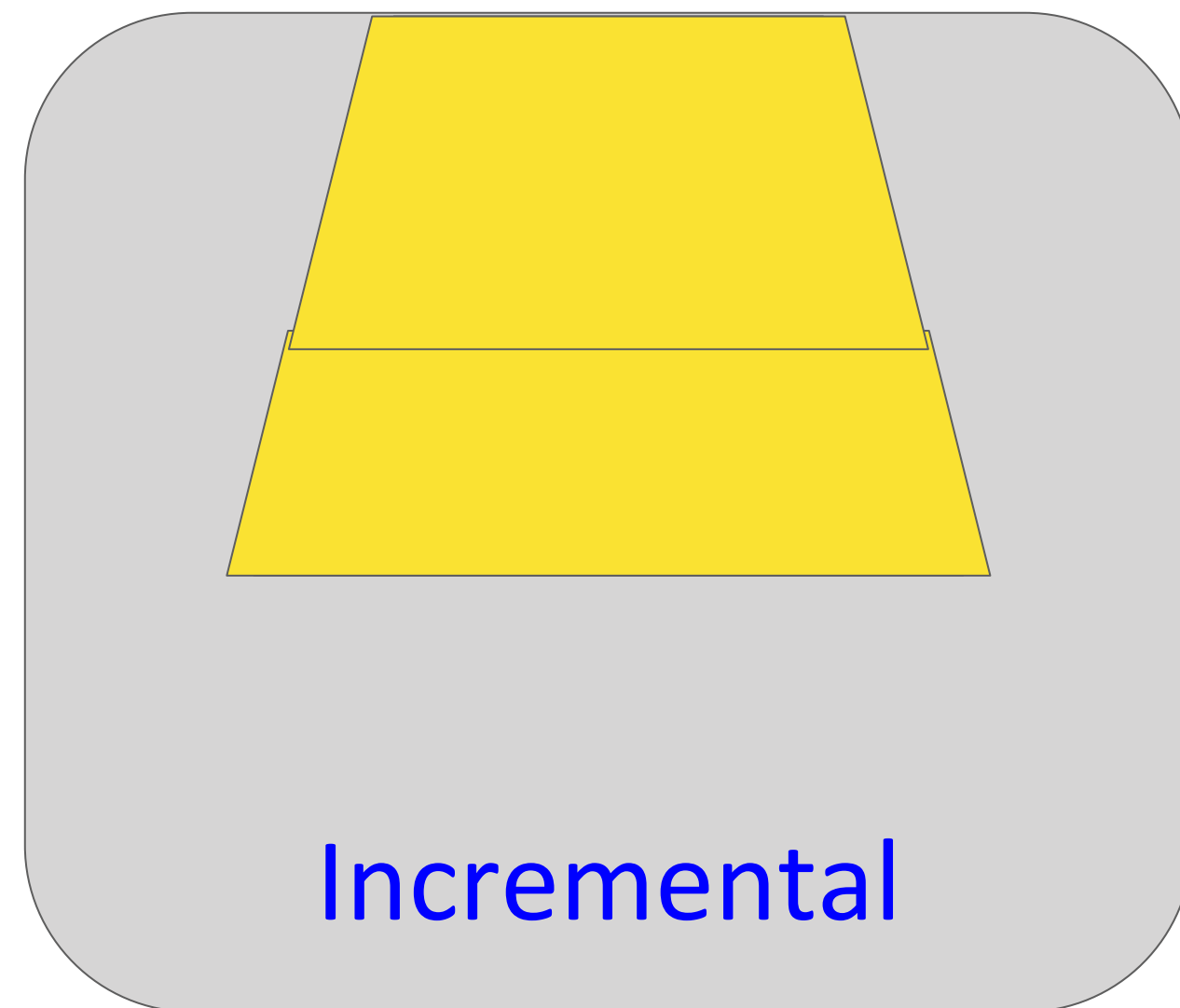
Epub 2019 Aug 14.

Clinical performance and chemical-physical properties of bulk fill composites resin –a systematic review and meta-analysis

- Conclusion - Clinical lifespan and performance of bulk fill composites were better or similar between 1 and 10 years.



Efficiency Conventional Fill



Almost impossible to be able to tell if the increment being placed is 2mm or 4mm, so there runs the risk of inaccurately placing increments and not curing the fill through

Efficiency Bulk Fill



Bulk fill is sought after, more often than not, for this one trait

Randomized Controlled Trial > BMC Oral Health. 2024 Dec 3;24(1):1461.

doi: 10.1186/s12903-024-05242-1.

Clinical evaluation of SonicFill bulk resin technique in the restoration of proximal deep caries of primary molars: a two-year randomized controlled trial

- Conclusion - More time-efficient in primary molars with deep proximal caries

Review > Dent J (Basel). 2025 Jul 15;13(7):320. doi: 10.3390/dj13070320.

Clinical Performance of Bulk-Fill Versus Incremental Composite Restorations in Primary Teeth: A Systematic Review of In Vivo Evidence

- Conclusion - Reduced procedure time by 2-4 min/restoration

Review > Polymers (Basel). 2020 Aug 10;12(8):1786. doi: 10.3390/polym12081786.

Clinical Effectiveness of Bulk-Fill and Conventional Resin Composite Restorations: Systematic Review and Meta-Analysis

- Conclusion - Bulk-fills are easy-to-use and use less chair-time to place



So what bulk fill do we have at the school?



None :(

Most dental composite manufacturers have a bulk fill product line

Apply a clinical decision tree to select appropriate placement techniques and materials for posterior composites, considering patient risk factors, cavity design, and occlusal demands.

Nathan Slocum

Search Strategy



P: Which composite to choose? conventional or bulk-fill

I: Conventional resin composite

C: Bulk-fill resin composite

O: Better clinical outcomes based on a variety of parameters

Databases used

Pubmed

Keywords

Conventional resin composite and
Bulk-fill resin composite

Pubmed search string

"Conventional resin composite"
AND "Bulk-fill resin composite"

Search results

137 found
3 were used
2 Systematic reviews
1 ADA Guideline

Highest level of evidence

Systematic reviews

Adhesives

- Single Bond Universal Adhesive
- Clearfil SE Bond
- Xeno III
- Xeno V
- AdheSE Bond
- OptiBond All-in-One
- Adper Single Bond 2
- Syntac Classic
- XP Bond
- Tetric N Bond Total-Etch
- Scotchbond Universal Adhesive
- Excite F
- Futurabond NR
- Peak Universal

CRC

- Filtek Z550
- Charisma Smart Composite
- Filtek Z350 XT
- Amelogen Plus
- Tetric EvoCeram
- Filtek Supreme Ultra Universal
- Tetric Ceram
- Filtek Z250
- Herculite Ultra
- Clearfil Photo Posterior
- Ceram X Mono
- Filtek Ultimate
- Grandio
- QuiXfil

Bulk 1 Step

- X-tra Fill Bull
- Filtek Bulk Fill Posterior Restorative
- Tetric EvoCeram Bulk-Fill
- Filtek Bulk Fill Flowable
- QuiXfil
- Filtek Bulk-Fill Restorative
- Tetric EvoCeram Bulk-Fill

Bulk 2 Step

- Filtek Bulk Fill Flow + Filtek Z350 X
- SureFil SDR + TPH3
- Filtek Bulk-Fill Flowable + Filtek P60
- SDR Flowable + Ceram X Mono



11 parameters:

- Absence of fractures
- Absence of discoloration or marginal staining
- Adequate marginal adaptation,
- Absence of post-operative sensitivity
- Absence of secondary caries
- Adequate color stability and translucency
- Proper surface texture
- Proper anatomical form
- Adequate tooth integrity (wear)
- Adequate restoration integrity
- Proper Occlusion

Including:

- Class 1/2/5 lesions
- Primary and permanent
- Liners and no liners



“This meta-analysis showed that there were no significant differences between conventional and bulk-fill resin compounds in terms of the type of restoration, the type of tooth restored, and the technique used.”

Arbildo-Vega et al., 2020

ADA guidelines reference use of bulk fill composites without caveat for fills of 4mm or more.



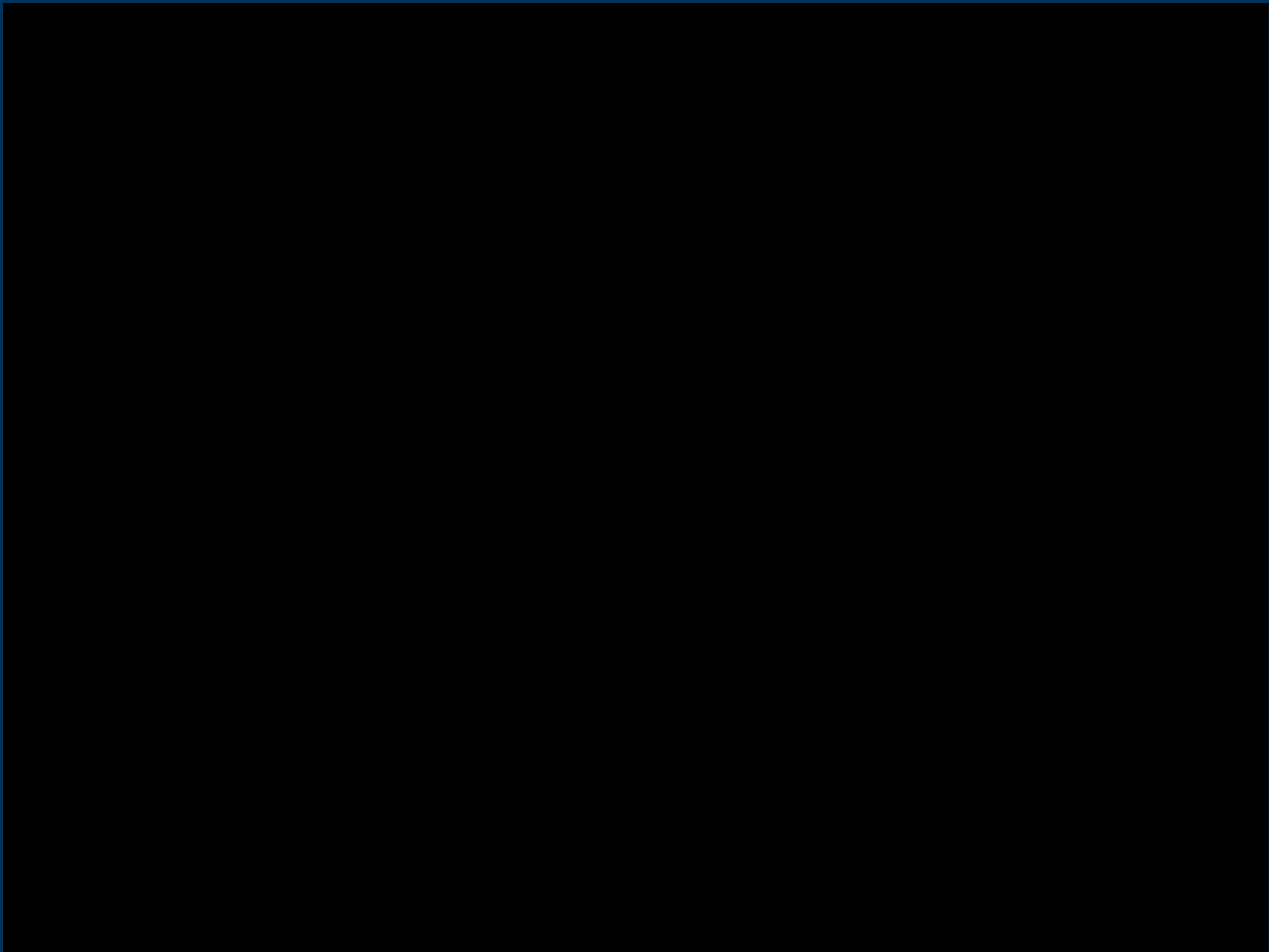


Making a decision tree

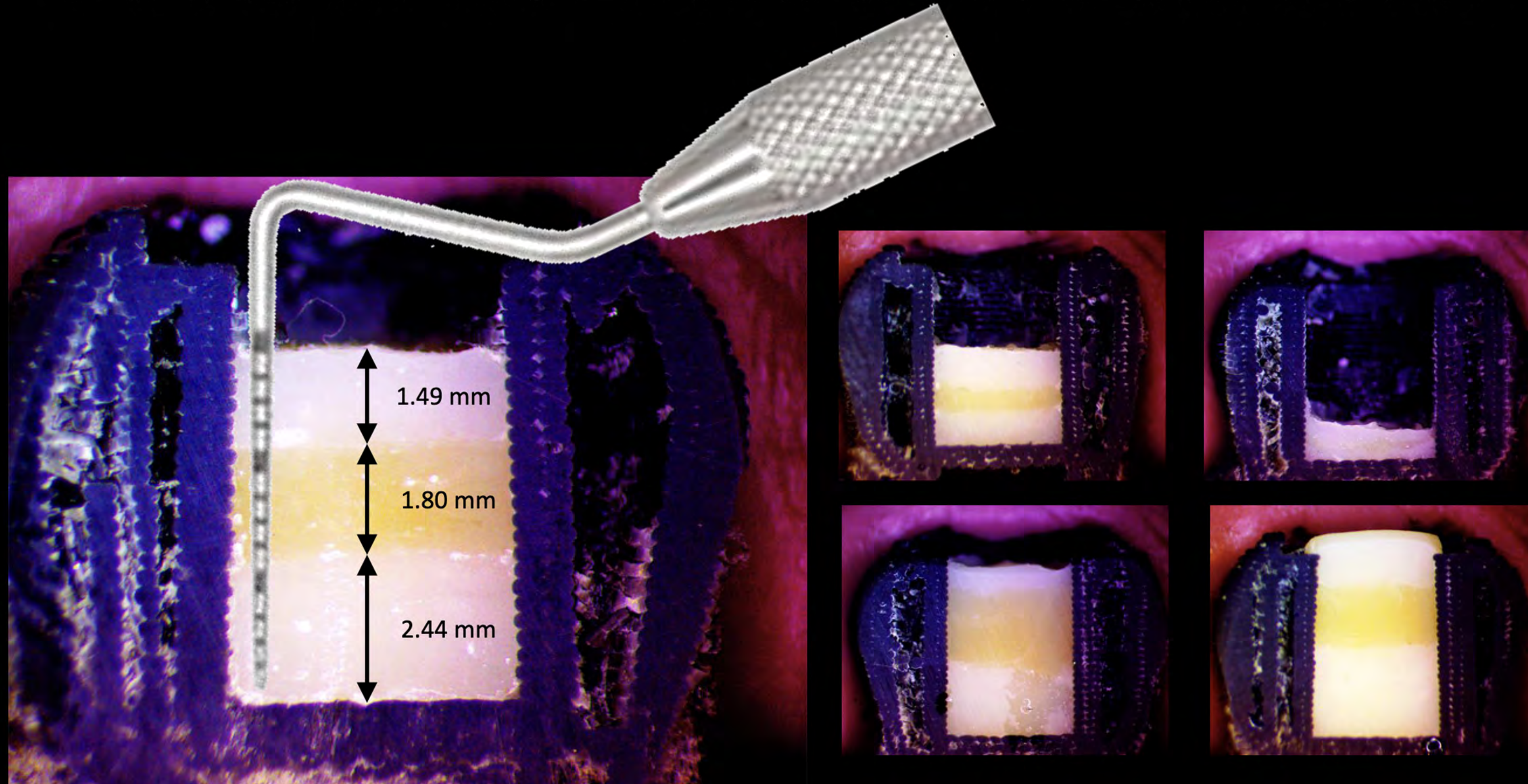
- We want the best choice
- Best according to what? Gold standard? Standard of care?
- Evidence based dentistry? Materials science? Clinical Experience?

There are no solutions,
only trade-offs

Thomas Sowell



FILLING IN 2-MM INCREMENTS



Dr. Rhoda Sword Dental students Dental Faculty

Even experienced clinicians cannot place increments within 0.5 mm thickness



So what's the best material?

- CRC because it's stronger and more esthetic

Should we always use CRC then?

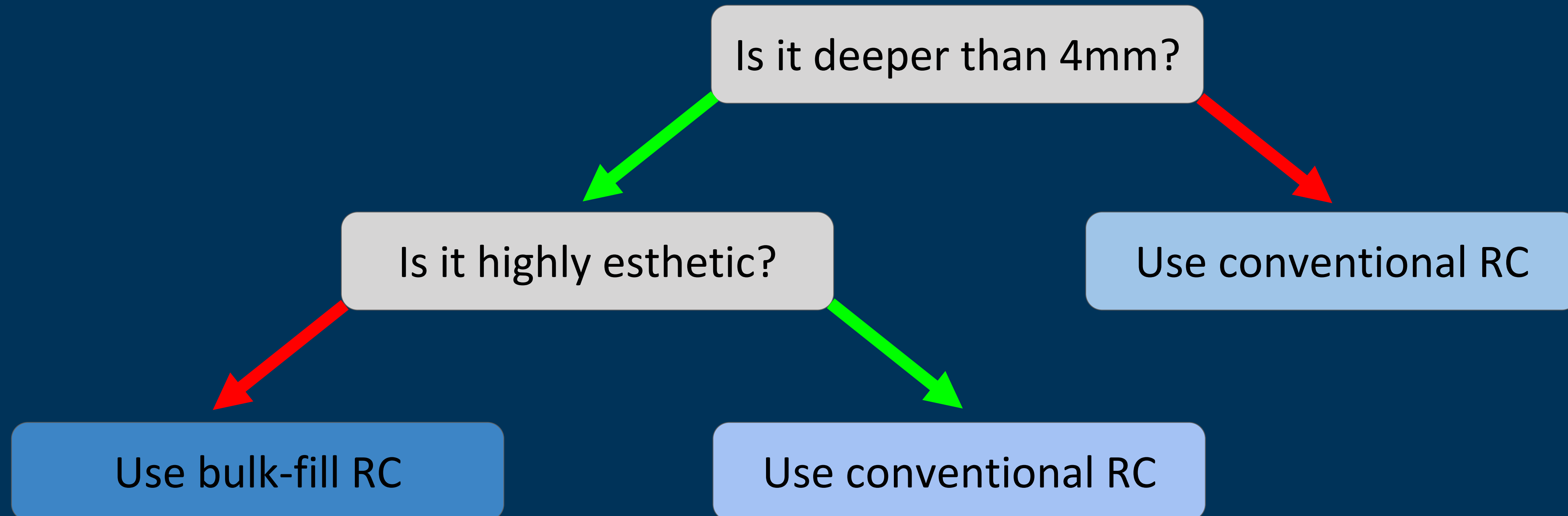
- Not necessarily, what is necessary to do the job. Think of a doorstep

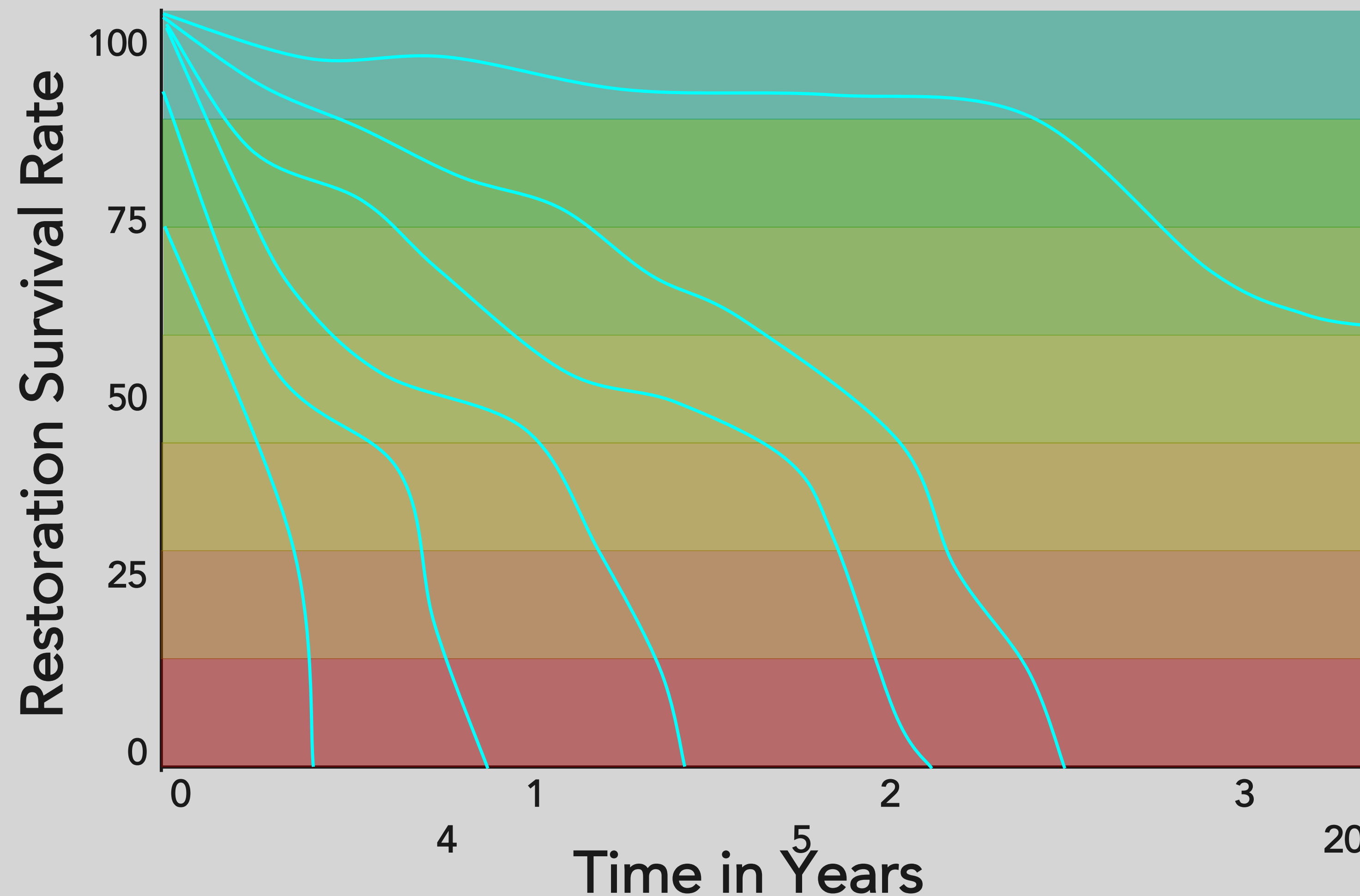
What's the best icecream flavor?

- The differences are more minute and lean more to do with personal preference
- What are you used to?
- What works well in my hands might not work well in yours



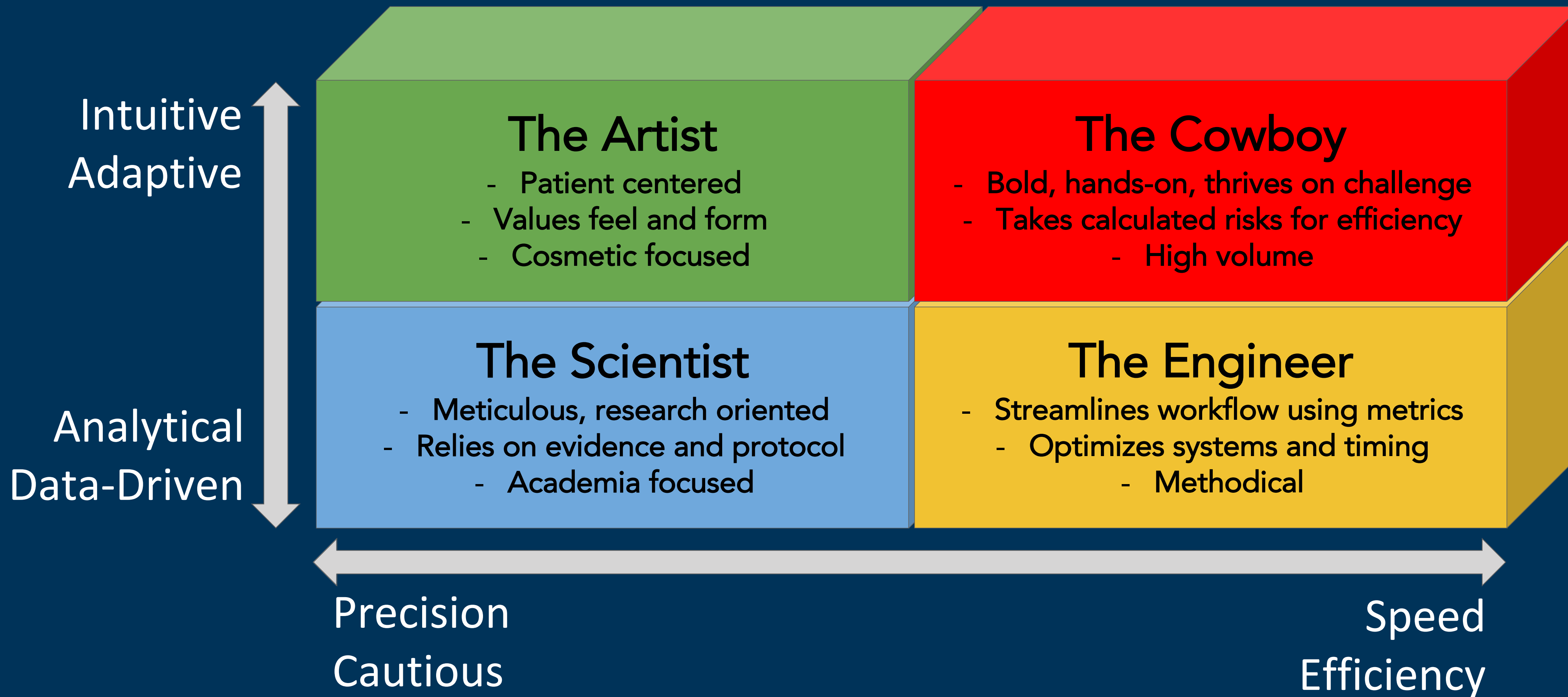
Decision Tree Time





- Didn't remove all caries
- A pinch of saliva got in during any of the steps
- Got a lil rubber dam in the preparation
- Toffelmier caused heme and slight contamination
- Insufficient etch
- Insufficient dry on prime
- Incomplete bond placement
- Void at base of fill
- Incomplete cure of material because of distance and/or thickness
- Void in between increments

Personality Tree Time



Based on the data you choose to emphasize you will derive your own unique solution

Case Presentation 1

Sarah-Lyn Saxon

Clinical Performance of Bulk-Fill Versus Incremental Composite Restorations in Primary Teeth: A Systematic Review of In Vivo Evidence



- In vivo randomized control trials- Children aged 3-12 with carious primary teeth
- Comparing bulk fill and conventional incremental composite restorations in primary teeth
- Primary Outcomes: Retention, 2-year survival rate, marginal integrity, and procedural efficiency
- Secondary Outcomes: Postoperative sensitivity, aesthetic appearance, and secondary caries

- Parallel groups and split mouth designs
- Follow up durations ranged from 12-60 months
- **“No consistent superiority of one approach over another was identified across the studies.”**
- **Primary Outcomes:**
 - Retention: Both bulk fill and incremental composite materials show high short-term retention
 - Survival Rate: Regardless of material, survival rates of restorations were consistently high
 - Marginal Integrity: Both materials showed good performance with MINOR deterioration over time
 - Procedural Efficiency: Several studies showed that bulk fill materials save time in pediatric settings (About 2 minutes)

Condition or domain being studied

P – Population

Primary teeth

Children

Pediatric dentistry

I – Intervention

Bulk-fill composite restorations

C – Comparator

Incremental composite restorations

Conventional composite restorations

O – Outcome

Restoration retention

Survival rate

Marginal integrity

Clinical performance

Clinical Evaluation of Self-Adhesive Bulk-Fill Composite Versus Conventional Nano-Hybrid Composite in Cervical Cavities—A Randomized Controlled Clinical Trial

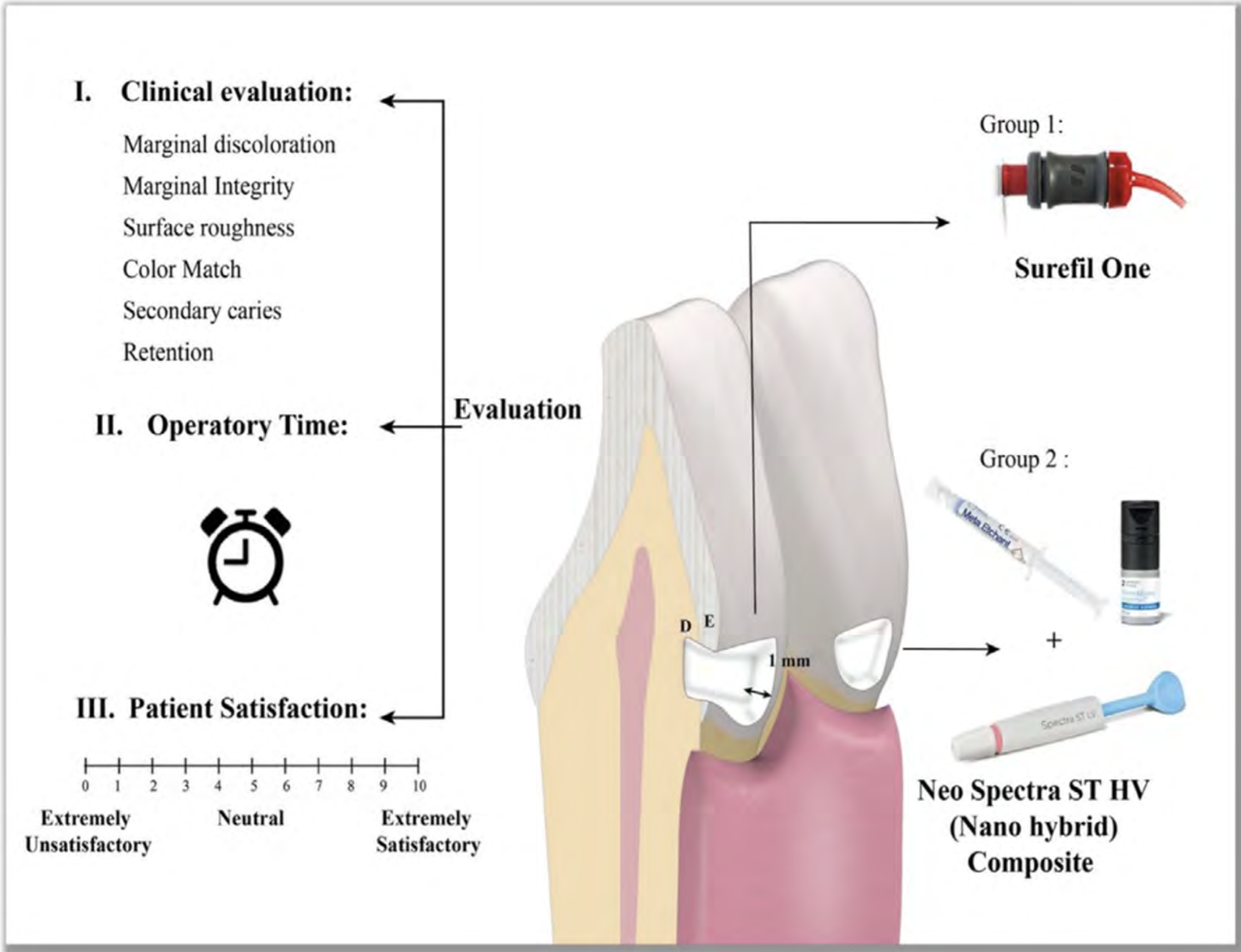


FIGURE 2 | Graphical abstract representing the flow of the trial's procedures.

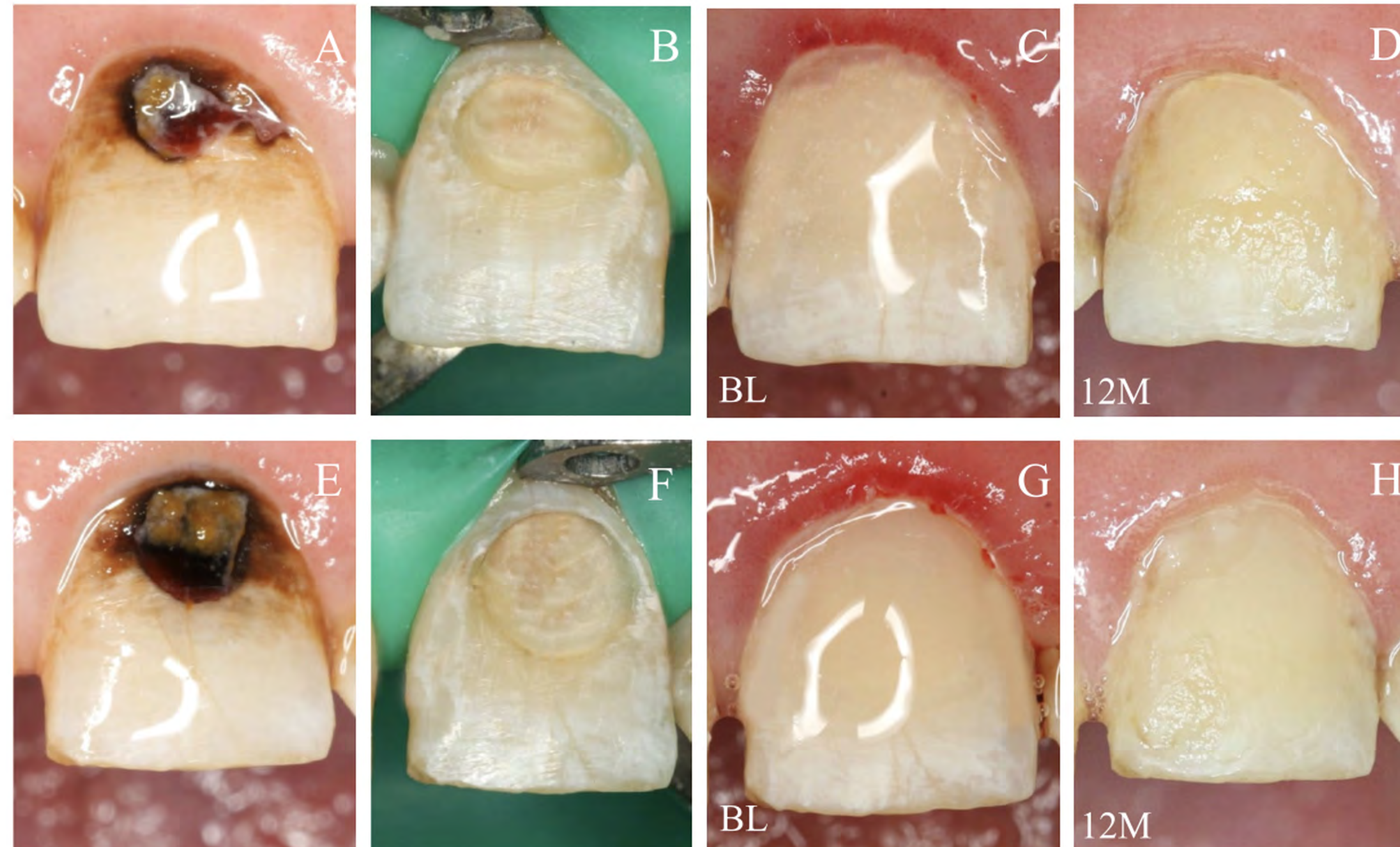


FIGURE 4 | Pre-clinical pictures of carious cervical lesions of upper central incisors (A, E). Cavity preparation pictures of carious cervical lesions of upper central incisors. (B, F) Differences between the Neo Spectra and Surefil One restoration at baseline and 12months. (C, D, G, and H) Neo Spectra restoration of upper left central incisor. (G, H) Surefil One restoration of upper right central (C, D).



TABLE 1 | Modified USPHS criteria.

Criterion	Score	Characteristics	Measuring unit	Methods of diagnosis
Marginal discoloration	Alpha	No discoloration between teeth structure and restorative material.	Ordinal	Visual and tactile examination
	Bravo	Non penetrating marginal discoloration which can polished away.		
	Charlie	Discoloration has penetrated margin in a pulpal direction.		
Marginal integrity	Alpha	Closely adapted, no detectable margin.	Ordinal	
	Bravo	Detectable marginal discrepancy, clinically acceptable.		
	Charlie	Marginal crevice, clinically unacceptable.		
Surface roughness	Alpha	No surface defects.	Ordinal	
	Bravo	Minimal surface defect.		
	Charlie	Severe surface defect.		
Color match	Alpha	Restoration color matches color of the tooth.	Ordinal	
	Bravo	Acceptable mismatch.		
	Charlie	Unacceptable mismatch.		
Secondary caries	Alpha	No secondary caries detected.	Ordinal	
	Charlie	Secondary caries detected.		
Retention	Alpha	No loss of restoration.	Ordinal	
	Charlie	Loss of restoration.		

TABLE 3 | Frequency (*n*) and percentage (%) for modified USPHS criteria for the intergroup comparison between materials within each follow-up and intragroup comparison within each material between different follow-up periods.

Outcome	Follow-up	Surefil		Neo Spectra				
		A	C	A	C			
Retention	Baseline	27 (100)	0 (0)	27 (100)	0 (0)	<i>p</i> = 1.0000		
	6 months	27 (100)	0 (0)	27 (100)	0 (0)	<i>p</i> = 1.0000		
	12 months	27 (100)	0 (0)	27 (100)	0 (0)	<i>p</i> = 1.0000		
	<i>p</i> value	<i>p</i> = 1.0000		<i>p</i> = 1.0000				
	Follow-up	Surefil		Neo Spectra				
		A	C	A	C			
Secondary caries	Baseline	27 (100)	0 (0)	27 (100)	0 (0)	<i>p</i> = 1.0000		
	6 months	25 (92.5)	2 (7.5)	27 (100)	0 (0)	<i>p</i> = 0.1534		
	12 months	24 (88.9)	3 (11.1)	27 (100)	0 (0)	<i>p</i> = 0.0774		
	<i>p</i> value	<i>p</i> = 0.097		<i>p</i> = 1.0000				
	Follow-up	Surefil			Neo Spectra			
		A	B	C	A	B	C	
Color match	Baseline	26 (96.3)	1 (3.7)	0 (0)	27 (100)	0 (0)	0 (0)	<i>p</i> = 0.3173
	6 months	17 (63)	10 (37)	0 (0)	24 (88.9)	3 (11.1)	0 (0)	<i>p</i> = 0.0273*
	12 months	5 (18.5)	18 (66.6)	4 (14.8)	20 (74)	7 (26)	0 (0)	<i>p</i> = 0.0001*
	<i>p</i> value	<i>p</i> < 0.001*			<i>p</i> = 0.005*			



	Follow-up	Surefil			Neo Spectra			
		A	B	C	A	B	C	
Marginal integrity	Baseline	27 (100)	0 (0)	0 (0)	27 (100)	0 (0)	0 (0)	<i>p</i> = 1.0000
	6 months	21 (77.8)	6 (22.2)	0 (0)	26 (96.3)	1 (3.7)	0 (0)	<i>p</i> = 0.0448*
	12 months	21 (77.8)	6 (22.2)	0 (0)	26 (96.3)	1 (3.7)	0 (0)	<i>p</i> = 0.0448*
	<i>p</i> value	<i>p</i> = 0.002*			<i>p</i> = 0.368			
	Follow-up	Surefil			Neo Spectra			
		A	B	C	A	B	C	
Marginal discoloration	Baseline	27 (100)	0 (0)	0 (0)	27 (100)	0 (0)	0 (0)	<i>p</i> = 1.0000
	6 months	15 (55.6)	12 (44.4)	0 (0)	23 (85.2)	4 (14.8)	0 (0)	<i>p</i> = 0.0182*
	12 months	14 (51.8)	13 (48.2)	0 (0)	22 (81.5)	5 (18.5)	0 (0)	<i>p</i> = 0.0221*
	<i>p</i> value	<i>p</i> < 0.001*			<i>p</i> = 0.015*			
	Follow-up	Surefil			Neo Spectra			
		A	B	C	A	B	C	
Surface roughness	Baseline	27 (100)	0 (0)	0 (0)	27 (100)	0 (0)	0 (0)	<i>p</i> = 1.0000
	6 months	3 (11.1)	24 (88.9)	0 (0)	18 (66.7)	9 (33.3)	0 (0)	<i>p</i> < 0.0001*
	12 months	2 (7.5)	25 (92.5)	0 (0)	18 (66.7)	9 (33.3)	0 (0)	<i>p</i> < 0.0001*
	<i>p</i> value	<i>p</i> < 0.001*			<i>p</i> < 0.001*			

Note: The asterisk (*) indicates a statistically significant difference with a *p*-value less than 0.05.

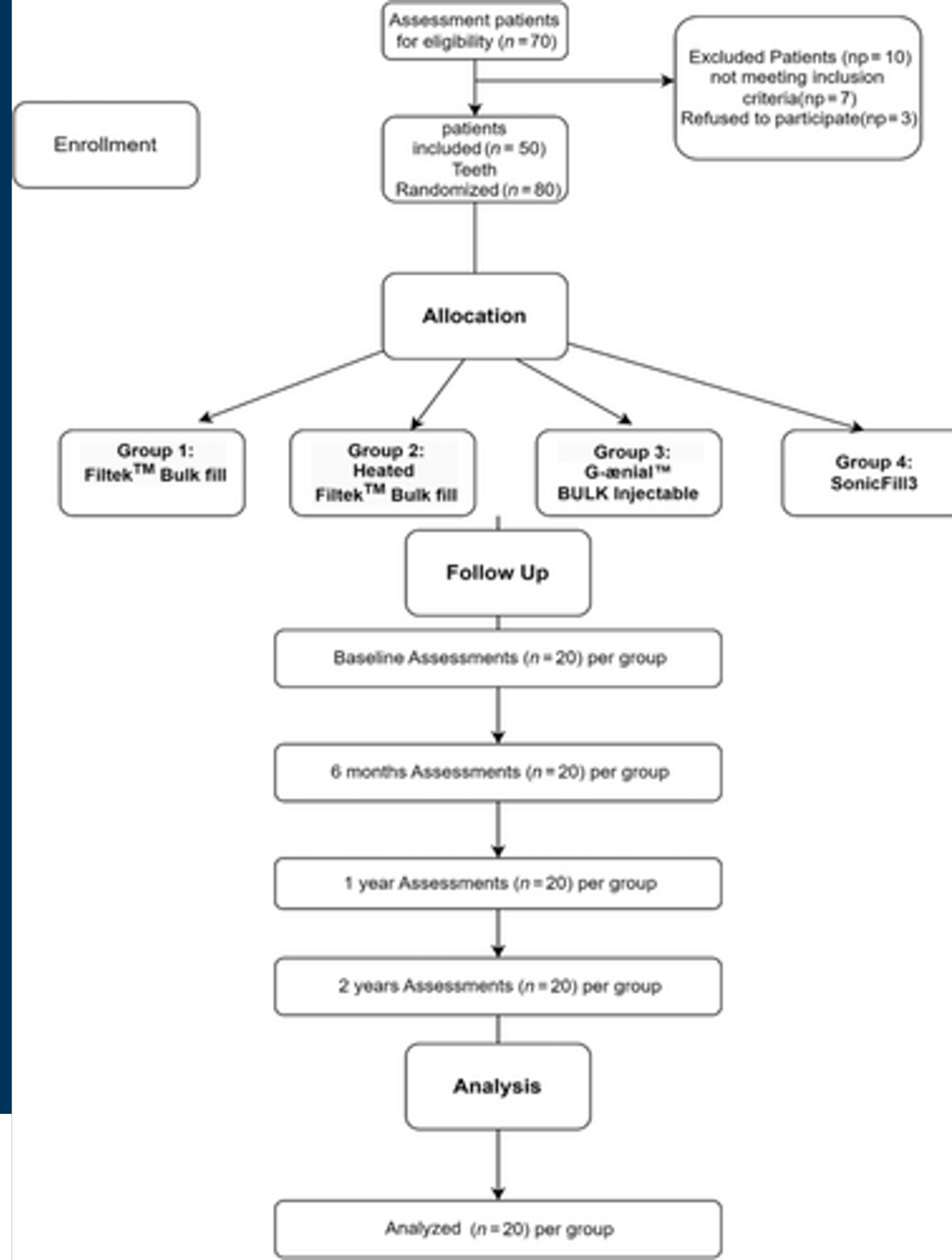
Case Presentation 2

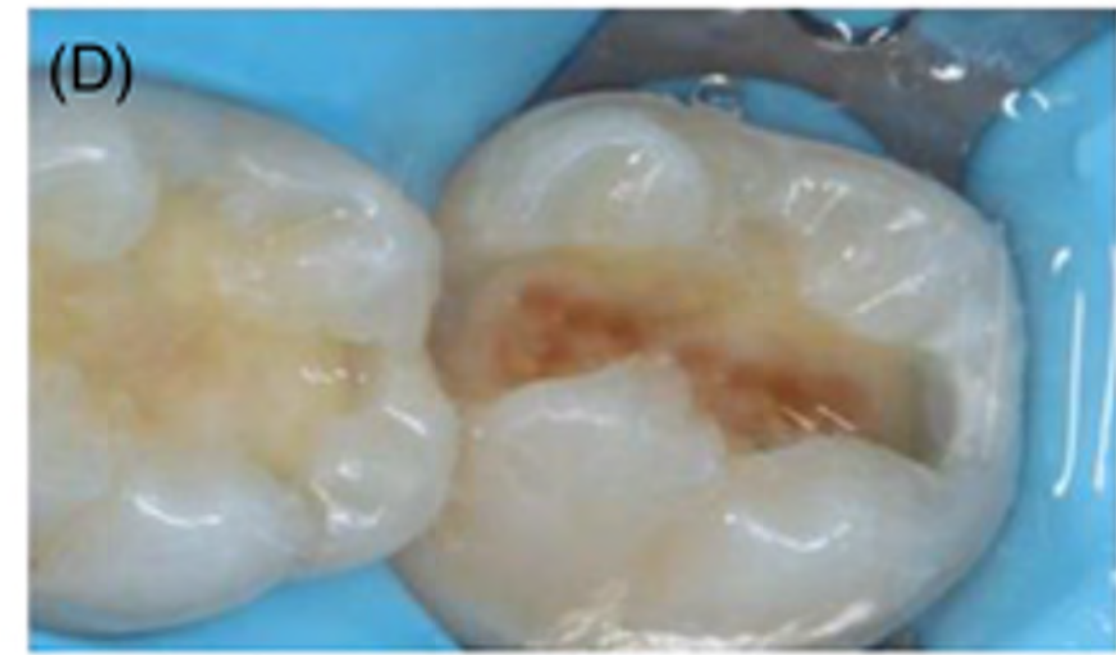
Morgan Reeves

Clinical performance of different bulk-fill composite resin systems in class II cavities: A 2-year randomized clinical trial



- 2 year in vivo randomized clinical trial including 50 patients with 80 carious molars between 18-35 years old.
- 50 patients were randomly divided into four groups:
 1. Filtek™ Bulk Fill Restorative System
 2. Heated Filtek™ Bulk Fill Restorative System
 3. G-ænial™ BULK Injectable Restorative System
 4. SonicFill3 Restorative System





Clinical Procedure:

- 80 Class II restorations.
- All restorative procedures were performed with rubber dam isolation.
- Preparation was limited to no more than $\frac{1}{3}$ intercuspal distance bucco-lingually.
- Gingival margins included sound enamel and did not violate biological depth.
- Preparations were at least 4mm in depth.
- All preps restored with garrison ring, pre-contoured metallic bands, and plastic wedges.

(A) A preoperative image of tooth no.36. (B): a preoperative image of tooth no.37. (C & D) cavity preparation.
(E & F) matricing system.



Evaluation: Esthetic properties

- Surface luster:

- 1= Luster comparable to enamel 2= Slightly dull

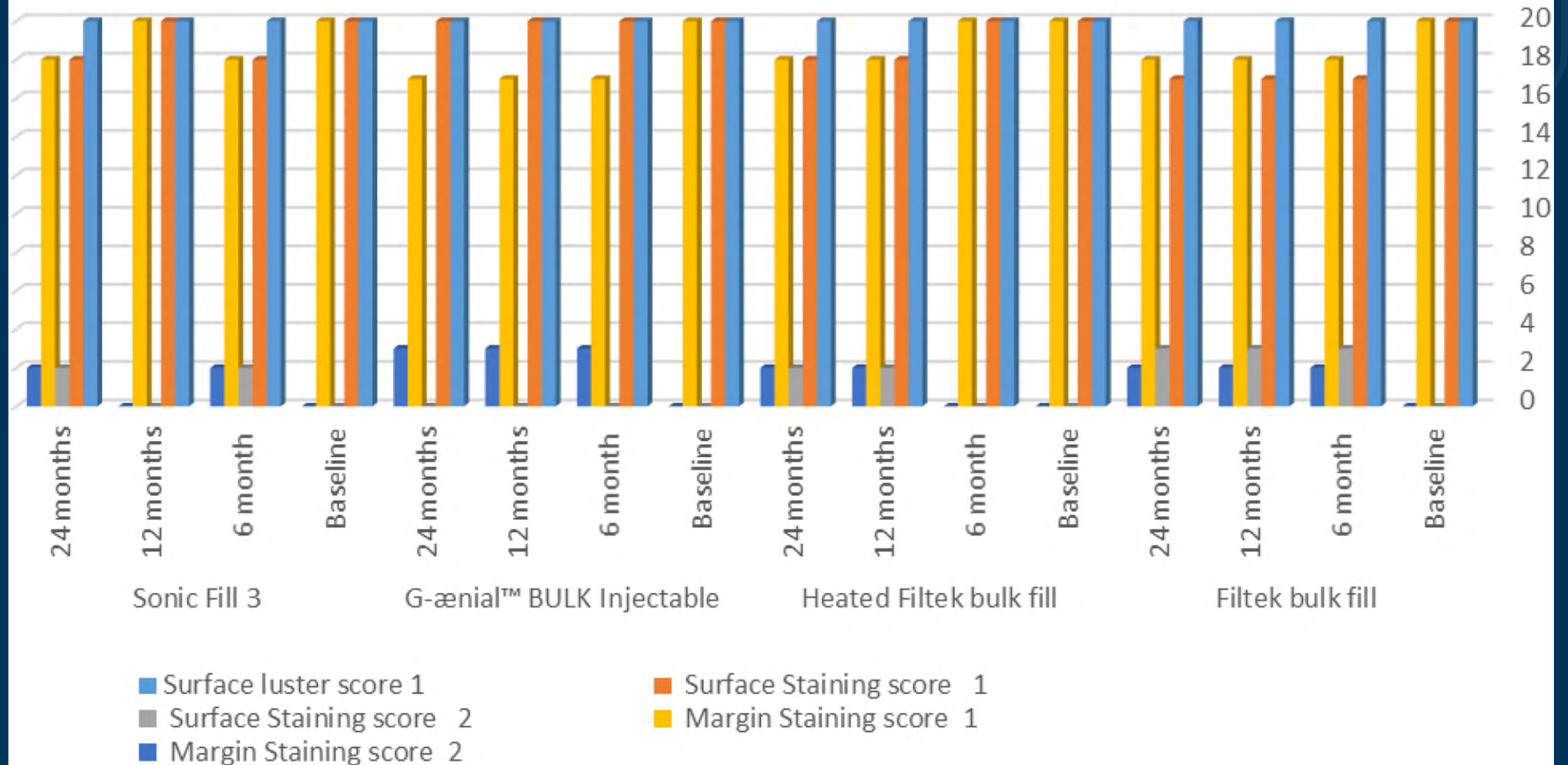
- Surface staining:

- 1= No surface staining 2=Minor surface staining, easily removable by polishing.

- Margin Staining:

- 1= No margin staining 2=Minor marginal staining, easily Removable by polishing.

Esthetic Properties



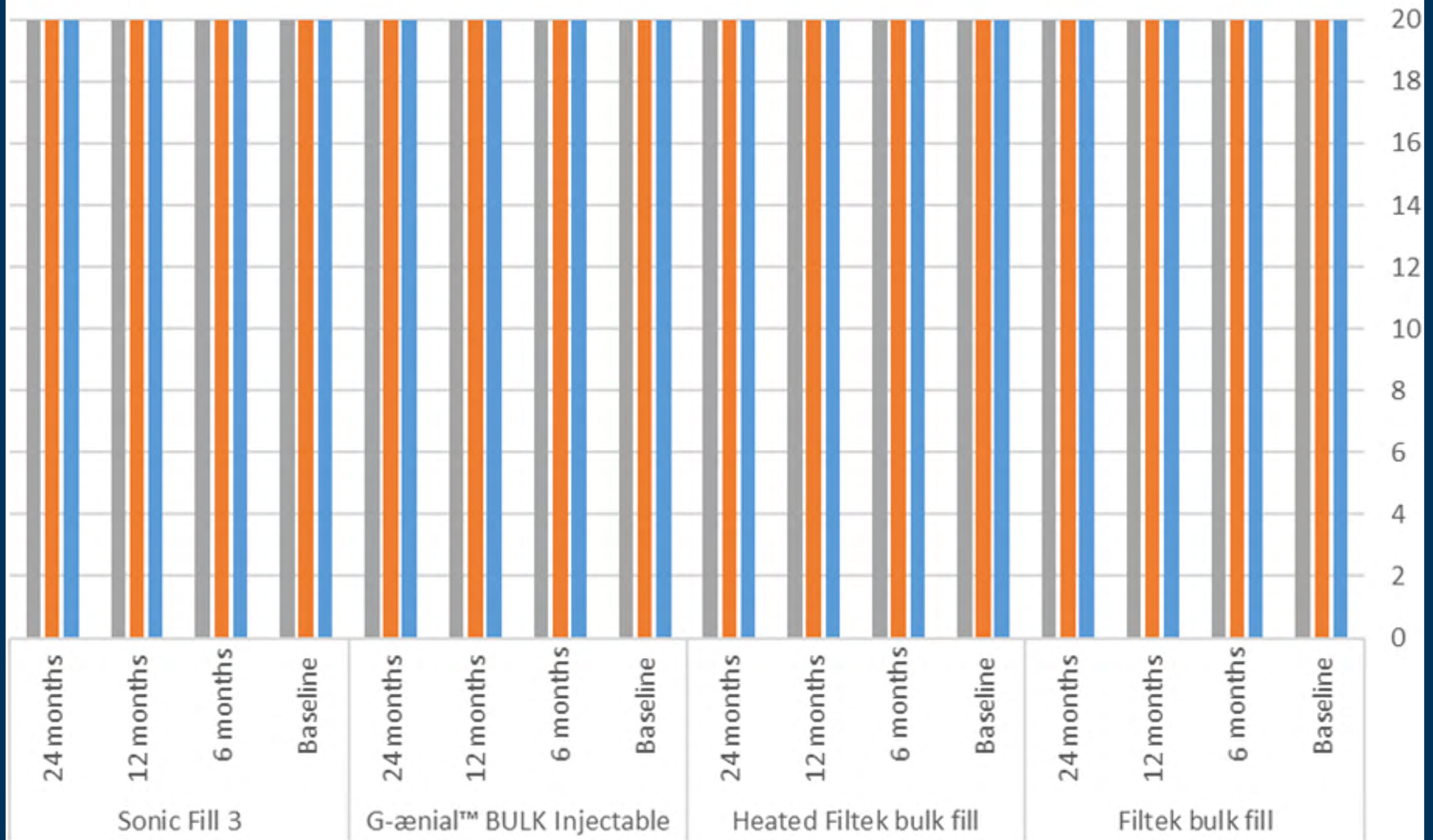


Evaluation: Functional properties

- Fracture of material:
 - 1= No fractures/cracks 2= Small hairline fracture
- Marginal adaptation:
 - 1= Harmonious outline/no gaps 2= Small margin gap, white lines
- Proximal anatomical form:
 - 1= Normal contact point/contour 2= Contact slightly too strong, deficient contour



Functional Properties



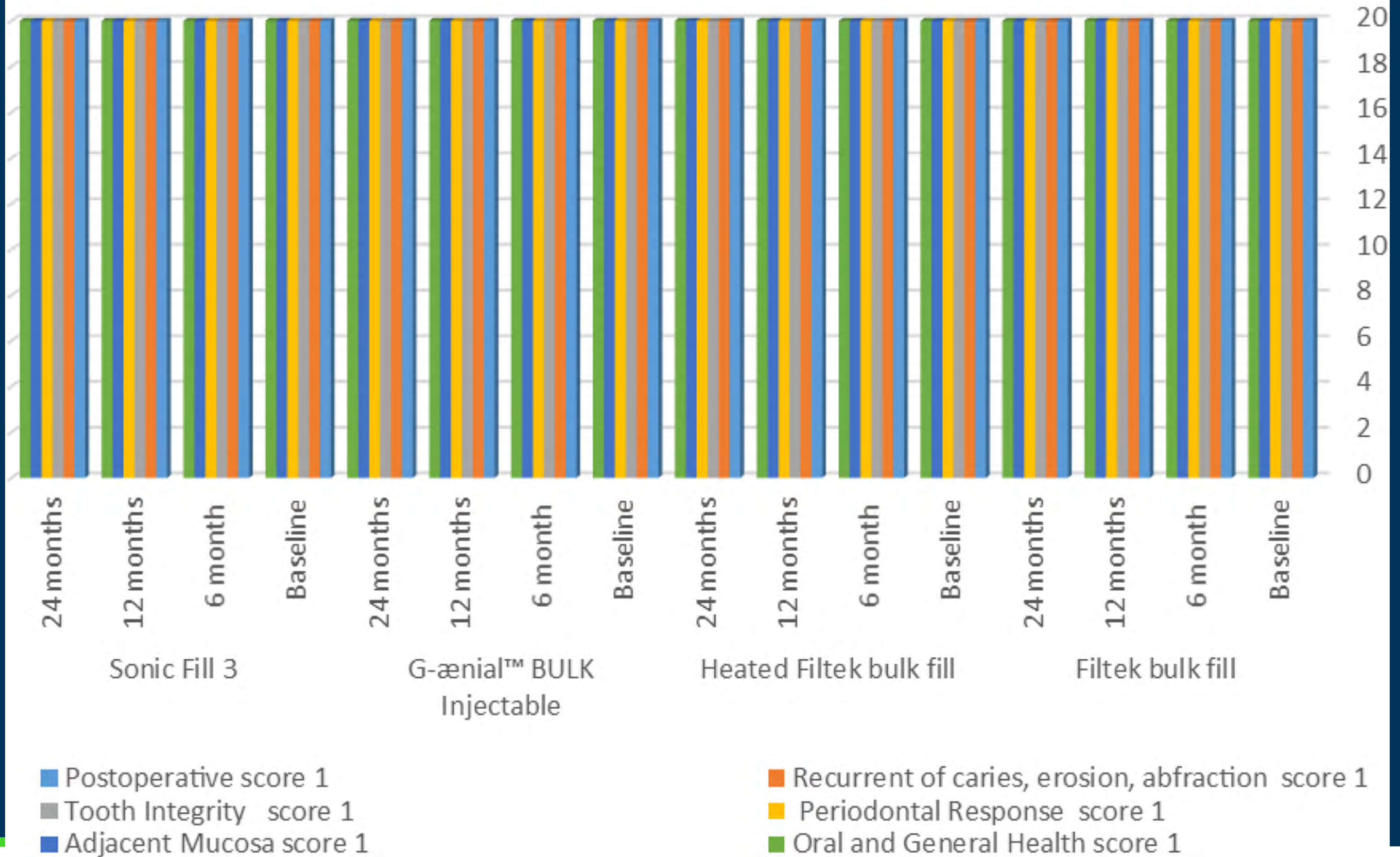
■ Fracture of material and retention score 1
 ■ Marginal adaptation score 1
 ■ Proximal anatomical form (contact) score 1

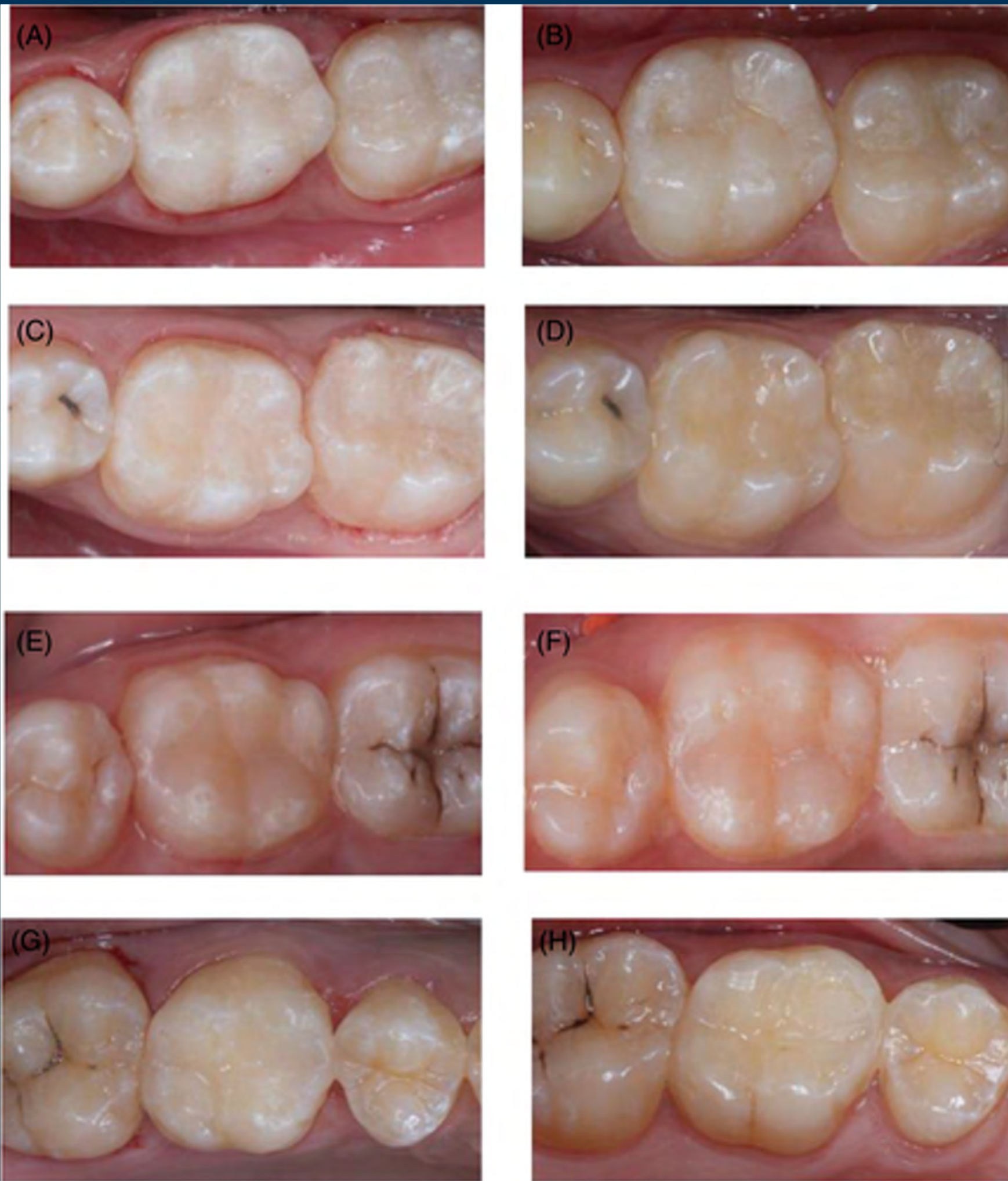


Evaluation: Biologic properties

- Post-op sensitivity
 - 1= No sensitivity 2= Minor hypersensitivity
- Recurrence of caries:
 - 1= No recurrence 2= Small and localized demineralization
- Tooth integrity:
 - 1= Complete integrity 2= Small marginal enamel fracture

Biological Properties





Conclusions:

- No significant difference in clinical performance between the different types of bulk fill composites.
- Bulk fill composite is an acceptable restoration in comparison to conventional composite.
- Longevity of bulk fill composites relies on patient- and operator- related factors.

(A) Baseline SonicFill3 bulk-fill restoration. (B) 24 months follow-up SonicFill3 bulk-fill restoration. (C) Baseline Filtek Bulk-fill restoration (D) 24 months follow-up Filtek Bulk-fill restoration. (E) Baseline G-aenial™ BULK Injectable composite restoration. (F) 24 months follow-up G-aenial™ BULK Injectable composite restoration. (G) Baseline Heated Filtek™ Bulk Fill. (H) 24 months follow-up of Heated Filtek™ Bulk Fill.



Behavioral Objectives



1. **Identify** the biological and mechanical factors that contribute to the failure of posterior composite restorations, including secondary caries, fracture, wear, and marginal breakdown.
 - a. Polymerization shrinkage during curing creates microscopic gaps between the tooth and the resin composite, allowing saliva, enzymes, and bacteria to infiltrate and degrade the tooth-resin interface, leading to marginal breakdown, sensitivity, and secondary caries. Secondary caries, the leading cause of failure, often arise from residual primary caries left during preparation or bacterial penetration along restoration margins, with high caries-risk patients being especially susceptible. Mechanically, restorations endure cyclic loading from mastication and bruxism, which can exceed the material's strength, resulting in microcracks, fatigue, and eventual fracture or wear. Over time, these biological and mechanical stresses interact, accelerating material degradation and the recurrence of carious lesions at restoration margins.
2. **Differentiate** between conventional incremental layering and bulk-fill composite techniques by explaining their effects on polymerization shrinkage, stress distribution, and marginal adaptation.
 - a. Incremental layering places composite in multiple ≤ 2 -mm layers, with each layer light-cured separately. This technique helps reduce polymerization shrinkage, better stress distribution and marginal adaptation because only a small volume of material is cured at a time.
 - b. In contrast, bulk-fill composites are designed to be placed in a single 4–5-mm increment. Bulk-fill materials control polymerization shrinkage and internal stress through modified resin chemistry and stress-relieving monomers rather than through layering. Bulk-fill demonstrates comparable clinical performance to incremental layering.



Behavioral Objectives



3. **Explain** how material properties (filler type, translucency, depth of cure, and modulus of elasticity) influence the long-term clinical performance of posterior composites.
 - a. For posterior composites, smaller filler type composites (micro-filled) lead to increased wear and fracture risk and a poor longevity, while larger filler type and mixed filler type composites exhibit greater mechanical strength to withstand masticatory forces reducing fracture risk. More opaque composites and composite layers deeper than 2mm (conventional)/4mm (bulk-fill) reduce the ability to fully polymerize the composite at the base of the restoration leading to a weaker restoration and increased recurrent decay, while more translucent composites and depths at or less than 2mm/4mm have a better long-term performance by allowing the base to be completely polymerized. Composites with an elastic modulus similar to dentin can withstand masticatory forces without fracturing and prevent polymerization shrinkage marginal gap formation (great longevity). An elastic modulus higher than dentin can lead to fractured restorations, while a lower elastic modulus will deform under masticatory forces.
4. **Compare and contrast** the advantages and limitations of bulk-fill and traditional layering techniques based on current scientific evidence, focusing on marginal integrity, restoration longevity, and clinical efficiency.
 - a. Research shows that bulk fill composites have similar longevity, margin integrity, and efficiency compared to conventional layered composite. Bulk fill should really be called bulk cure as this is the primary attribute of these composites—their ability to be cured beyond the conventional 2mm. Many properties of bulk fill are highly technique sensitive just like conventional composite. Bulk fill's use is highly dependent on the clinicians skill, level of risk, type of bulk fill, level of isolation, and the patient case and type of restoration. When all strengths and weaknesses are considered, the potential best use for bulk fill composites would be to use them in the same manner as conventional incremental composites. This decreases clinician increment thickness error while retaining all of the benefits of modern composites.



Behavioral Objectives



5. [Apply](#) a clinical decision tree to **select** appropriate placement techniques and materials for posterior composites, considering patient risk factors, cavity design, and occlusal demands.
 - a. The most current meta-analysis agrees that there is no clinical difference between the use of bulk-cure and conventional resin composites. The choice of whether to use each material falls into finer details of optimisation rather than necessity. Optimisation of time, esthetics, cost, ease of use. As far as a longevity difference, there has been none noted as of yet.



Six Thinking Hats®



White Hat (Faculty): Provides objective data and scientific evidence about the dental material.



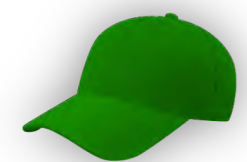
Red Hat (Patient): Expresses emotions, concerns, and preferences regarding the material.



Black Hat (Dentist): Evaluates potential risks, challenges, and clinical implications.



Yellow Hat (Sales Rep): Highlights the benefits and advantages of the material.



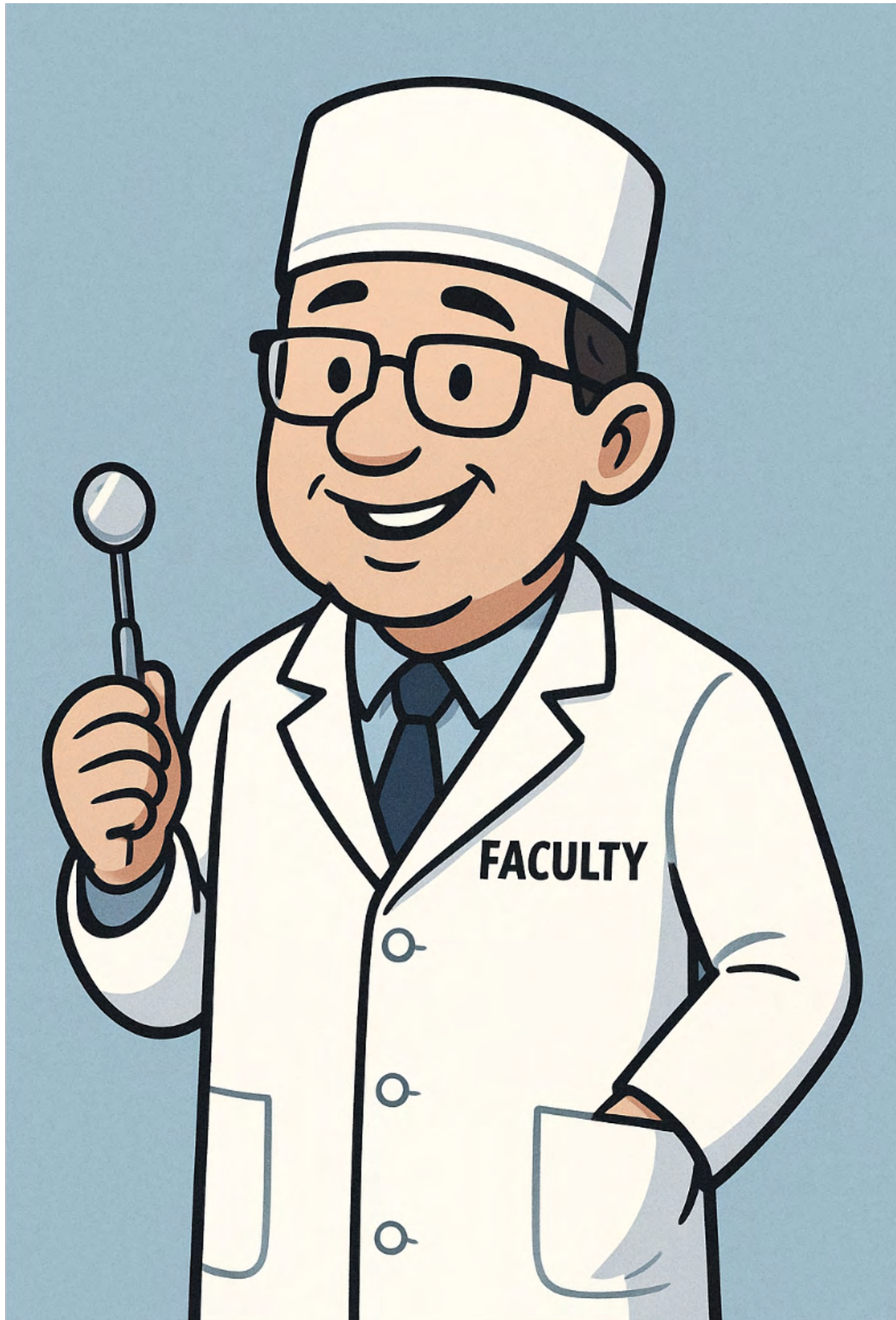
Green Hat (Manufacturer): Explores innovation, and future possibilities for the material.



Blue Hat (Board of Dentistry): Ensures all perspectives and standards are considered.

The Kahoot! logo is centered on a dark blue background. The word "Kahoot!" is written in a white, bold, sans-serif font. A large, light blue arrow points from the left towards the text, partially overlapping it.

Kahoot!



"The research shows that there are pros and cons to both bulk fill material and traditional composite materials. Many in vitro studies show us that bulk fill composites can cure up to 4 mm deep when cured under proper conditions. Because we are not accurate enough to measure 2mm for traditional composite compared to 4 mm for bulk fill composite, this is a huge plus. However, traditional composites give us the advantage of layering the material to give us proper shade matching. It is important for students to be taught multiple methods so that they can choose the proper material based on the clinical scenario."



"I'll be honest, I totally love the idea of this procedure being fast, but is this bulk fill material actually the best for my filling? I trust you I guess but I want what is going to last the longest"



"I choose to use both bulk fill composite material and traditional composite. While I use bulk fill material more so with posterior teeth and traditional composite with anterior, esthetic cases, I like to have options when it comes to treating patients. I remain cautious when using the bulk fill materials so that I am properly curing of the material, using appropriate bonding systems, and handling the product as the manufacturer desired. I also want to make sure I am using the appropriate material based on what the procedure implies. One example being, I will not use bulk fill material on an anterior, highly esthetic case. I need to be more precise and have more accurate control of the material."



"Do you want to be able to see MORE patients and have SHORTER appointments? Look no further than bulk fill composite material! This material allows you to cure up to 4-5 mm deep in ONE increment. This material also ensures well adapted margins therefore delivering consistency in your restorations."

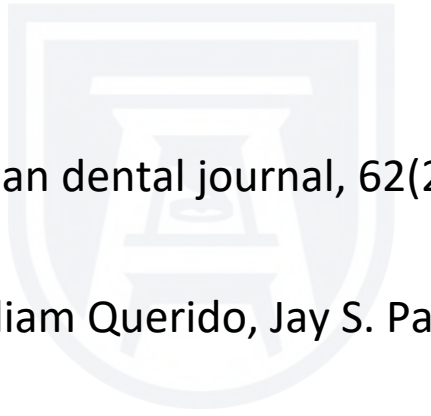


"Due to the desire to make dentists as efficient as possible, bulk fill composites have entered the dental world. There is constant research being done in order to address the challenges that dentists face when it comes to restoring teeth. We hope that one day we will be able to incorporate digital dentistry into composites so that you may properly shade match and efficiently polish materials being used. Overall, we aim to develop products that simplify procedures while giving the patient the best outcome."



"At the end of the day, the highest standard of patient care is the expectation. Although the conversation continues about the comparison of bulk fill composites and traditional composites, there is really no "right" answer as to which material is the "best." We expect that you practice based on the standard given by the manufacturer, and we also hold you to the expectation that you remain a lifelong learner. As a dentist, you must use the evidence that is provided by thorough research in order to use the proper material based on the patient's need."

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<https://www.youtube.com/watch?v=fad108koEBU&t=283s> (omnichroma composite)

<https://www.youtube.com/watch?v=WpRiWopnhUA&t=13s> (class 2 placement technique)



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